# BASICS OF ENVIRONMENTAL SCIENCE

**(BS PROGRAMME)**

**Course Code: 9389** **Units: 1–9**

Department of Nutritional Sciences & Environmental Design
Faculty of Sciences
ALLAMA IQBAL OPEN UNIVERSITY

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**Department of Environmental Sciences**
**Faculty of Sciences**
**Allama Iqbal Open University**
**Islamabad**

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# CONTENTS


***Page #***

**Foreword** ..... v

**Message from the Dean** ..... vi

**Acknowledgements** ..... vii

**Introduction of the Course** ..... viii

**Objectives of the Course** ..... viii

**Unit-1:** Introduction to Environmental Science ..... 1

**Unit-2:** Basics of Physical Components of Natural Environment ..... 13

**Unit-3:** Basics of Biological Components of Natural Environment ..... 43

**Unit-4:** Basics of Climatology ..... 53

**Unit-5:** Aqueous Ecosystems ..... 65

**Unit-6:** Land Ecosystems ..... 81

**Unit-7:** Introduction to Environmental Chemistry ..... 97

**Unit-8:** Introduction to Environmental Physics ..... 107

**Unit-9:** Environmental Conservation ..... 117

**References** ..... 139

# FOREWORD

The world is facing an unprecedented environmental crisis. Industrialization led to global warming and climate change issues which affected us in many ways. There is a rise in extreme weather events, floods, droughts, rising sea levels, and glacier melting. To safeguard our environment, there is a need for public awareness and a deeper understanding of the environmental problems to build a sustainable future. Keeping in mind the significance of environment and environmental education it is pertinent to impart a basic understanding of environmental structure and its components in different schools of thought. The significance of environmental education is further raised by keeping in mind the current environmental challenges that are being faced by the world, particularly Pakistan.

This book, Basics of Environmental Science, aims to provide an understanding of the environment, its basic structure and its role as a multidisciplinary field. It explains the interactions between natural systems and human influence to offer greater insights into how complex environmental issues should be addressed. The book is specifically designed to provide sufficient information to its readers about the environment and challenges that are being faced in different environmental mediums. We hope that this work will inspire the readers to actively engage in protecting the environment, make informed decisions, and contribute to a healthier and sustainable future.

**Prof. Dr. Nasir Mahmood**
**Vice-Chancellor**

## MESSAGE FROM THE DEAN

On behalf of the course team of "Basics of Environmental Sciences" I appreciate and welcome you to this course. The environment and environmental challenges are no longer a new term to the society. Environmental Sciences stands as the critical interface between human and earth's delicate physical and biological systems. As the growing concerns about the environment encompass every field of life, therefore it is pertinent to impart basic knowledge about the environment and environmental systems. Endowed with ideas of environment and environmental systems we believe that you will be able to understand and apply the critical balance of man and nature in your field of work. Environmental science is recognized as an interdisciplinary study; therefore, this course is designed to give an overview of the different environmental systems. I hope that you will find this course interesting, useful and inspiring so that together we can achieve the goal of a safe and healthy environment for all.

Best Wishes

**Prof. Dr. Hajra Ahmad**
**Dean Faculty of Sciences**

# ACKNOWLEDGEMENTS

We extend our sincere gratitude to all the contributors who have authored units in this book, each drawing on their expertise in their respective fields. We also acknowledge the valuable work of those whose research has been cited throughout this book. The authors of this book are Dr. Sofia Khalid, Dr. Zahidullah, Dr. Samia Qadeer, Dr. Sapna, and Dr. Bibi Ilmas from the Faculty of Sciences, Department of Environmental Sciences, Allama Iqbal Open University, Islamabad.

We are deeply grateful to Dr. Rohama Gill, Dr. Azeem Khalid, and Dr. Muzammil Anjum for their keen interest in this book and their guidance and support.

We are also grateful to Professor Dr. Hajra Ahmad, Dean of the Faculty of Sciences, and Professor Dr. Nasir Mahmood, Vice-Chancellor of Allama Iqbal Open University, Islamabad, for their encouragement and support in the development of this book.

## INTRODUCTION OF THE COURSE

Welcome to the fascinating world of the environmental sciences. Keeping in mind the beautiful nature and natural systems around us, this course is articulated to impart general information about the environment and environmental systems for readers from any other field. Being an interdisciplinary field of study, the course basic of environmental sciences is particularly designed for the students of general education, so they can have a better understanding of their role in structuring the natural environmental systems. The course will provide a comprehensive understanding of the natural world and its intricate systems which are critical to sustain life on this beautiful planet. The course is designed to lay the foundation for understanding the complex interactions of the environment in a simplified way. Through structured unit design, each unit is specifically placed to strengthen the previous knowledge systematically. This is particularly articulated by the contents that provide basic but sufficient information about the earth's physical features and, the role of diverse living components in sustaining the earth's whole ecosystem. Considering the significant need for climate education, the course also entails basic information about weather, weather patterns climate systems and their effect on our daily lives. Different principles and underlying environmental phenomena of our daily life are embedded in this course, which can be considered as part of general knowledge for the students adopting this course from the other stream of studies. The overall goal of environmental education is to provide a concept of environmental conservation, preservice and sustainable development. We believe that this course will equip the students with sufficient information about the environment that can effectively be applied in their field of study.

## OBJECTIVES OF THE COURSE

The course is designed to equip the students with knowledge related to:

- Basic Concepts of Environment
- Structure of the Environment
- Different Environmental Systems
- Physical and Biological Features of the Environment
- Introduction to Climate Change
- Environmental Challenges we are Currently Facing

**Unit–1**

# INTRODUCTION TO ENVIRONMENTAL SCIENCE

**Compiled by: Dr. Sofia Khalid**

**Reviewed by: Dr. Rohama Gill**

# CONTENTS

***Page #***

Introduction ..... 3

Objectives ..... 3

1.1 Introduction to Environment and its Components ..... 4

1.1.1 Natural Environment ..... 4

1.1.2 Social Environment ..... 5

1.1.3 Built Environment ..... 5

1.1.4 Components of The Environment ..... 6

1.2 Environmental Science as a Multidisciplinary Field ..... 7

1.3 Definitions of Terms and History of Environmentalism ..... 8

1.4 History of Environmentalism ..... 9

Summary ..... 12

Self-Assessment Questions ..... 12

## INTRODUCTION

Environmental science studies the environment, its components, and interactions between living and nonliving organisms. The environment comprises a complex system of physical or biological materials or conditions and their interactions with each other. Environmental science is a multidisciplinary field with biology, chemistry, and sociology mainly dominating other disciplines. Many activities of humans remove materials from the natural environment in a non-sustainable way. These activities disturb and degrade the environment. It is important to study environmental science to get to know about all activities of humans that degrade the environment and the reasons behind them. An understanding of all this can help to protect and conserve the environment.

## OBJECTIVES

The main objectives of this unit are:

1. To make you aware of the environment and its components.
2. To clarify the multidisciplinary nature of the Environment.
3. To familiarize you with a few terms currently commonly used in environmental science.
4. To introduce you to the concept of environmentalism.

## 1.1 INTRODUCTION TO ENVIRONMENT AND ITS COMPONENTS

*"Environner"* is a French word which means to surround or to encircle. The word Environment has its origin in this word. The environment is defined as the surroundings of a person or the surroundings of any living or non-living organism.

The environment of a person could be the place where a person lives, makes a living, or moves to for any purpose. All living and nonliving elements at a particular place have an impact on the life of organisms and constitute the environment of that organism.

The Pakistan Environmental Protection Act 1997 describes the Environment as water, land, and air with all atmospheric layers, having organic and inorganic components. It also mentions the environment as having living organisms and the interaction between them. The infrastructure and economic as well social conditions impacting a community are also included in the definition of the Environment.

The quality of the environment is the characteristics of the environment, if the environment is clean and can help in the healthy survival of the organisms living in it, then the quality of the environment is considered good. A good environment provides greater rates of survival for organisms that could be an individual, a population, or a community living in it. If, however, the environment of the organism is not supporting a healthy life due to several reasons then the quality of the environment is not regarded as good or the environmental quality is bad and is not suitable to support a healthy and long-lasting life support system.

The environment can be categorized as Natural, Social, and Built Environment.

### 1.1.1 Natural Environment

Any place or interaction where living and non-living things occur and form relationships is known as the natural environment. Generally, it is considered that the natural environment is relatively less disturbed by human activities. The different flora and fauna, the different weather patterns and the different natural landscapes with water, mountains, and forests all play a crucial role in the environment. The value of this environment is not only through providing a habitat for interaction between species but, also by providing clean water, air and natural climate regulation. The natural environment consists of an array of different interactions between living and nonliving whether it may be in an aquatic environment (oceans, seas, lakes, ponds, springs, rivers, etc.) or a land environment (forests, grasslands, mountains, hills, deserts, plateaus, plain areas, North and South

Poles of earth etc.) each has a specific role and service to provide to its inhabitants. Different climate patterns that occur naturally determine the specific conditions of a natural environment in an area.

### 1.1.2 Social Environment

Wherever a group of living organisms live, they form communities, and these communities are determined by specific environmental conditions and interactions between living organisms. The Social Environment in an area is the interactions between communities in an area. These interactions can sometimes influence the natural environment. It should be noted here that there are animal and plant communities also present in any environment, especially in the natural environment, so when the term Social Environment is coined, it does not necessarily mean only human communities. Generally, however, the social environment mostly talks about human communities and their interactions. Some different societal factors and structures shape the behaviour and interactions of living organisms in a community. In human communities, the social environment mostly considers three ideas economic, cultural and environmental values. Technology and activism have created awareness about the effect of human activities on the environment and have also provided solutions to overcome the problem.

### 1.1.3 Built Environment

In any physical setting where human beings live in the form of communities, there is a need to build structures that provide human beings with suitable places to live, work, move and interact with each other. The Built Environment describes how man-made structures influence the work, living standards, and interaction of humans.

The built environment in any community or human population includes houses, offices, places of meditation, educational institutes, places of festivals or other places of sports or amenities such as public parks etc.

Generally built environment is created by disturbing the natural environment, e.g. to construct any building land clearing is done by disturbing the plants growing in that area. The presence of plants in the area indicates the presence of other living organisms that directly or indirectly depend on them for their survival. Through the clearing of land, all the interactions of living organisms in that area are lost. Building structures for human settings in the area also require the construction of further associated infrastructure e.g. roads, bridges, and systems for the provision of basic amenities such as water, gas and electricity distribution networks.

### 1.1.4 Components of the Environment

The environment is divided into two main components:

Abiotic Components and Biotic Components

Abiotic components of the Environment are the nonliving components present at a particular place.

Biotic components of the environment consist of the living organisms present at a particular place at a particular time.

**Abiotic Components**

The nonliving components of an environment can further be classified as physical and chemical components of the environment.

Physical components will have soil (Lithosphere), water (Hydrosphere) and air (Atmosphere).

**Physical components of the Environment** are responsible for the presence of biotic components of the environment e.g. if the soil is present then plants will grow on it, water is a basic component of life on earth and air is needed for the exchange of gases in living organisms. Any change in the physical components of the environment will affect the life present in that environment.

**Chemical components of the environment** include all chemicals present in our environment. Living organisms, especially human beings use so many chemicals in their daily life and some of these chemicals are generally not good for human health and cause pollution in the environment. Some of the chemicals called Green Chemicals will not do much harm to the environment.

**Biotic Component of the Environment**

The living organisms make the biotic or biological component of the environment. The living organisms vary from very small microorganisms that cannot be seen without using a microscope to giant and large mammals. There is so much diversity in living organisms present in different regions of the earth. The living organisms are present in the water, air and soil. The biotic components make the sphere of the Earth called as "**Biosphere"**. The plants and animals of an area are greatly affected by abiotic components of the environment, e.g. less availability of water in an environment constitutes a desert ecosystem where a specific type of plants and animals can be seen with modifications in their body structure to store water e.g. leaves of plants are not expanded and are needle-like to conserve water, similarly, camels are often found in a desert ecosystem where they store water in their bodies.

## 1.2 ENVIRONMENTAL SCIENCE AS A MULTIDISCIPLINARY FIELD

As we have discussed earlier environment is our surroundings and includes many biological, social, and physical components of the Environment. The study of environmental science includes many disciplines both from life sciences as well as social science. Many interrelated disciplines are part of environmental science. Some of these are given below:

**Environmental Biology:** To study living organisms, we need to understand the processes in living organisms for which Biology (the study of life) is important. So, in Environmental Biology, we study the impact of the environment on the life processes and interactions of living organisms among themselves.

**Environmental Chemistry:** There are many chemicals present in our environment some of them can be beneficial and some can be harmful study of their distribution in the environment, their impact on the environment, and interactions in an environment is an important discipline of Environmental Science.

**Environmental Physics:** It applies the principles and concepts of Physics to understand and solve environmental problems. It combines the physical processes in the atmosphere, hydrosphere, biosphere, and geosphere. It explains solar radiation and climate, wind energy, water cycle and hydrology, ocean currents and thermal dynamics, atmospheric physics and weather forecasting, greenhouse gas emissions, noise pollution, soil physics, and remote sensing in environmental monitoring.

**Environmental Psychology:** The surroundings of a person have a greater impact on his/her mental state that shapes a person's behaviour towards certain actions either to improve or degrade the physical, social, biological, or built environment. In environmental psychology, all such interactions between a person and the environment are studied.

**Environmental Sociology:** In this field, the interactions of different societies with their environment are studied. There may be many cultural things in a society that either degrade or improve the physical or social environment of an area. There may be certain social movements in a society to protect the environment.

**Environmental Economics:** This is an important discipline of environmental science in which the valuation of environmental resources is being carried out.

**Environmental Geography:** The distribution of plants, animals, and other forms of life as well as physical features of different landscapes on the surface of the earth that are mainly formed by different weather patterns are studied in this discipline.

**Environmental Geology:** In this discipline interactions between humans and different natural processes taking place on or below the surface of the earth are studied. It may involve the impact of different human activities as well as natural phenomena on the layers of the earth and the formation of different minerals. The weathering of rocks and the formation of soil in different regions and the impact of climate and other related factors are studied in environmental geology.

**Environmental Policy and Law:** To maintain the sustainable use of natural resources and to prevent environmental degradation, policies and laws are needed. In this discipline, international as well as local policies and laws related to the environment are discussed.

**Environmental Management:** In this discipline, different strategies related to the management of environmental resources are discussed so that environmental degradation can be prevented.

## 1.3 DEFINITIONS OF TERMS AND HISTORY OF ENVIRONMENTALISM

**Definition of Terms:** A few of the commonly used terms in environmental sciences, these days are given below:

**Biodiversity:** The variety of life forms present on earth.

**Bioremediation:** The process of reducing or controlling pollution with the help of living organisms, mainly microorganisms.

**Biomes:** Different regions of the world that can be differentiated based on distinct physical and biological environments.

**Blue Economy:** The blue word is used for water, so this term means the valuation of resources obtained from different aquatic ecosystems.

**Circular Economy:** An economic system in which materials are consumed in such a way that no waste is produced. It means an economic system in which materials are recycled or reused.

**Contaminant:** Any substance introduced into the environment that is detrimental to living organisms.

**Climate Change:** Changes in weather patterns over the years in an area.

**Climate Change Adaptation:** In this, we change the current processes or practices so that climate change may not have greater impacts in an area.

**Climate Change Mitigation:** In this, we try to avoid or stop the processes or practices that bring about climate change.

**Carcinogenic:** Any cancer-causing substance.

**Carbon Footprint:** It is the impact of an organization or human on the environment. It is calculated as total greenhouse gas emissions produced by a person or an organization through various activities.

**Green Economy:** It is an economy with many sustainable environmental practices

**Greenhouse gases:** These are gases that are responsible for climate change or global warming.

**Ecotourism:** It is responsible for travel and sightseeing to places that are environmentally important because of the presence of resources that may degrade without sustainable use.

**Habitat Fragmentation:** Break down of living places of organisms into different areas, isolated from each other due to a physical barrier e.g. building of a road etc in a forest. This may lead to the loss of biodiversity in that area.

**Net Zero:** It relates to zero greenhouse gas emissions in an area. The amount of greenhouse gas entering an area equals to absorbed greenhouse gas leading to zero emissions. The term Carbon Neutral is also used for it.

**Nature-Based Solutions:** These are local solutions to reduce pollution, cut down emissions and lower the impacts of climate change. These are generally old and traditional methods in harmony with nature.

**Pollutants:** Those substances that may cause harm to the living organisms in water, soil, or air.

**Sustainability:** It is an environmentally friendly practice in which resources are used in such a way that they are not eliminated from a particular environment and remain available.

## 1.4 HISTORY OF ENVIRONMENTALISM

A movement that contributes towards the belief that the environment must be protected from all human activities that harm it and its inhabitants is known as environmentalism. It involves a constant effort to reduce or mitigate human

activities that are a threat and do not preserve natural resources. This movement includes all social, political and environmental spheres. It is essentially based on the concept of advocacy for the protection, restoration and enhancement of the environment. There are two kinds of motives behind this movement, some people support the ecocentrism idea which is to fundamentally protect the environment for its good while others focus on protecting the environment for the benefit of humans (anthropocentrism). These two motives cause a division among the real causes of environmentalism. The long-term goal of environmentalism should be to safeguard natural resources and find a promising way to use them sustainably.

In the United States, environmentalism began much earlier in the Conservation Movement at the end of the 19th century. This was the period when attempts to protect the areas of natural beauty and reasonable use of the great potential of natural and non-renewable resources were made. Some of the pioneers included John Muir who established the Sierra Club in 1892 and Gifford Pinchot who advocated for correct forestry. The setting up of National parks, for instance, Yellowstone in 1872 was a sign of emerging consciousness that there was a need to preserve natural areas for future generations. Findings made after World War II raised appreciable consciousness of environmental pollution because of increased industrialization. Rachel Carson's book "Silent Spring" printed in 1962 is extensively considered the book which initiated the modern environmental movement in the United States. The book painted a picture of the risks of pesticides, especially DDT and other consequences of human actions on the environment. It also marked the formation of the Environmental Protection Agency (EPA) in 1970, not forgetting legislative statutes such as the Clean Air Act and the Endangered Species Act to protect quality air and wildlife respectively. More recently, however, the concept of environmentalism as practiced in the U. S., has diversified to cover new challenges such as global warming, energy, and equity. Today's activism is much more diverse, especially regarding the advocates of communities negatively impacted by environmental issues. It has also been witnessed that the U. S has experienced increased activism at the grassroots level and there is a rise in the country's involvement in various global environmentalism or global environmental protectionism thus indicating the shift towards much of environmentalism.

Environmentalism in Europe has its roots going around the colonial period and the continent's industrialization and social reforms. During the 1800s, industrialization occurred around the world, and questions about air and water pollution came up, especially in Great Britain and Germany. European attempts at conservation were firstly based on the protection of cultural landscapes and diversity; with Switzerland and Sweden being the first countries to initiate the process of protection of the natural lands. In Europe, people became aware of environmental concerns in the 1960s and 1970s in the same manner as in the U. S. There were famous ecological

accidents, such as the Torrey Canyon oil slick in 1967. With the formation of the European Union (EU), there was an environment common policy which encompassed all member states of the union. There were important legal interventions in the form of the EU Birds Directive of 1979 and the Habitats Directive of 1992. Europe has recently been very much involved in going green, especially in combating climate change. The EU's Green Deal which was initiated in 2019 sought to make Europe the first continent that will be climate neutral by 2050. EU countries have also been pioneers of renewable sources of energy and shift towards the circular economy, thus being indicative of European countries' commendable, sustainable orientation.

Environmentalism in Pakistan has evolved somewhat later in comparison with the U. S. and Europe thus being more of a response to Pakistan's environmental problems. The Shehla Zia case also known as Shehla Zia v. WAPDA (1994) is one of the most famous cases in the legal and constitutional history of Pakistan and has the greatest implication regarding human rights and environmental law. This case came to light when Shehla Zia with the other residents of Islamabad approached the apex court through a petition against WAPDA (Water and Power Development Authority). The residents had apprehensions about the high-voltage grid station being constructed in the residential area for the electromagnetic radiation could prove deleterious to health and was capable of leading to the development of cancer. They said that this construction infringed on their constitutional provisions on rights to life and dignity. The petitioners argued that the right to life in Article 9 of the Constitution prescribes the fundamental right to a healthy environment. WAPDA, however, said that there was no scientific evidence to show that electromagnetic radiation produced by the grid station endangered the residents. It was for the first time that the Supreme Court of Pakistan gave its verdict in favour of the petitioners. The Court understood that the right to life as a constitutional right under Article 9 of the constitution means the right to a healthy environment.

The process of developing environmental governance efficiency in Pakistan started with the adoption of the Pakistan Environmental Protection Ordinance (PEPO) in the year 1983. PEPO was the first legal structure of the country concerning environmental problems and formulated the Pakistan Environmental Protection Agency (Pak-EPA) for the execution of environmental measures and laws. However, PEPO lacked some specific regulatory frameworks as well as a mechanism to ensure compliance with the laws. This led to the formation of the Pakistan Environmental Protection Act (PEPA) in 1997 which replaced PEPO and made strict laws for environmental control in Pakistan. PEPA broadened the roles and authorities of the Pak-EPA to be much more assertive in terms of implementing existing environmental legislation and policies and in the process of formulating and implementing national environmental quality standards. The Act also created

the structures for the formation of the provincial environmental protection agencies hence permitting more decentralization of environmental programs in the provinces. The change from PEPO to PEPA remained a vital move in the organizational process of introducing environmentalism in Pakistan.

## SUMMARY

This unit introduces the concept of the environment and the importance of the quality of the environment. Categorization of the environment along with components of the environment is given. Environmental science as a multidisciplinary field is explained. Definitions of a few terms commonly used in the field of environmental science are provided and at the end history of the environmental movement is given.

## SELF-ASSESSMENT QUESTIONS

1. Define Environment in your own words.

2. What are other disciplines apart from the ones given in this unit, that can be related to environmental science?

3. Give a few examples of the natural environment.

4. Differentiate between pollutant and contaminant.

5. Describe any case of environmentalism in Pakistan.

6. Hint: Please use terms “environmental tribunals in Pakistan” in the search engine).

**Unit–2**

# BASICS OF PHYSICAL COMPONENTS OF NATURAL ENVIRONMENT

**Compiled by: Dr. Zahid Ullah**

**Reviewed by: Dr. Azeem Khalid**

# CONTENTS


***Page #***

Introduction ..... 15

Objectives ..... 15

2.1 Abiotic Factors of Environment ..... 16

2.2 Earth's Physical Environment ..... 16

2.3 Topographic and Physio-Graphic Factors ..... 20

2.4 Material ling in the Environment ..... 22

2.4.1 Hydrological Cycle ..... 23

2.4.2 Carbon Cycle ..... 25

2.4.3 Oxygen Cycle ..... 28

2.4.4 Nitrogen CycleCyc ..... 30

2.4.5 Phosphorus Cycle ..... 32

2.5 Macro-Nutrients and Micro-Nutrients ..... 35

2.5.1 Macro-Nutrients ..... 35

2.5.2 Micro-Nutrients ..... 37

2.6 Cycling and Balance of Nature/Material Cycles and Balance of Nature. 39

Summary ..... 42

# INTRODUCTION

Since the late 19th century environmental problems have become one of the major concerns initially in Europe and the USA then in the whole world. At present human society is facing a multitude of environmental problems that are threatening to life in general and human life in particular.

At present the earth is the only known planet that can sustain life. It provides all the necessities of life i.e. shelter, air, food and water. The whole life on Earth derives these necessities of life from Earth's environment. As you know these essential elements (abiotic elements or factors) plus the life on earth (biotic factors) constitute the earth's environment.

Any disruption in the supply or degradation of the quality of these essentials i.e. the abiotic factors can affect the whole life on earth. Unfortunately, human activities both at individual and societal levels are causing a constant deterioration of the environment. It is therefore important that we know about the nature and importance of these elements and how we at the individual level play a role in the degradation of the environment.

In this unit, you would study abiotic factors, and how natural systems work to keep a constant flow of essentials in the environment through the cycling of materials such as water, nitrogen and carbon dioxide. How cycling of materials helps in balancing the natural environment. You will also study about the macro and micronutrients and their importance for plants.

# OBJECTIVES

After completing this unit, you will be able to:

1. Enlist the abiotic and biotic factors of the environment.
2. Describe the abiotic components of the environment.
3. Describe biotic factors.
4. Explain the need for and importance of cycling of materials.
5. Describe the steps involved in the cycling of water, nitrogen, oxygen, carbon and phosphorus.
6. Give an overview of the impact of human interventions on various material cycles.
7. Define macro and micro-nutrients and explain their importance for living organisms.
8. Explain how material cycling keeps a balance in nature.

## 2.1 ABIOTIC FACTORS OF ENVIRONMENT

The physical environment is not static. It is a constantly changing system comprising some elements which are closely linked with the biotic elements i.e. living things including human beings. Abiotic elements as mentioned earlier include besides the physical components i.e. air, water, soil, all the climatic factors and natural disasters. Light and temperature are climatic factors which play a crucial role in changing and shaping the Earth's environment. Whereas natural disasters e.g. earthquakes, floods, forest fires, storms and cyclones. Natural disasters destroy habitats of living organisms and kill many organisms. In this way, they play a critical role in changing the environment as they affect both biotic and abiotic factors.

## 2.2 EARTH'S PHYSICAL ENVIRONMENT

Earth's physical environment has been categorized into four domains or zones. These zones are named based on their predominant features. These are as follows:

- Lithosphere
- Hydrosphere
- Atmosphere
- Biosphere

Let us discuss the first three categories in some detail.

**Lithosphere**

The term Lithosphere refers to the dry rocky zone of the earth's environment. It is a rigid and brittle top layer of the earth. It mainly consists of soil, minerals, rocks, mountains, plateaus, and valleys. This layer is 8 – 40 km thick. The crust is covered by soil. Soil is the most important part of the lithosphere as it supports land animals and plants by providing food and habitat. Soil is formed through the weathering of the crust and contains inorganic and organic materials essential for plant growth. That is why the Lithosphere supports forests and grasslands and provides a conducive environment for agriculture as it helps in the fixation and development of plants. It also provides minerals. Minerals are derived from rocks. Soil acts as the medium for nutrient and mineral cycles (We shall learn about some natural cycles in the following section). Soil is also a habitat for many organisms such as bacteria, fungi, nematodes and arthropods. The lithosphere provides for human settlement. It is the main source of food for both man and animals.

**Hydrosphere**

The hydrosphere is the total amount of water in the earth's environment (hydro means water and sphere refers to layer or zone). Water is present on the earth's surface in the form of water bodies e.g. lakes, rivers, seas, oceans and estuaries.

Water is present not only on the surface of the earth but also underground as well as in the air. It exists in solid, liquid and vapour forms. It is present in vapour form in the air, liquid form in various water bodies, solid form on mountain peaks, and glaciers as ice. Seventy-five percent of the Earth's surface is covered with water therefore the earth is sometimes referred to as a water planet. Ninety-five percent of the total water present in the rocks is chemically bound with other molecules in various combinations. Of the remaining five percent, 97.5 percent is in the oceans 21 percent is in the ice form and the rest is in the form of water bodies, soil water and in the atmosphere as vapors.

Water is critical for sustaining life on the earth. Water can dissolve more substances than any other liquid which is why it is called a universal solvent (solvent is the substance in which other substances are dissolved). Being the universal solvent, it is used in nature to transport other materials through air, ground and bodies of living things. Wherever water moves it takes along valuable chemicals such as nutrients and minerals.

Water also provides habitat to many aquatic plants and animals. The fact that it is transparent makes it a suitable medium for the growth of green plants as light can penetrate to the bottom of the ocean. Ocean water keeps the environment cool as it does not allow the temperature to increase to extremes.

**Atmosphere**

Atmosphere refers to the thin layer of air that envelops the earth. It is mainly composed of gases including carbon dioxide, oxygen, nitrogen and many other gases. The atmosphere is composed of 21 % oxygen, about 0.04% carbon dioxide and 78% nitrogen.

The atmosphere is categorized into five zones: Troposphere, stratosphere, mesosphere, ionosphere and exosphere. This division is based on the temperature in different levels of the atmosphere. Each zone has its characteristic features.

**1. The Troposphere**

The lowest zone of the atmosphere is the part of the earth we live in. Its height is about ten kilometres. This zone contains 75 percent of all of the air in the atmosphere and it also contains almost all water vapours of the atmosphere. Most of the climatic processes such as rain, thunders, storms, snow and cyclones take place in this zone. In the upper part of this zone, the temperature is low because as the air moves upwards air pressure decreases and air expands resulting in low temperature. The lower part of this zone which touches the earth’s surface is called boundary layer e. In this part of the troposphere, air movement depends upon the properties of the earth’s surface. The top of this zone is called the tropopause.

## 2. The Stratosphere

The area extending from the tropopause to about fifty kilometres above is called the stratosphere. This is the part of the atmosphere which contains the ozone layer. Ozone is an odourless and colourless gas. The ozone layer absorbs the ultraviolet rays of the sun which is why the temperature is higher in this part. The ozone layer protects us from the harmful rays of the sun. Ultraviolet rays can cause cancer, eye disease and many other health problems and are damaging to all life forms. Unfortunately, many human activities result in the release of harmful gases which enter the atmosphere and cause depletion of the ozone layer. Chemicals (chlorofluorocarbons or CFCs and halons) used in refrigerators, fire extinguishers, spray cans etc. have damaged the ozone layer.

## 3. Mesosphere

The third zone of the atmosphere is the mesosphere. The region above the stratosphere is called the mesosphere. The temperature in this part of the atmosphere is minus eighty to minus ninety degrees. The temperature increase is due to the absorption of ultraviolet rays and X-rays from the sun.

## 4. Ionosphere

The ionosphere is the region in which solar radiations remove electrons from atoms and molecules of gases. These electrons are positively charged. The ionosphere absorbs and also reflects radio waves which helps in communication between different parts of the world.

## 5. Exosphere

This region is about 500 km and expands above ionosphere. It contains mainly hydrogen and oxygen atoms. But their number is too small to have any possibility of collision.

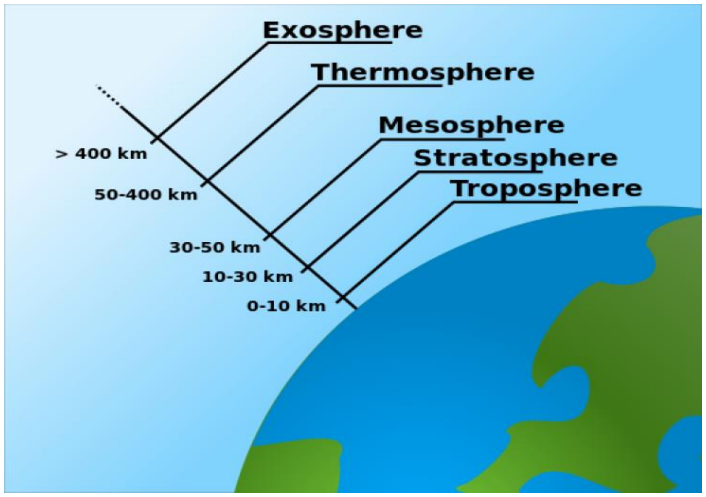


**Figure 2.1:** Structure of atmosphere

**Biosphere**
It is the part of the earth where water air and the earth's upper dry layer are interconnected and support life. Hence it is where we find animal and plant kingdoms.

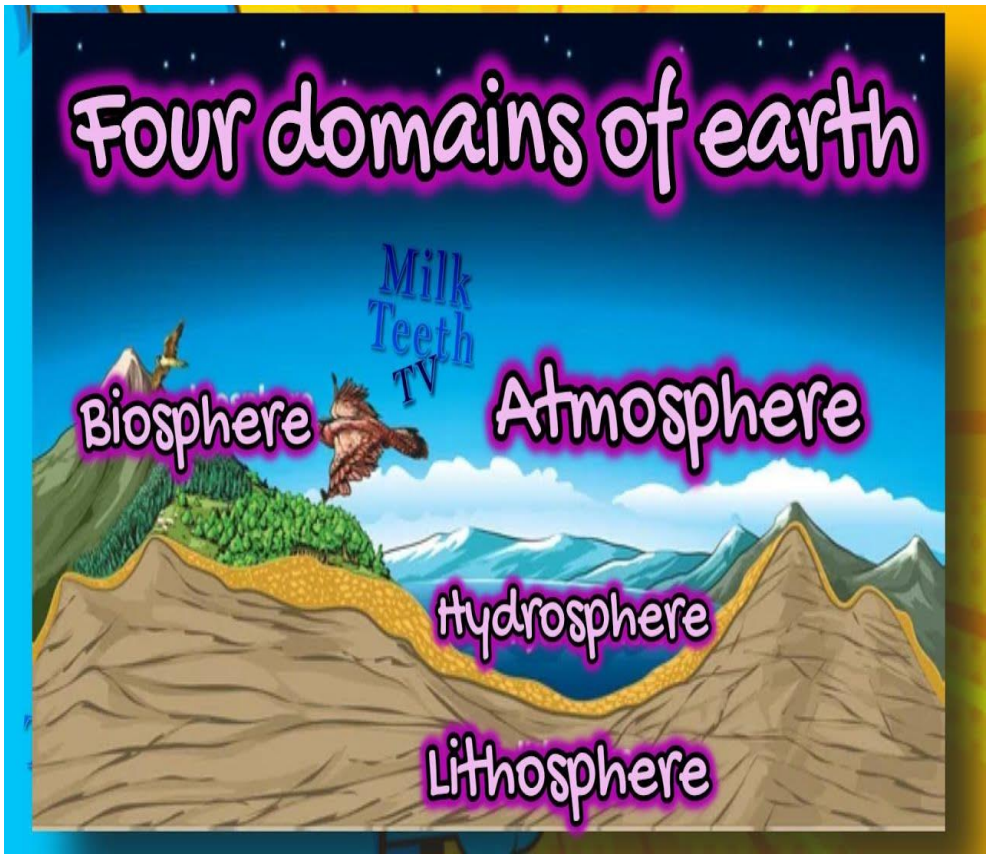


**Figure 2.2:** The four major zones/components of the environment

## Self-Assessment Questions

1. The earth's environment is classified into four major zones or spheres. Which of these is home to land animals and plants?

2. What are the characteristic features of each sphere of Earth's environment?

3. Name the sphere which protects Earth's environment from the ultraviolet rays of the sun. Briefly describe how.

## 2.3 TOPOGRAPHIC AND PHYSIO-GRAPHIC FACTORS

Topography is the study of physical features of the earth's surface hence topographic factors include description or mapping of mountains, lakes, plains, plateau etc. Whereas physiography is the study of the physical nature of an area. Physio-graphic factors include slope, altitude and aspects. These factors determine the climate of an area which in turn affects the kind and distribution of flora and fauna in that area.

Let us briefly review the physiological factors and how they affect life.

**Altitude**

Altitude is the height of land from the sea level. At high altitudes temperature is low, the wind's velocity is high, air pressure is also low, there is less rainfall and these areas experience snowfall. At high altitudes intensity of light increases.

Low temperature, dry weather, and exposure to ultraviolet light affect the flora and fauna of these areas. Animals and plants that can adapt to harsh environmental conditions are found in these areas. Therefore, the flora and fauna at high altitudes are very different from those at low altitudes. Altitude is measured in meters or feet.

*The altitude of Murree ranges from 1600 to 1700 feet. Whereas Karachi is located at an altitude of 26 feet. Using the information from internet compare the climate and types of animals and plants which are found in these cities of Pakistan.*

**Latitude**

Latitude is the distance between a surface on the earth and the equator about the

North and South Pole. Latitude is an angular distance hence it is measured in degrees for example latitude of Pakistan is 30 degrees north and that of Uk it is 55.3781 degrees north which means both are located in the northern hemisphere. If we simply take the distance from the equator ignoring the poles, the UK is at a distance of 6157.77Km while Pakistan is located at a distance of 3335.85Km.

**Food for Thought**

*Which of the two countries is located nearer to the equator? What is the difference in the climate of UK and Pakistan? Do you think the distance from equator has any effect on climate? Why? Consider the fact that surfaces of the earth nearer to the equator gets direct sun light and for longer period.*

The earth's surfaces near the equator get more direct sunlight and are warmer. Whereas the regions that are farther from the equator are cooler. The position relative to the equator determines the climate of a region. Near the equator, sunlight is more intense resulting in more evaporation which in turn causes heavy rainfall all the year. A humid and warmer climate helps the vegetation to thrive resulting in rain forests. Amazon is one of the largest rainforests in the world. It spans on 6000,000 Km area.

**Slope**

It is a common observation that water flows faster on steep surfaces than is slopes. That is why the roofs of houses are built with a slant so that water runs easily and does not accumulate on the roof. The same happens when any surface on the earth is steep. The slope increases the rate of water run-off. However, it has a negative effect. Fast water runoff takes with it soil which is called soil erosion. Soil is a medium for plants' growth as it is the source of minerals and nutrients. Steep surfaces have shallow layers of infertile soil. Therefore, steepness and slope adversely affect plants. Slopes are exposed to light, wind and rain. The direction and steepness of the surface therefore affect the temperature and climate of the area. Physiography describes the features or landscape of a place. If you look at the landscape of Pakistan, you will realize that we have a rich diversity in physical features. Based on a variety of landscapes. We have a high mountain range in the north and, a low mountainous region in the west. If you move towards Balochistan you will see highlands and plateau, then we have mountainous uplands of Pothohar and plain fertile regions of Sindh and Punjab. Our flora and fauna and climate in different parts of the country are also as varied as the landscape. This indicates that physiography and topography play a pivotal role in determining the nature of the environment including both the biotic and abiotic elements of a place on the earth.

## Self-Assessment Questions

1. What is the difference between topography and physiography?

2. What is the important physio-graphic feature of the earth which affects its topography?

3. What are the characteristic features of areas having a high altitude?

4. What is the difference in the climate of regions located near the equator and those located farther from the equator and why?

5. How does slope affect plants?

## 2.4 MATERIAL CYCLING IN THE ENVIRONMENT

Living things require a constant supply of essentials of life i.e. air, water and food. The earth's environment is the only source of materials that are crucial for life. However, the environment does not have infinite reservoirs of these materials. Nature has an elaborate system which makes it possible to use and reuse these materials to keep a constant flow of materials in the environment. These materials are constantly flowing between biotic and abiotic components of the earth's environment in the form of cycles. The cycling of materials like water, carbon and nitrogen helps in creating and sustaining environmental conditions including the availability of materials for living things including human beings.

Cycling of materials is the continuous flow of materials from non-living to living and back to the non-living factors of the environment in the form of cycles which are referred to as biogeochemical cycles, bio means living things, Geo refers to geosphere i.e. soil, sediments and rocks whereas chemicals are the materials that are cycled e.g. carbon, oxygen, water and nitrogen.

There are six most important elements which comprise some organic and inorganic chemicals. These are; Carbon, Phosphorus, Sulphur, Oxygen, Nitrogen, and hydrogen. These elements are basic constituents of the bodies of living things. Therefore, are essential nutrients for plants, animals and human beings. These elements are a part of our constitution and are also used in different body functions. We also use them in some ways in various activities. They exist in many different forms in the environment e.g. carbon exists in the form of coal, oil, natural gas, etc. which are important sources of energy.

Water is an essential abiotic component of the environment. It is crucial to sustain life on the earth and to maintain the environment. Seventy percent of the earth's surface is water which gives the illusion that water is present in infinite amounts. But this supposition is far from reality as we witness that in many parts of the country, water is scarce. We see people struggling to access clean drinking water. The problem of diminishing resources of water has become a global problem. To create awareness about the increasing problem of water shortage and the need to take measures to conserve water resources the United Nations has declared March 22 as World Water Day.

Like other factors of the environment water is constantly circulating between different zones of the earth's environment. So we use the same water again and again. However,

*Cycling of materials creates and maintains the conditions for humans and nature to co-exist in a productive manner. Biogeochemical cycles connect the biological, geological and chemical aspects of biosphere.*

overpopulation, pollution and climate change have adversely affected this cycle and we are facing severe shortages of clean water all over the world.

## Self-Assessment Questions

1. We know that natural resources of the earth e.g. water are not infinite yet from billions of years living things are consuming these material sources (it has been estimated that life on this planet is about 3.7 billion years old). Why these materials are still available? Explain.

2. Explain the importance of material cycles for the sustenance of life on the earth.

### 2.4.1 Hydrological Cycle

The hydrological cycle or water cycle is the continuous circulation of water on, below and above the surface of the earth. This complex process is completed in many steps during which water is changed from one state to the other. You know that the change of water from one state to the other involves energy loss and energy gain. We heat water it changes into vapors and when it cools down it changes back to liquid and on further cooling it changes to ice. The water cycle is also driven by the sun's energy. The heat energy from the sun evaporates the water from oceans and the surface of the earth. The water vapours move to the atmosphere lose energy and change from vapour form to liquid again and form clouds. When clouds become too laden with water it rains. The major steps of the water cycle include evaporation, condensation, precipitation, infiltration and runoff. Let us look at each step in some detail.

**Evaporation**

As you know the sun is the major source of heat energy on the earth. When sun rays fall on the surface of water its temperature is raised, and water vapors are formed. The water vapours rise in the air. Water also evaporates from the surface of plant leaves. This process is called transpiration. Evaporation also occurs on snow-covered peaks of mountains. Dry winds and low humidity caused the snow to change into vapours. The process where a solid changes directly into gas form skipping the liquid state is termed sublimation. Evaporation from these sources results in the accumulation of a large percentage of water in the atmosphere.

**Condensation**

Condensation is the process by which changes vapours into water. At high altitudes the temperature is low, and the air is cooler. When vapors rise into the atmosphere due to low temperature they lose their heat and change back into water. You might have observed that sometimes when we put a glass of ice-cold water outside, tiny drops of water appear on its outer surface, it happens because when vapours in the air touch the outer surface of glass, they lose their heat and change into water

droplets. In the morning, we often observe dew on the grass it is also formed through the same process. In this process, vapours attach themselves to dust particles or other impurities in the air. So, particles act as anchors and without them, condensation cannot take place. Condensation results in cloud formation. Cloud formation is crucial as without clouds there will be no rain. Clouds are categorized into four types. This classification is based on their height, density, transparency and expanse. These types are; nimbus, stratus, cirrus and cumulus. Dew, frost, fog and clouds are all types of atmospheric water.

**Precipitation**

Precipitation or rainfall results in returning the water to the earth's surface. When water droplets merge and form bigger drops, they fall on the earth due to the pull of gravity. They may fall in the form of rain, hail, or sleet. Or snow.

**Infiltration**

The process by which a liquid permeates into a substance by filtration is called infiltration. When it rains rainwater permeates into the soil this process is called infiltration. The rainwater moves in the layers of soil and reaches the water under the surface of the earth i.e. groundwater. The amount of water that infiltrates depends upon the surface on which it falls. There is more absorption in the soil than in the rocks. Due to urbanization most of the earth's surface has been covered with concrete which has reduced the amount of water that infiltrates and reaches the ground water. The upper level of groundwater is called the water table. In many parts of Pakistan, the water level of groundwater has become very low resulting in reduced access to groundwater.

**Runoff**

Runoff water is the excess water which is not absorbed by the soil. It flows across the land and flows into lakes, streams and creeks which finally fall into rivers and rivers fall into the sea.

Runoff is one of the major causes of water pollution. Runoff water flows through drains and ditches and finally falls into streams, rivers, lakes and the ocean taking with it pollutants like pesticides, bacteria and many other pollutants.

Runoff causes soil erosion. Soil erosion is a gradual process which causes the removal of soil from the earth's surface due to rainwater or high winds. The erosion process has three stages namely; detachment, transport and deposition. Under the impact of water soil particles detach from the earth's surface and are carried away with runoff water. In the areas where the earth's surface is covered with vegetation, or another protective cover the rate of soil erosion is less. Any activity that leaves the surface of the earth exposed for long periods makes the land vulnerable to erosion. Deforestation to clear

the land, or to use wood for various purposes is a major cause of soil erosion. Plant roots hold the soil together and protect it from erosion.

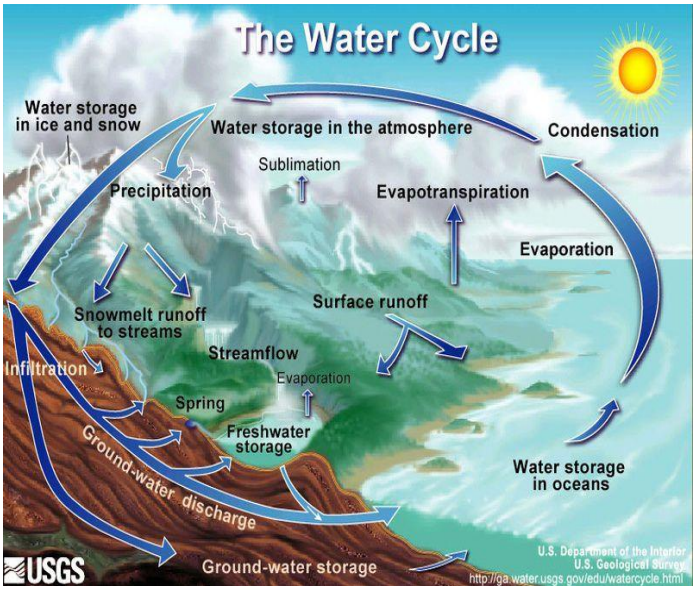


**Figure 2.3:** Water cycle

Figure 2.3 shows the process of the water cycle. Due to high-temperature water from the earth's surface (rivers, streams and lakes) evaporates and moves to the atmosphere to form clouds. The same water comes back to the earth's surface in the form of rain. Thus, the circulation of water in the environment continuously renews and operates.

## Self-Assessment Questions

1. What is the role of the sun's energy in the water cycle?
2. List the different stages/steps of the water cycle.
3. Describe the process of condensation.
4. What is run-off and what are its effects on soil?

### 2.4.2 Carbon Cycle

All the elements that are part of the bodies of living organisms are continuously cycled between living and non-living components of the environment. Carbon is the base of life as it is the basic component of the materials of which bodies of living organisms are constituted. Being the basic constituent of living organisms it is an essential component of food that we and all other living beings eat. We eat carbon compounds in the form of proteins, fats, and carbohydrates. Green plants use carbon to make their food.

We also use carbon as fuel (coal, oil, and natural gas are forms of carbon), as it is the main source of energy to run factories, and automobiles and to keep our houses and buildings warm.

Carbon is one of the most abundant elements on the earth. It is present in our environment in element as well as compound form. It is found in solid and gaseous form. The atmosphere and bodies of living organisms, oceans sediments and rocks are the major reservoirs of carbon. Carbon is constantly exchanged between these reservoirs, so it is the same carbon that we use again and again. Let us see how nature has made it possible to keep this cycle going and how carbon circulates between different factors of the environment.

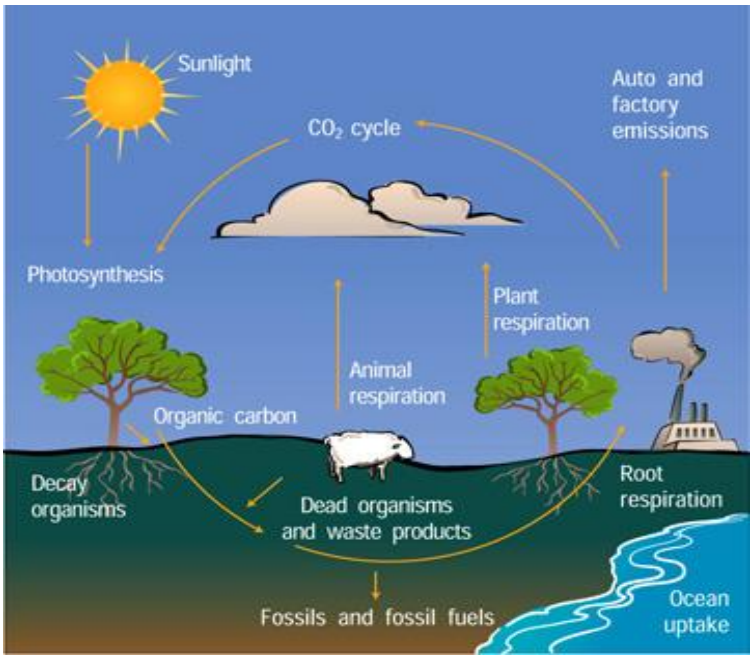


**Figure 2.4:** The Carbon cycle

Look at the Figure of the carbon cycle and carefully trace the route of the carbon cycle. Try to make a flow chart to show different channels through which carbon moves between various factors of the environment then read the following description.

If you look at the diagram given above, you will notice that carbon that is present in the atmosphere in the form of carbon dioxide is captured by green plants to make their food for growth through the process of photosynthesis. Photosynthesis is the process through which plants make their food from carbon dioxide and water with the help of sunlight (photo means light and synthesis are making). In this process,

carbon is transferred from the atmosphere to plants. When human beings or animals eat plants, carbon is transferred to their bodies, from plant-eating animals i.e. herbivores it is transferred to meat-eating animals i.e. carnivores. We eat plants and animal meat, so we get carbon from both these sources. When living things die their bodies decay and decompose the carbon stored in their bodies is added to the soil. These deposits of carbon when remaining in the soil for millions of years are converted into fossil fuel. We use this fuel for several purposes e.g. to cook. To keep warm and as an energy source to run our factories and automobiles. The atmosphere does not have infinite reserves of carbon. How is the atmospheric carbon replenished? All living things respire that is they take in oxygen and excrete carbon dioxide in the air and become a part of the atmosphere. The other source of adding carbon is human activities. When we burn oil, natural gas and wood or coal carbon dioxide is formed which is ultimately become a part of the atmosphere. In this way, the carbon cycle is completed.

Along with other gases, carbon dioxide is present in the atmosphere. As stated earlier atmospheric gases protect the planet from ultraviolet rays and also trap heat to keep the planet warm. Without carbon and other gases, the planet will freeze. Carbon and other gases that trap the heat area are called greenhouse gases. To understand it think of a greenhouse. The greenhouse is built of glass or some other material which can trap heat. In cold regions, plants are grown in the greenhouse to protect them from the cold. During the day heat is absorbed at night when the temperature is low the glass or plastic cover acts as an insulating sheet preventing the heat from escaping into the air. Greenhouse gases of the atmosphere work in the same manner. They preserve the heat absorbed by the earth's surface during the day so that when at night there is no heat source the temperature does not fall to extremes.

If Greenhouse gases are produced and added to the atmosphere they are harmful to the environment in general and life in particular. Greenhouse gases are added to the environment through natural sources and human activities. Natural resources include respiration in animals, humans and plants, oceans emit large amounts of carbon dioxide into the atmosphere, decay and decomposition of dead organisms, natural forest fires and volcanic eruptions. There are human activities which cause an increase in the amount of greenhouse gases in the environment. Burning fuel for domestic or commercial purposes, agricultural practices, transportation, and industrial sources. Human activities contribute about 55.46 percent of total greenhouse gas emissions. This means that the ratio of gases emitted through natural resources and human activities is about 0.8. So, our activities are putting extra pressure on the otherwise naturally balancing ecosystems.

**Food for Thought**

*Observe all the activities in your home and make a list of activities which result in adding carbon dioxide in the environment. Leave out the natural sources like breathing.*

## Self-Assessment Questions

1. Carbon is an essential element for the survival of living organisms, explain.
2. Present the carbon cycle in the form of a diagram.
3. What are the major sources which add carbon dioxide to the environment?
4. Why access to carbon in the environment is a threat to life on the earth?
5. Why is carbon dioxide called a greenhouse gas?
6. What is the function of greenhouse gases in the atmosphere?

### 2.4.3 Oxygen Cycle

Oxygen is the second most abundant element in the atmosphere. It is present as a free element as well as in compound form. It is the most important element for the sustenance of life on the earth. It is continuously used in respiration by animals, plants and human beings. It is also consumed in natural processes like decay and decomposition of organic matter. We use it in many activities and processes like respiration, burning fossil fuel, and generating energy-generating processes. However, despite its heavy consumption, the level of oxygen in the environment remains quite stable because it is constantly being replenished in many ways. It is circulated between the atmosphere, biosphere, lithosphere and hydrosphere in both element and compound form. To understand the whole process let us look at the route of oxygen through different components of the environment.

**Steps of the Oxygen Cycle**

**Atmosphere**

Oxygen is added to the atmosphere through different sources. One of the sources is the breaking down of water molecules and nitrous oxide present in the atmosphere. Sun's ultraviolet radiation breaks these molecules which results in the release of oxygen. This process is called photolysis, photo means light and lysis are breaking down.

**Biosphere**

In the biosphere, oxygen is exchanged between animals and plants and the atmosphere through the process of photosynthesis and respiration. Plants produce oxygen which is used by other organisms in respiration. Growing plants can help to increase the amount of oxygen in the environment.

**Respiration**
Both animals and plants consume oxygen in respiration and release carbon dioxide.

**Photosynthesis**
Plants use carbon dioxide in the process of photosynthesis and release oxygen as a byproduct which is added to the atmosphere and again used by living organisms.

**Lithosphere**
The lithosphere is the hard crust of the earth. It contains the most amount of oxygen content in the compound form like silicon and oxide. As a result of many weathering processes oxygen trapped in these compounds is released and added to the atmosphere.

**Hydrosphere**
The hydrosphere contains about 33% of the total oxygen in the environment. It is present in the hydrosphere in dissolved form, as a component of water molecules and as a part of compounds like carbonic acid. Aquatic organisms use oxygen dissolved in water.

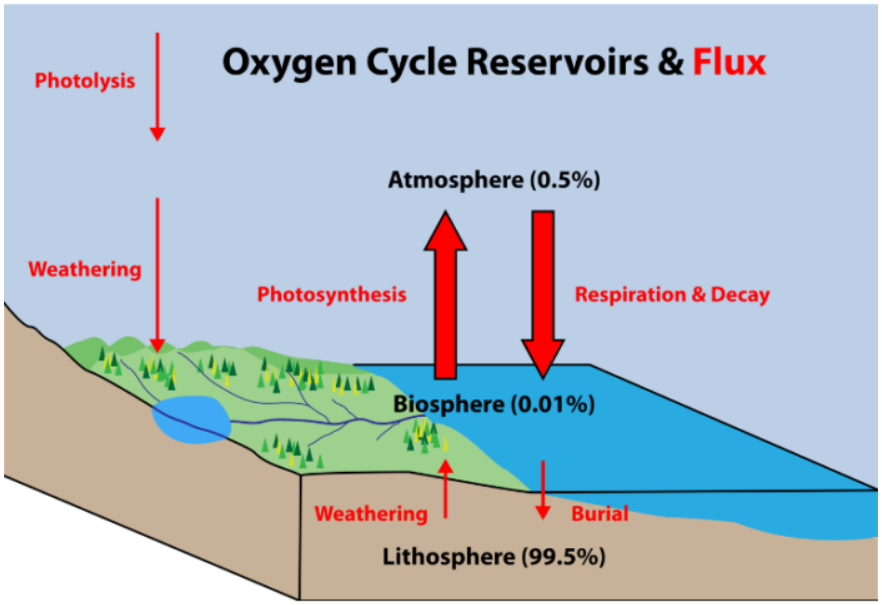


**Figure 2.5:** Oxygen cycle

**Food for Thought**

*Look at the stages and steps of water, carbon and oxygen cycles and think how these independent cycles are codependent. (Hint: Look at the consumption and addition of carbon, oxygen and water in the environment.)*

## Self-Assessment Questions

1. Describe the route of oxygen in the environment.

2. What is photolysis?

3. Name the important sources through which oxygen is added to the environment.

4. What steps we can take at the individual level to decrease consumption and increase the addition of oxygen in the environment?

### 2.4.4 Nitrogen Cycle

Nitrogen is the most abundant gas in the environment, making up 78% of the total gases in the environment. Like other materials, nitrogen is also constantly consumed and replenished in the environment. It is one of the essential elements for the sustenance of life on the planet. It is an important component of proteins which are the fundamental building materials used in the bodies of animals and plants. In humans and animals' bodies proteins are used to build muscles and blood cells. In plants protein is an essential component of chlorophyll. Without it, plants cannot make their food as it absorbs sunlight, a source of energy used in photosynthesis.

Nitrogen in the air cannot be used by the plants as such so it needs to be changed into usable form. Like other biogeochemical cycles nitrogen changes into multiple forms during its movement from one sphere to the other in cycle form. The nitrogen cycle is….

*Biogeochemical cycle in which nitrogen moves from one sphere to the other in multiple forms. The conversion of nitrogen involves both physical and biological processes.*

Now read on to explore various stages of the nitrogen cycle.

**Stages of Nitrogen Cycle**

The nitrogen cycle is completed in four major stages:

- Nitrogen fixation
- Nitrification
- Ammonification
- Denitrification

Let us study each stage in some detail.

**Nitrogen Fixation**
Nitrogen is present in the atmosphere and soil, but plants cannot use it unless it is converted to usable form. The conversion of inert nitrogen to usable form is called nitrogen fixation. It is converted by biological and physical processes.

**Physical process**
Lightning strikes water and nitrogen molecules in the atmosphere and as a result, they are electrified and converted to ammonia ($NH_3$) and nitrates ($NO_3$). With rain, these compounds fall on the ground. Plants through their roots take these compounds and use them in their biological processes.

**Biological Process**
Bacteria living in the soil and the roots of leguminous plants like peas and peanuts, in symbiotic relationships, convert atmospheric nitrogen into ammonia and ammonium.

**Nitrification**
This is also a biological process through which ammonia present in the soil is converted to nitrates and nitrites. Special types of bacteria and archaea convert ammonia to nitrite NO2. Bacteria and archaea both are single-cell organisms and lack a nucleus i.e. are prokaryotes. However, archaea due to their unique properties are classified as a separate group. At the final step of this stage, another type of prokaryote (single-celled organisms lacking a nucleus) oxidizes nitrites to nitrates, $NO_3$. Land plants absorb these nutrients through their roots.

**Food for Thought**

*The microorganisms present in the soil make essential elements available for plants. Can you think of the human interventions which would affect these processes adversely?*

**Ammonification**
When living organisms die and decompose the corps become a part of the soil. Bacteria in the soil turn the nitrogen in the decayed material into ammonium which is taken back by the plants.

**Denitrification**
It is the process by which nitrates are converted back to nitrogen gas which is then added to the atmosphere. This process is also biologically carried out by special types of bacteria and archaea. This is how the nitrogen consumed by living organisms is replenished. However, human activities could disrupt the cycle which will have severe effects on life. Use of fertilizers, sewage leaks, burning fuel and pollution from wastewater are the major factors causing an increase in nitrogen and phosphorus levels in the environment. Fertilizers contain nitrogen and phosphorus.

Excessive use of fertilizers results in leaching of these elements into water. Human and animal wastes contain large amounts of these elements. [3] Leaking sewerage lines could add these elements to clean water. When excessive amounts of nitrogenous materials are added to soil or water the micro-organism converts them into nitrous oxide which enters the atmosphere. Nitrous oxide is more potent as a greenhouse gas than carbon dioxide. Excess nitrogen in the soil adversely affects the health of plants. It can result in a decline in plant diversity, and hinder flower and fruit formation as it increases vegetative growth.

## Self-Assessment Questions

1. What is the importance of nitrogen for living organisms?

2. The form in which nitrogen is present in the environment is not usable for plants. What are the main processes of converting it into usable form? Describe the processes.

3. What is the impact of human activities on levels of nitrogen in the environment?

   What steps are needed to control it?

4. Enlist various stages of the nitrogen cycle.

### 2.4.5 Phosphorus Cycle

Phosphorus is a part of the structure of living organisms. Phosphorus is an essential component of molecules of DNA, RNA and ATP. DNA or deoxyribonucleic acid carries all the instructions for the development and functioning of all organisms, ribonucleic acid (RNA), is important for protein synthesis in plants and animals.

Phosphorus and its compounds are found only in the rocks, sediments and soil that is in the lithosphere. It is mainly transferred to living organisms through the uptake of plants. It moves between the lithosphere, hydrosphere and biosphere. The atmosphere has no role in the phosphorus cycle. The phosphorus cycle is completed in four stages:

- Weathering
- Absorption by plants
- Absorption by animals
- Return to the environment through decay and decomposition of dead organisms.

**Weathering**

Breaking down of rocks into smaller particles is called weathering. It is the first step in soil formation. Breakdown or weathering of rocks happens in three major ways: physical, chemical and biological. The details are as follows.

**Physical Weathering**
It is a type of weathering in which rocks break down into smaller particles without any change in their chemical composition. Here are some factors which cause the physical wear down of rocks. Water, ice, winds, temperature and gravity are major factors causing mechanical weathering.

Spontaneous changes in temperature are one of the major causes of weathering. If the temperature fluctuates between below zero to high the water collected in crevices and cracks of the rock will freeze and thaw alternately and over and over again.

The main causes behind physical weathering include spontaneous fluctuations in temperature like too high or too low heat or cold. Variations would be the causal element of freeze-thaw action and this occurs when water gets collected in the fissures in rocks all through the day and during the temperature drops at night that would be sufficient to freeze water into ice. When water freezes it expands resulting in expansion of the gap or crack in the rock which would split the rock. It happens in regions where the day and night temperatures are very different. Water is a major weathering agent. Sometimes water flowing with pressure or glaciers remove and carry away big pieces of rock. A big piece of rock removed from its original position by water, or other natural forces is called a boulder. When a big piece of rock falls on its way to the ground it damages other rocks.

Gravity is also an important factor causing weathering. Under the force of gravity loose parts of rock. High winds move the pebbles and sand particles on the surface of rocks causing abrasion of the rock.

**Chemical Weathering**
In chemical weathering, the composition of minerals of the rock changes and new minerals are formed. Chemical weathering occurs in many ways.

Carbon dioxide dissolved in rainwater forms carbonic acid. This process is called carbonation. This slightly acidic rainwater acts on minerals and changes their composition. So, carbonation is one way in which rocks are weathered chemically. Water is the major agent in weathering. It acts on minerals in different ways and changes them into new materials. Hydration and hydrolysis are two different ways of the action of water on minerals which result in chemical weathering.

**Biological Weathering**
Living organisms are also responsible for weathering rocks. Plants grown in cracks of rocks split the rock as they grow and their size increases, Rodents like rabbits and other animals which dig burrows also cause weathering.

Weathering and other processes like mining and leaching release the phosphor tied up in rocks and sediments.

**Absorption of Phosphorus by Plants and Animals**
Land plants absorb phosphorus dissolved in water. Animals get phosphorus from plants directly or indirectly i.e. either by consuming plants or eating plant-eating animals.

**Return of Phosphorus Back to the Environment**
When animals and plants die their bodies decompose and decay in this process phosphorus trapped in their bodies as a part of organic compounds changes into inorganic phosphorus by the action of certain bacteria. It is then added to the soil and water.

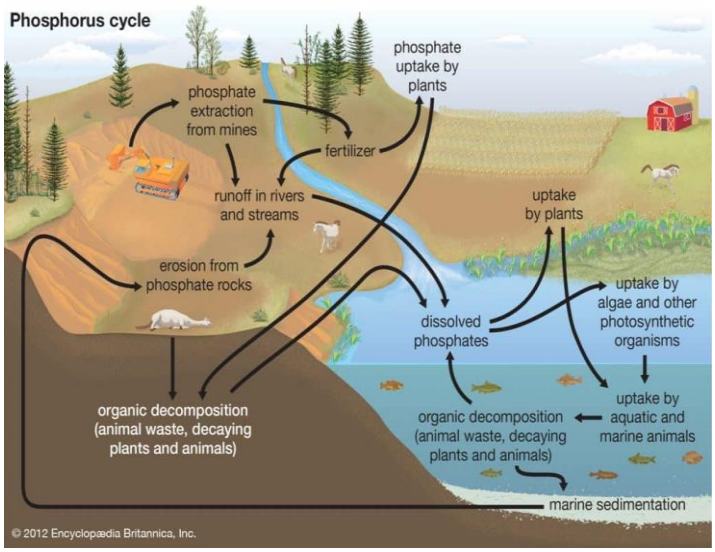

**Figure 2.6:** Phosphorus cycle

## Self-Assessment Questions

1. What is the main process by which phosphorus is made available to plants and animals?

2. What are the main reservoirs of phosphorus in the environment?

3. Which of the spheres of the environment has no role in the phosphorus cycle?

4. What is weathering? What are its different types? What is its role in the phosphorus cycle?

## 2.5 MACRO-NUTRIENTS AND MICRO-NUTRIENTS

Cell is the basic unit of both the structure and function of living organisms. Cells are formed of molecules called biomolecules or organic molecules. These include carbohydrates, lipids, nucleic acids and proteins. Biomolecules are structural and functional materials of cells. They constitute the structure of living things and also play a pivotal role in the functions or physiology of animals, plants and micro-organisms. They are compounds of carbon, oxygen, hydrogen, nitrogen, phosphorus and many other elements. Based on the quantity in which these elements are required, they are categorized into two major groups: macro-nutrients and micro-nutrients. The elements required in larger quantities are called macro-nutrients while those required in smaller quantities are termed micro-molecules. Think about the material cycles and the elements that are continuously moved between biotic and biotic factors of the environment and their importance. Now read on.

### 2.5.1 Macro-nutrients

The elements required in larger quantities are nine and include carbon, calcium, hydrogen, magnesium, nitrogen, oxygen, phosphorus, potassium and sulphur.

**Carbon, Hydrogen and Oxygen**

These are essential elements as they are part of many compounds which are part of the structure and function of the cells. If you recall material cycles you will realize that living things obtain them from air, water and soil. What is their function in living organisms? These elements are essential components of many structural and functional compounds in the body of living organisms. Carbohydrates, fats, proteins and lipids are compounds of these elements.

**Carbohydrates**

Carbon is the fundamental element of most biomolecules. It can combine with many elements forming a variety of compounds unique to living things. There is a rich variety of carbon compounds which perform diverse functions in the body of plants, animals and human beings. Carbon compounds vary from very simple to complex compounds.

Carbohydrates are the most familiar carbon compounds. Sugars (glucose, fructose galactose, and lactose), and starch a complex carbon compounds found in grains, roots plants grains and plants roots and stems of plants, cellulose is another carbohydrate found in the plant body. Different carbohydrates have different functions. Starch and sugar provide energy while cellulose is a structural compound. It forms a cell wall in plants which gives support and shape to the plant parts.

**Lipids**

Lipids include oils, fat, waxes and steroids. Lipids perform a variety of functions. Fats stored in the cells act as energy sources for long-term use. Lipids provide insulation and protection from environmental factors. The layer of fat under the skin helps to conserve body heat and its layer around vital organs protects them from shock. Phospholipids are important components of the membrane of cells as they act as a barrier to the entry and exit of molecules from cells. Lipids are also included in the constitution of hormones. Hence perform controlling and regulating functions in the body. Cholesterol is a lipid which is formed in the liver makes many hormones, vitamin D and also forms cell membranes. Cholesterol is also a part of bile and therefore helps in the digestion of fats.

**Proteins**

Proteins are the most abundant biomolecules in the body of living organisms. Amino acids are the basic units of structure of proteins. They perform many functions in the body. Proteins' functions include structural, regulatory, controlling and protective functions.

Proteins are building materials for many body parts. Collagen is present in the skin, ligaments, and tendons. It gives strength and elasticity.

Enzymes act as catalysts for almost all the biochemical reactions within the cells. Most of the enzymes are proteins. Enzyme amylase for example, in saliva is an enzyme that digests starch. Hormones are either steroids or proteins. Hormones coordinate, regulate and control different functions in the body. Hemoglobin is also a protein. It contains iron and helps in the transport of oxygen throughout the body.

**Nitrogen, Phosphorus and Sulphur**

Nitrogen is present in all organisms as a part of biomolecules. It is an essential component of biomolecules like proteins, nucleic acids, and energy transfer molecule adenosine diphosphate (ADP). Deficiency of nitrogen in plants can cause retarded growth and suspended flowering in plants.

Phosphorus is an important component of adenosine diphosphate and adenosine triphosphate. These compounds function together to provide cellular energy. Phosphorus is an essential element for the growth, repair and maintenance of cells and tissues. It is also required for making DNA and RNA. Phosphorus plays an important role in balancing and use of vitamins like vitamin D and minerals for example zinc, iodine and magnesium. In human beings' phosphorus deficiency can cause anaemia, fragile bones, fatigue, loss of appetite and low immunity.

**Secondary Macro-Nutrients**
Sulphur, calcium, and magnesium are categorized as secondary macro-nutrients. They are essential but are needed in less quantity than the other macro-nutrients.

**Sulphur**
Sulphur is an important element for the proper functioning and health of plants and animals. In plants, it is used to form chlorophyll which is essential for photosynthesis. It is also necessary for protein production. It is needed to activate enzymes which help in the metabolic activities of plant cells. It is also essential for oil formation in seeds. Its deficiency can cause chlorosis i.e. the lack of chlorophyll resulting in the yellow color of leaves. Its deficiency can affect plant growth and seed oil formation.

In the human body sulphur is needed to build DNA, to keep skin, ligaments and tendons healthy and for metabolic reactions. Its deficiency could result in anaemia, low immunity, stiff joints, fragile bones, fatigue and anxiety.

**Calcium**
Calcium is essential for both plants and animals. In plants, it is deposited in cell walls and gives rigidity to cell walls and plant tissues. Calcium plays a crucial role in regulating and stabilizing many processes in plants. It neutralizes toxic materials in plants. It is important for root formation and growth.

**Magnesium**
Magnesium is an important element for plants and animals. It has both structural and functional roles in the body of living organisms. It is an important constituent of the structure of molecules of chlorophyll. Its deficiency results in yellowing of leaves and poor growth of the plant. It helps in the storage of sugar in the plant body. Magnesium assists in many functions in plants. It activates many enzymes and, in this way, helps in regulating many biochemical reactions. It also helps in the absorption and transport of phosphorus.

Magnesium is an important nutrient for us as it facilitates many biochemical processes in our body. It plays a key role in enzyme systems which regulate and control a variety of biochemical reactions in our body such as protein synthesis, nerve function, and regulation of blood sugar. It is also essential for muscle and nerve functions and regulation of blood pressure.

**2.5.2 Micro-nutrients**
Micro-nutrients are required in smaller quantities than macro-nutrients. However, their effect on health is very critical and their deficiency can result in serious health problems. There are eight micro-nutrients essential for the health and well-being of living organisms. These are boron, Chloride, copper, iron, manganese, molybdenum, nickel and zinc.

**Iron**

Though required in small quantities Iron is crucial for the health and bodily processes of all living organisms. Like many other biomolecules iron is also an important part of the structure as well as function of living bodies.

Green pigment or chlorophyll in leaves and stems of plants is crucial for photosynthesis as it absorbs the sun's energy which is then used in the process of synthesis of food. Chlorophyll is present in a cell structure called chloroplast. Plants synthesize chlorophyll with the help of iron.

Iron deficiency therefore results in yellowing of leaves called chlorosis. Chlorosis affects plants' health adversely. Iron facilitates the action of many enzymes which control and regulate biochemical reactions in plant cells.

In our body, 70% of iron is found in red blood cells in the form of a protein, haemoglobin. Haemoglobin carries oxygen from the lungs to all parts of the body. Iron is used to make haemoglobin. Iron deficiency results in reduced production of haemoglobin in cells causing anaemia. Symptoms of anaemia include weakness, fatigue, and low immunity. Iron plays an important role in biochemical reactions because it activates some enzymes needed for this reaction.

**Manganese**

It is an important element for all living organisms including man. It plays a crucial role in the activation of many enzymes that control and regulate biochemical reactions in the cells.

Twenty-five to forty percent of the total manganese present in our body is stored in bones, brain, pancreas, liver and kidneys. It activates enzymes which regulate cholesterol, carbohydrates and sugar metabolism in our body. It helps in blood clotting.

In plants, it is essential for the process of photosynthesis and nitrogen metabolism. It assists in the assimilation and utilization of nitrogen. It is essential for many processes in plants as it activates enzymes.

**Boron**

It is required in very small quantities in plants. In plants, for hormone development and sugar transport. In cell wall development. It also helps in water absorption, calcium metabolism, carbohydrate and fat metabolism. Its deficiency affects root growth, flowering and new leaves.

Its use in the human body is not very clear. However, it is said to have some role in the metabolism of vitamins and minerals. It is found in leafy vegetables.

**Copper**

Copper is found throughout our body. It helps in the production and development of red blood cells. It is also important for keeping nerve cells and the immune system healthy. Copper is essential for making collagen which is a protein in bones and connective tissue. Copper may also help in reducing free radicals that can damage cells and DNA.

In plants, it is essential for photosynthesis as it activates enzymes. It helps in the production of oxygen during photosynthesis.

**Zinc**

Zinc is an important constituent of many enzymes. It acts as an activator of enzymes which control and regulate many physiological functions in our body. Zinc plays an important role in cellular growth and tissue differentiation. Zinc deficiency causes low immunity.

In plants, zinc is an important activator of many enzymes which is why it is important for some bio-physiochemical functions ranging from protein synthesis to the metabolism of carbohydrates.

**Molybdenum**

Molybdenum is important for plants and humans however it is required in very small quantities. It plays a role in regulating many functions of plants. Molybdenum plays a critical role in converting nitrate into usable form. Hence it is important for plants.

In the human body, it helps in the breakdown of toxic substances like drugs, and it also helps in the processing of protein and DNA synthesis. Effects of its deficiency are very rare.

## 2.6 CYCLING AND BALANCE OF NATURE/MATERIAL CYCLES AND BALANCE OF NATURE

In the natural world, living things coexist with each other and with their environment in a relatively stable state or a state of equilibrium. Let us explore how nature maintains the state of equilibrium. You have read about biogeochemical cycles in previous sections. These are crucial for the sustenance of life on the earth as they keep a continuous flow of energy and nutrients between biotic and abiotic

factors of the environment through the recycling process. The biogeochemical cycle makes it possible for the same materials to be used again and again. Materials like carbon, hydrogen, oxygen, nitrogen phosphorus and sulphur circulate through different spheres of the environment. Material cycles are essential for sustaining life as they ensure the recycling and redistribution of essential nutrients and energy in ecosystems creating a state of equilibrium. Recycling of materials and energy is fundamental to the stability of ecosystems and the survival of living organisms. It keeps the ecosystems in a state of dynamic equilibrium. It means that despite disturbances caused by human activities or natural causes the system maintains an equilibrium through its stabilizing factors like; feedback loops, diversity of life forms and recycling of materials and energy. Feedback loops are natural mechanisms which maintain the state of equilibrium in nature. For instance, carbon dioxide is added to the environment through many sources and at the same time it is absorbed by plants, and dissolved in oceans thus its amount in the environment remains fairly stable. This is an example of the negative loop which checks the change before it disturbs the balance. A positive feedback loop increases the change. Biodiversity which is a variety of species also plays a critical role in keeping the balance of nature. For example, predators help to check the population of the population. Similarly, there are microorganisms which decompose waste materials in the environment thus helping in reducing the amount of waste materials.

Material cycles are the major factors in keeping a stable environment. Let us look at the The Carbon cycle involves the movement of carbon among different spheres of the environment. Plants use carbon dioxide from the environment in the process of photosynthesis to make their food in the form of organic compounds. Plants are consumed by animals and in this way, carbon is transferred to animals. Through respiration and decay and decomposition of dead organisms' carbon is added back into the environment. Hence the amount of carbon in the environment remains stable. However, human activities, such as burning fossil fuels, can disrupt this system, leading to an increase in the amount of carbon dioxide in the atmosphere leading to global warming. Similarly, in the nitrogen cycle, inert nitrogen is converted to usable compounds by bacteria in a process called nitrogen fixation. These compounds are then taken up by plants and transferred to animals through the food chain. When dead organisms are decomposed by micro-organisms nitrogen is added back to the soil, and through the process of denitrification, it is released back into the atmosphere. Human activities may add excessive amounts of nitrogen in the environment resulting in disturbance of the nitrogen cycle which keeps a check on the amount of nitrogen in the environment. An increase in nitrogen pollutes air, water and land resulting in the depletion of the protective ozone layer and global warming. The water cycle involves the continuous circulation of water

in the earth's atmosphere. The hydrological cycle plays a key role in the distribution of heat in the environment and also in ensuring the availability of fresh water for living organisms. Disturbance and disruption in the water cycle can lead to floods or droughts, affecting the balance of ecosystems. As you might have noticed different biogeochemical cycles are interconnected. Any disruption in a cycle can affect other cycles which can have far-reaching effects on the balance of ecosystems. For example, an excess of nitrogen and phosphorus can cause eutrophication, a process which results in depletion of oxygen in water bodies which leads to the death of aquatic life. Therefore, these cycles and their interactions play a key role in maintaining the health and balance of the ecosystems on the planet Earth.

## Self-Assessment Questions

1. Explain the concept of balance of nature.

2. Material cycles are interconnected and interdependent explained with the help of examples.

3. Briefly explain the role of feedback in balancing the ecosystem.

4. A disturbance in one biogeochemical cycle can affect the state of equilibrium in an ecosystem.

**Practical Activity**

1. Watch a video on YouTube about nitrogen cycling and draw a diagram showing the different stages of the cycle.

2. Prepare a chart of daily micro-nutrient requirements by human adults and children.

## SUMMARY

This unit explains the abiotic and biotic factors of the environment. It discusses the significance of the cycling of materials. It describes the steps involved in the cycling of water, nitrogen, oxygen, carbon and phosphorus. It also gives an overview of the impact of human activities on various material cycles and defines macro and micro-nutrients. Finally, it discusses how material cycling keeps a balance in nature.

**Unit–3**

# BASICS OF BIOLOGICAL COMPONENTS OF NATURAL ENVIRONMENT

**Compiled by: Dr. Sofia Khalid**

**Reviewed by: Dr. Azeem Khalid**

# CONTENTS

***Page #***

Introduction ..... 45

Objectives ..... 45

3.1 Biotic Components of Living Organisms ..... 46

3.2 Bioenergetics ..... 49

3.3 Interactions between Organisms ..... 49

Summary ..... 52

Self-Assessment Questions ..... 52

## INTRODUCTION

In any natural environment, the biological or biotic components make life exist. The living organisms or biotic components have an impact on the physical components of the environment e.g. if plants are present in an environment, they will help in lowering the temperature of the area and reducing global warming. In this unit, main groups of living organisms will be discussed along with their distribution in different parts of the world. An introduction to energy transfer at the cellular level in living organisms will be discussed and the unit will end with a discussion of various types of relationships that exist between living organisms in an area.

## OBJECTIVES

- To have an understanding of the living components of the environment based on their classification
- To know about energy, transfer at the cellular level in living organisms
- To recognize the relationships that exist between different living organisms in an area

## 3.1 BIOTIC COMPONENTS OF LIVING ORGANISMS

Bio means life and all living organisms present in an environment constitute the Biotic or Biological components of that environment. There is so much variety of living organisms present on the earth, and they are found in all physical components of the environment i.e. water, soil, and air. The living organisms vary from several micrometers to several feet in size.

The living organisms are generally classified into 3 main groups as follows:

1. Plants
2. Animals
3. Microorganisms

**Plants:** These are classified based on the presence or absence of vessels (tubes) that make a system (Vascular system) for the transfer of food and water to all parts of plants. These vascular and nonvascular plants are grouped further. The vascular plants are grouped into 3 categories, the first one has broad leaves second is the ones with narrow needle-like leaves and the third category is plants that do not produce seeds rather they have spores. Those with broad leaves are called Angiosperms and are divided into two main groups based on seed leaves (cotyledons): Dicotyledons and Monocotyledons. Mainly they are dicots. Monocotyledons are characterized by features like leaves with parallel venation and flowers with parts in thrice. On the other hand, plants that have needle-like leaves are classified under gymnosperms which have seeds that are not enclosed but on cones as opposed to flowers. Some examples include; pines and spruces which are normally conical and can grow in areas where environmental conditions are harsh.

Plants without a vascular system are called nonvascular plants and are comparatively smaller in size. These can simply absorb water and nutrients directly through their surface. This group consists of bryophytes which are sub-divided into mosses, liverworts, and hornworts. Mosses grow in rich masses and are commonly found in areas with high humidity and liverworts are characterized by flatted lobed structures and hornworts by horn-like sporophyte. These plants are very important to various ecosystems, as they are the first to colonize inhospitable areas and take part in the formation of soil.

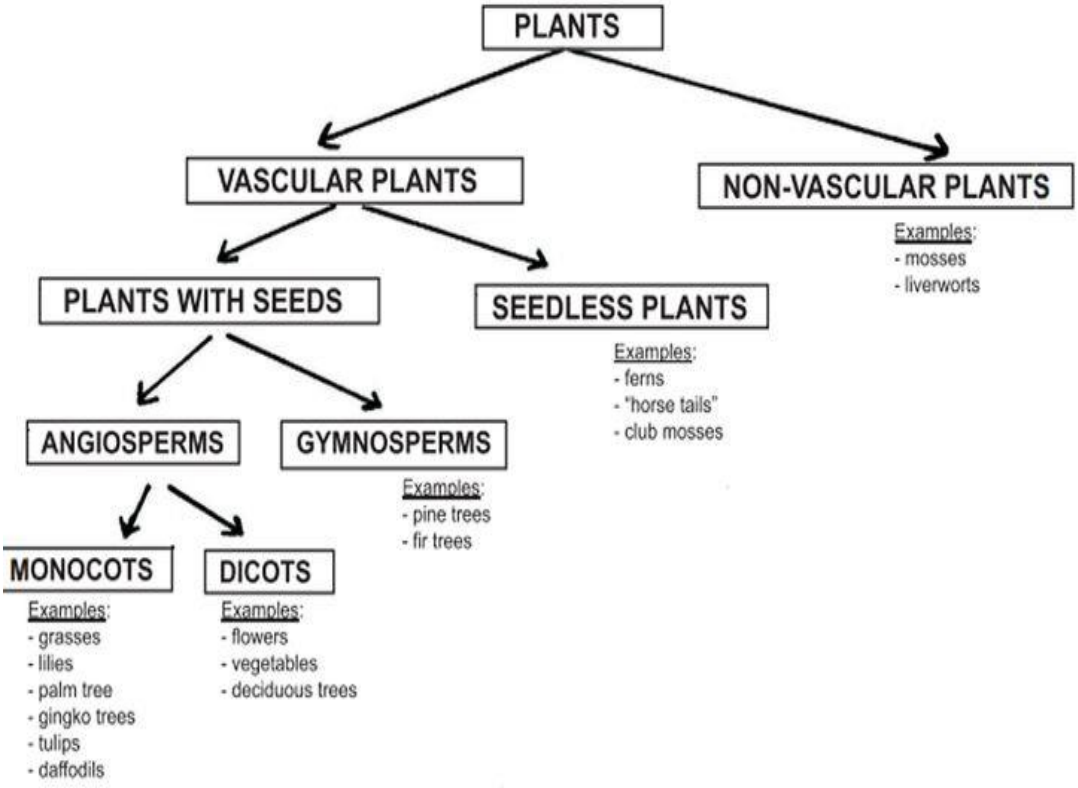


**Figure 3.1:** Classification of flowering and non-flowering plants (Source:https://www.pinterest.com/pin/classifications-of-flowering-and-non-flowering-plants--257690409905098886/)

Plants present in an area give us an understanding of the environmental conditions of that area e.g. desert plants have certain adaptations to save loss of water; similarly, some aquatic plants have adapted to support floating leaves.

**Animals** are subdivided into the groups known as invertebrates as well as vertebrates. A backbone is absent in invertebrates, and these are classified into arthropods, molluscs, annelids, cnidarians, and echinoderms. Arthropods possess an exoskeleton and segmented bodies, molluscs have soft bodies and hardcover, annelids are known to be segmented, cnidarians are symmetrically arranged with stinging cells and echinoderms are symmetrically arranged with a water vascular system. They can be categorized into fish which includes fishes such as salmon and shark, amphibians which include animals like frogs and salamanders; reptiles which include animals such as snakes and turtles, birds which can be eagles and sparrows and mammals including humans and whales.

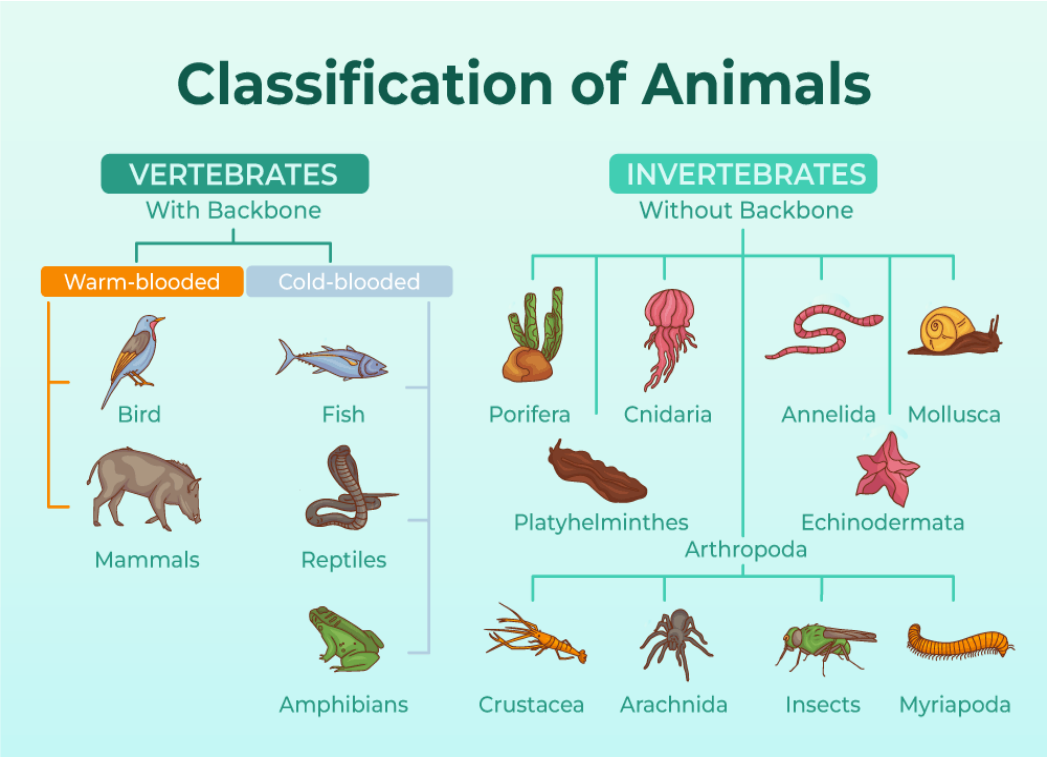


**Figure 3.2:** Classification of animals (Source: https://www.geeksforgeeks.org/classification-of-animals/)

**Microorganisms** are way too diverse, and they are predominantly single-celled or simple compact structured organisms. These bacteria are a large group of microorganisms that are very important in nutrient cycling and decomposition. Another group of microorganisms is called archaea. They have features similar to bacterial ones but have different biochemical and genetic features, this kind of microorganisms often inhabit hot springs or areas with high salinity. Protists are eukaryotic microorganisms that include plants like algae, animal-like protozoa, and fungus-like slime moulds compose their food from organic matter, and reproduce asexually and sexually in different ways according to their classification. Finally, the yeast and moulds, as well as the bigger fungi including mushrooms, partake in decomposition and nutrient recycling. Composing nearly one-third of all microbial biomass, these microorganisms play important roles in ecosystem functioning and many biotechnological processes.

## 3.2 BIOENERGETICS

Bioenergetics is a branch of biology that is concerned with the ways and means through which energy is procured, transformed and metabolized in an organism. It starts with energy procurement whereby organisms acquire energy from the environment they live in. For instance, plants trap light energy in the form of photosynthesis where light energy is transformed to chemical energy in glucose molecules. This process not only synthesizes the glucose which is rich in energy but also liberates the oxygen. Whereas animals get their energy through the oxidation of organic matter whereby energy is released from the matter.

The following step is energy transport where obtained energy is converted to a form that can be utilized in the cells. This is well illustrated in the production of adenosine triphosphate (ATP) in plants and animals which functions as the chief energy currency. Cellular respiration is a process where glucose is metabolized in the cell's organelle known as mitochondria through various metabolic processes such as; glycolysis, the citric acid cycle, and oxidative phosphorylation. This breakdown process creates ATP which is involved in storing and providing energy needed for specific activities within the cell.

Last, of all, energy utilization is concerned with how the various processes in cells use ATP. This encompasses tasks like muscle contraction, transport of molecules across the cell membranes as well as synthesis procedures that include the formation of bigger molecules like protein and nucleic acids. Energy management is very important as it determines cellular functions, growth, and reproduction.

In this way, bioenergetics is linked greatly with the environment as it affects the traffic of energy in ecosystems. The energy received by plants through photosynthesis is used by nearly all other living organisms and the metabolic activities in animals and microorganisms form cycles of nutrients and energy exchange in ecosystems. Hence by understanding bioenergetics, one is in a better position to understand how an organism relates to the environment or how an ecosystem is balanced.

## 3.3 INTERACTIONS BETWEEN ORGANISMS

These relations involve exchanges of materials and energy among organisms and are the essential driving forces in ecosystems involving survival, reproduction, and equilibrium. Such interactions can be classified as follows and each of them has certain impacts on the species.

## Types of Interactions

**Mutualism:** It is advantageous for both the organisms which have such a relationship. For instance, bees and flowers are partners in a mutualism since the bees will benefit from the nectar that they get from the flowers while the flowers are protected by bees from being trampled on by animals. Plants are fertilised by bees while the bees collect nectar from the flowers in the process benefiting both plant and bee.

**Commensalism:** This allows one organism to gain something while the other continues to remain indifferent to the occurrence of the process. Some of them include barnacles on the whales. The barnacles get good areas for feeding as the whale moves, and the whale on the other hand is not so affected.

**Parasitism:** This is one example of a parasite in ecological systems whereby a species derives some benefits out of a loss incurred by another one. In mammals, an example of parasitic existence is the tapeworms; although they feed on the host's food or nutrients, they are usually a nuisance to the host's health.

**Competition:** In a negative feedback system, many organisms are fighting over the same amount of resources. Lions and hyenas are enemies in the savannah region as both seek the same things that they need in the wild; food for instance.

**Predation:** Predation on the other hand involves one organism (the predator) actively searching for, and/or consuming another organism (the prey). In line with this, lions targeting zebras are the best example of how some lions benefit food-wise while soliciting change on the Zebra.

**Amensalism:** This interaction happens when the organisms of one pair suffer or are damaged in some way but the organisms of the second pair remain unaffected. Cows will stomp on plants and destroy them, but the same cows will not be affected by the plants which they have crushed.

**Neutralism:** In neutralism, the two organisms are said to be in harmony such that, none negatively impacts the other. Squirrels and frogs in a forest may interact little because one represents earthworms and the other represents birds.

### Mimicry

Mimicry on the other hand is a process whereby one organism evolves similarity with another to benefit. There are several types:

i) **Batesian Mimicry:** A type of organism is a model of a dangerous one in order not to be eaten. For example, the viceroy butterfly looks like a monarch butterfly when it is not.

ii) **Müllerian Mimicry:** Two or more toxic species are somehow able to evolve mechanisms of mimicking, and this kind enhances general avoidance by predators. Worker bees and hornets carry out functions that will be protective in nature Honeymooners and wasps both possess warning coloration.

iii) **Aggressive Mimicry:** A predator or parasite looks or behaves like another organism which is advantageous, or at least, non-threatening, to its victim. For instance, in the example mentioned above, where anglerfish has a lure that resembles a small fish to lure its prey.

The types of interaction and mimicry adaptations elucidate the relation and dependency of species in an ecology.

## SUMMARY

This unit explains the biotic components of living organisms and their categorization as plants, animals, and microorganisms. It also discusses bioenergetics and the interactions between organisms as classified into mutualism, commensalism, parasitism, competition, predation, amensalism, neutralism, and mimicry.

## SELF-ASSESSMENT QUESTIONS

1. How by looking at plants in an area we talk about the environmental conditions of that area?

2. What are bioenergetics and why do living organisms need it?

3. Why is mimicry required by living organisms in certain environments?

**Unit–4**

# BASICS OF CLIMATOLOGY

**Compiled by: Dr. Sapna**

**Reviewed by: Dr. Muzammil Anjum**

# CONTENTS


***Page #***

Introduction ..... 55

Objectives ..... 55

4.1 Introduction to Climatology ..... 56

4.1.1 Weather and Climate ..... 56

4.1.2 Meteorology vs Climatology ..... 56

4.2 Atmosphere of the Earth ..... 57

4.2.1 Earth's Atmospheric Composition ..... 57

4.2.2 Atmospheric Layers ..... 57

4.2.3 Environmental Issues Related to the Atmosphere ..... 57

4.3 Atmospheric heat Trappers and their Impact on Earth's Temperature ..... 58

4.3.1 Sources and Types of Greenhouse Gases ..... 58

4.3.2 Global Warming ..... 58

4.4 Climate Change ..... 59

4.5 Impacts of Climate Change ..... 60

4.6 Studying Climate: Data & Tools ..... 61

4.6.1 Ground-Based Data ..... 61

4.6.2 Remote Sensing Data ..... 61

4.6.3 Introduction to Climate Models ..... 62

Summary ..... 62

Self-Assessment Questions ..... 62

# INTRODUCTION

The concept of Climatology is often mixed with Meteorology but there is a big difference in both terms. One is the long-term changes in the weather and the other refers to the short-term changes in weather. Both represent different fields.

Climate change is a big issue the world is facing right now. Humans' anthropogenic activities are causing changes in the climate by adding Greenhouse gases into the atmosphere thereby changing the earth's temperature. Climate change is impacting different places in the world differently. Developing countries, though not contributing much to Greenhouse gas emissions are more vulnerable than the developed countries.

Keeping in view the impacts of climate change, there is an urgent need for research. Other than traditional techniques, modern ways of research are also developing with the advance of technology. Advancement in satellite remote sensing helps study the climate change effects on different regions of the world and at the same time predict what can be observed in future. Climate models are helping in the prediction of climate change by giving a future forecast.

Sustainable practices and policies need to be developed to tackle this issue. Focus is needed on the innovations in green technology and sustainable energy, climate governance, and the role of international agreements and cooperation.

This unit looks more closely at the climate, the role of greenhouse gases in climate change, and climate change impacts. We will also get an overview of satellite remote sensing usage in climate change studies and the climate models will be introduced.

# OBJECTIVES

The objectives of the unit are to:

1. Differentiate Climatology and Meteorology
2. Understand the atmosphere of the earth and atmospheric heat trappers
3. Discuss the climate change and its impacts
4. Know the utilization of ground-based data, remote sensing data and climate change models

## 4.1 INTRODUCTION TO CLIMATOLOGY

### 4.1.1 Weather and Climate

Are the weather and climate the same? The answer is no, weather is what you are experiencing, and climate is what you expect.

**Weather**

Weather is the short-term changes in the atmosphere. It is what you experience each day within minutes, hours, days, and months. These changes happen in the Earth's troposphere. Many different factors such as rainfall, humidity, temperature, sunshine, air pressure, wind speed and direction, and much more lead to changes in the atmosphere. Each location on Earth has different weather conditions.

**Climate**

Climate refers to the long-term changes in the atmosphere. When you talk about the climate of any location, you talk about its average temperature, rainfall, humidity and other factors over long periods. A specific time frame for these averages is internationally accepted as 30 years. These 3 decades of averages of weather are known as the climate normal. For example, when we describe the climate of a location; it might be how windy is that location, how often the rainfall falls, and what usually is the temperature in different seasons. Every location's climate is different from the other locations on Earth.

### 4.1.2 Meteorology vs Climatology

Meteorology and Climatology are both sub-fields of atmospheric science. The scientific principles and phenomena of both fields are similar, but they usually differ in their approach, application, and time scale. Meteorology is the study of changes in weather patterns in the atmosphere over a short time (e.g. minutes to months). It studies the atmospheric variables (e.g. rainfall, humidity, temperature, wind speed and direction) current and near-range forecasting. Climatology studies the long-term changes in atmospheric conditions. It is more focused towards the patterns of climate and fluctuations. The detailed discussion is on the interactions between the Earth and the atmosphere system. For example, a statement describing the weather would be that now the outside temperature is 15$^{o}$C and the climate would be that the temperature is mild with an average value of 25$^{o}$C. Scientists working on weather patterns are referred to as the "Meteorologists" and those working on the climate, its variations and effects are referred to as "Climatologists".

## 4.2 ATMOSPHERE OF THE EARTH

### 4.2.1 Earth's Atmospheric Composition

Look up in the sky. What do you observe? A protective layer around the earth which keeps us safe from outer space and without which we would not exist is known as the atmosphere of the earth. The sunshine, the air that you breathe, the clouds that you see, and the wind that moves the trees, all are due to our Earth's atmosphere. This layer of gases is held together by the gravitational pull of the earth.

Oxygen is most important and is available in the atmosphere, but our earth's atmosphere is not oxygen. It is comprised of gases such as Nitrogen (78%), Argon (0.9%), Oxygen (21%), and some other available gases (0.1%). Other gases include methane, water vapor, carbon dioxide, and neon which make up our atmosphere's remaining 0.1%.

### 4.2.2 Atmospheric Layers

Five layers of the atmosphere exist including the troposphere, mesosphere, stratosphere, exosphere, and thermosphere which are divided based on their temperature changes. The nearest layer to Earth is the troposphere where most of the atmosphere's mass lies. Then there is another layer known as the stratosphere that contains the ozone gas molecules which are made up of three oxygen atoms that protect life on earth from potentially harmful ultraviolet radiation. In the middle of the troposphere and stratosphere, a thin boundary is available which parts the troposphere from the stratosphere and is known as the tropopause. The layer after the stratosphere is the mesosphere and its temperature declines with altitude reaching a minimum temperature of about -90ºC at the top. The boundary separating the stratosphere from the mesosphere is the stratopause which is the top of the stratosphere. The thermosphere is the fourth layer, here temperature increases with altitude. The upper regions of this layer are very hot due to their closeness to solar radiation. The boundary between mesosphere and thermosphere is known as mesopause. The final layer is the exosphere which consists of mainly hydrogen and oxygen atoms, but they are very few. Normally the pull of gravity is so small that the gas molecules escape into outer space. The thin boundary between the thermosphere and exosphere is thermopause.

### 4.2.3 Environmental Issues Related to the Atmosphere

Increasing human activities are creating environmental problems such as emitting greenhouse gases which cause global warming, hence making the earth an unlivable place for human beings and wildlife. Further details are given in below sections:

## 4.3 ATMOSPHERIC HEAT TRAPPERS AND THEIR IMPACT ON EARTH'S TEMPERATURE

### 4.3.1 Sources and Types of Greenhouse Gases

Greenhouse gases are the atmospheric heat trappers which exist in the lower atmosphere. They maintain high temperatures thus allowing less escape of heat back to space. Greenhouse gases make our Earth a habitable place for living because without these gases the temperature of the Earth would be negative. However, due to the Industrial Revolution, these gas concentrations increased and surpassed the set international standards.

Common greenhouse gases in the atmosphere include chlorofluorocarbons (CFCs), carbon dioxide ($CO_2$), hydrofluorocarbons (HFCs), ozone ($O_3$), nitrous oxide ($N_2O$), water vapour, and methane ($CH_4$). Four gases such as $SO_2$, $CH_4$, $CO_2$, and $N_2O$ are attracting serious global attention due to their significant impact on global warming, climate change and air quality. $CO_2$ is the most abundant greenhouse gas released by human activities. Sources of $CO_2$ are primarily burning fossil fuels, certain industrial processes, and deforestation. $CH_4$ has 20 times more global warming potential than $CO_2$. It is emitted during the process of production and transportation of fossil fuels, agricultural activities, livestock, and the decay of organic waste in landfills. Also, $N_2O$ has more global warming potential than $CO_2$ and it also contributes to the ozone layer depletion which is important and protects Earth from harmful ultraviolet (UV) radiation. $SO_2$ is released when fossil fuels are burned, and volcanos erupt.

### 4.3.2 Global Warming

Any kind of human activities whether they are at home, in the field, in industries or using transport, all raise greenhouse gas emissions which increase the temperature of the Earth by absorbing the heat and act like a blanket around the Earth hence keeping it warm. An increase in the temperature also leads to more water vapors present in the atmosphere which again is like another blanket around the Earth, trapping more heat. This heating would be good for the people who are living in cold places having cold climates but would be problematic for those living in warm places having warm climates because the globe is warming, not particular spaces. If the Earth's temperature continues to increase, it is estimated that it will increase the average temperature by about 3 degrees a century which may not sound very much but it will have significant impacts on the regions already warm. It is due to this change that many human communities and ecosystems will find it difficult to adapt, especially in developing and least-developed countries.

## 4.4 CLIMATE CHANGE

Long-term weather alterations in a region are known as climate change. Here comes a question in mind who is responsible for climate change? Natural variability (orbital drifts, volcanic eruptions, solar activity, ocean currents, etc.) influences the Earth's climate but most certainly human activities have given rise to greenhouse gases. The growth of the world population is affecting $CO_2$ emissions.

In the 21st century, one of the greatest threats to human health is climate change. The temperatures are increasing globally, and the frequency and intensity of extreme weather events are rising such as floods, storms, heatwaves, and droughts. The below graph (Figure 4.1) manifests the fluctuations of the world's surface temperature which is compared to the 1951 to 1980 long-term mean. We observe that it has crossed 1°C and it is found that the Earth's overall temperature was about 1.36 $^{C}$ warmer during 2023 than in the late 19$^{th}$ century (1850 to 1900) mean.

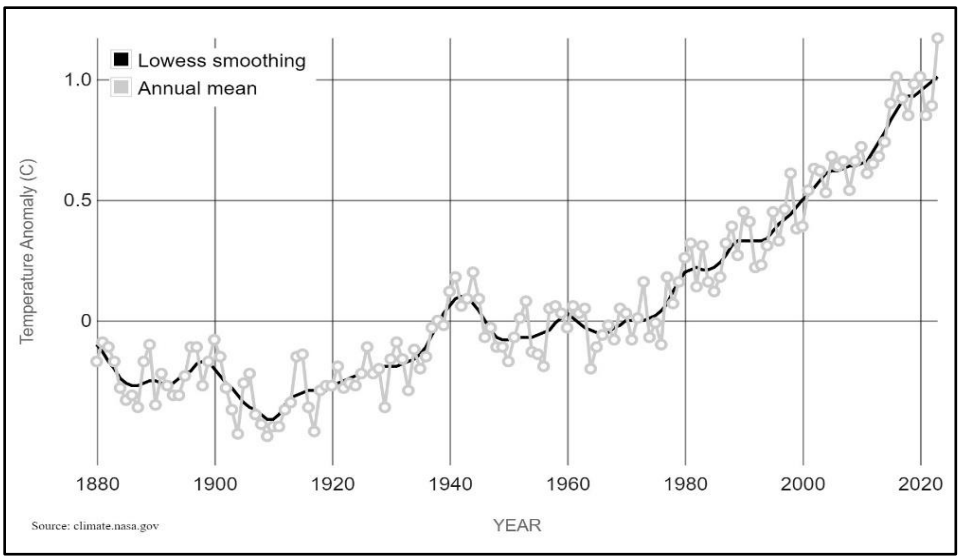


**Figure 4.1:** Fluctuation observed in world surface temperature in comparison to the long-term average (1951-1980) (Source: NASA/GISS)

Figure 4.2 manifests the warming over the past 30 years (1994-2023) which is much faster than the long-term trend (1901-2023). The change in temperature per decade is highest from 1994 to 2023 with the year 2023 being recorded as the warmest year.

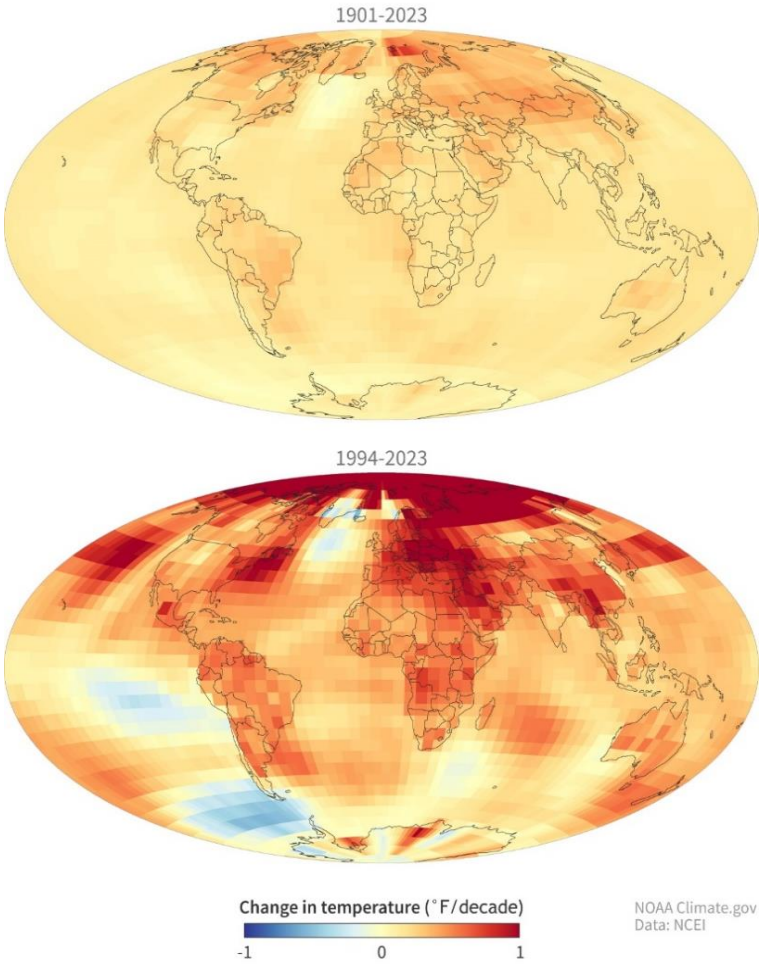


**Figure 4.2:** Past 30 years (1994-2023) global warming compared to the long-term record (1901-2023) (Source: Climate.gov Data: NCEI)

## 4.5 IMPACTS OF CLIMATE CHANGE

The environment, humans, and animals are being affected by climate change in different ways. Some places are experiencing floods due to extreme rainfall while some are affected by droughts due to scant rainfall. Sea levels are rising due to the melting of glaciers and expansion of warmer water which may affect the coastal communities, coral reefs, and wetlands. Sea water is becoming acidic due to the increase in the absorption of $CO_2$. Temperature rise in larger lakes is increasing algal blooms and favouring the invasive species. Ecosystems are also changing, which is forcing different species to migrate from their habitats to colder places or die. Flowering events and the time of growing seasons are also changing. Crop

productivity is declining due to climate change's significant effects on water resources and agriculture. Future generations will face food security issues. Figure 4.3 illustrates the climate change impacts.

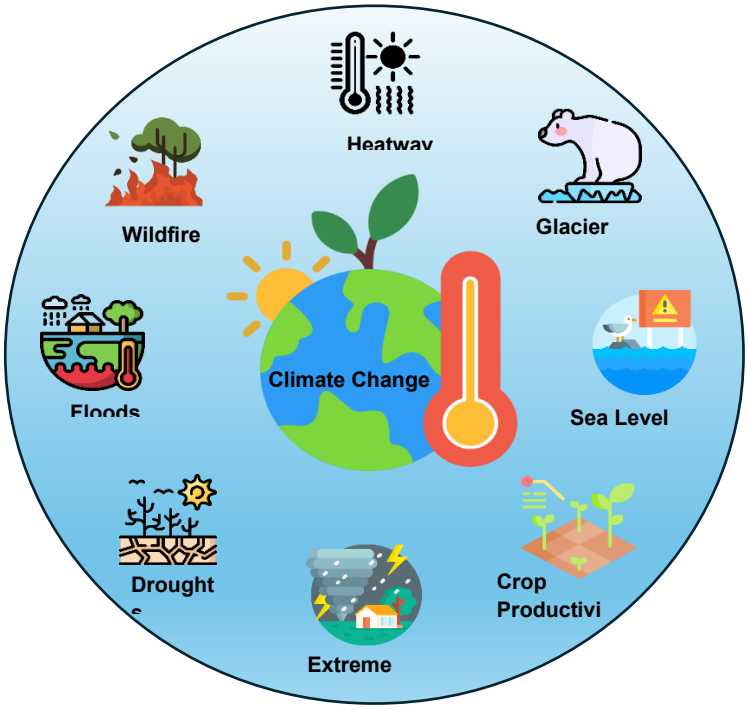


**Figure 4.3:** Impacts of climate change

## 4.6 STUDYING CLIMATE: DATA & TOOLS

### 4.6.1 Ground-based data

Recording and keeping accurate historical ground-based weather data is important in studying the climate of a region and climate modelling. This data is also known as observed data and station-based data. It is used as a benchmark for comparison with remote sensing datasets to ensure that satellite observations accurately represent the ground data and for validating the models. Ground-based weather stations, temperature sensors and rain gauges provide accurate and reliable data on several weather parameters including precipitation, air pressure, temperature, wind speed, and humidity. These measurements often have long-term records which are crucial for identifying trends and changes in climate over time. It is useful for studying local and regional patterns of climate, specific weather events, and microclimates.

### 4.6.2 Remote Sensing Data

Data that is acquired through satellites without being in physical contact with the ground. It provides a detailed view of the atmosphere of the Earth, land and ocean using sensors. This allows for monitoring climate variables on a global scale such as land

surface temperature, cloud cover, sea surface temperature, ice cover, and vegetation. Other than these variables, they also provide data for greenhouse gas concentrations, aerosol, humidity, air temperature, air pressure, precipitation, wind speed, latent and sensible heat fluxes over land, solar radiation, and nighttime light data. These datasets are important because of their extensive spatial and frequent temporal coverage, making it possible to observe the climate over larger areas and over different timescales. For example, they are crucial for studying the large-scale phenomena of El Niño and La Niña. Their importance is also in cases where there is a need for data from inaccessible and remote areas such as in polar regions, deserts, and oceans.

### 4.6.3 Introduction to Climate Models

What are climate models? Climate models are essential tools in Climatology, utilized to understand climate systems and predict future climate changes. For example, if someone wants to project the future climate scenario based on different levels of greenhouse gas emissions, climate models will be used. Or if someone wants to observe how the temperature or precipitation of the region can change in future based on previous historical data, climate models can help. They aid scientists and policymakers in evaluating the effects of climate change and designing strategies for adaptation and mitigation. The types of climate models are Energy Balance Models (EBMs), Earth System Models (ESMs), General Circulation Models (GCMs), and Regional Climate Models (RCMs).

## SUMMARY

This unit explained the basics of Climatology. It is very important to understand the difference between Climatology and Meteorology. In the first part of the unit, the two fields have been differentiated. In the second part, the atmosphere and its layers have been discussed in detail. In the third part, the types and sources of greenhouse gases have been explained. Further, the role of these gases in climate change has been discussed. In the last section, the significance of data and tools for studying the climate have been explained.

## SELF-ASSESSMENT QUESTIONS

1. Differentiate between Climatology and Meteorology.
2. What do you know about the Earth's atmosphere?
3. What are the sources of greenhouse gas emissions?
4. Write about the impacts of climate change.
5. What is the importance of ground-based data and remote sensing data in climate change studies?
6. How do climate models predict climate change and its impacts?

**Unit–5**

# AQUEOUS ECOSYSTEMS

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# CONTENTS


***Page #***

Introduction ..... 65

Objectives ..... 65

5.1 Introduction to Aqueous Ecosystems ..... 66

5.2 Ocean Ecosystems ..... 67

5.2.1 Distribution of Oceans in The World ..... 67

5.2.2 Ocean as Habitat of Different Organisms ..... 68

5.2.3 Light Penetration and Oceanic Zones ..... 68

5.2.4 Different Ecosystems of The Ocean ..... 70

5.2.5 Ocean and Environment ..... 72

5.3 Lake and Pond Ecosystem ..... 72

5.3.1 Lake Ecosystem ..... 72

5.3.2 Pond Ecosystem ..... 73

5.4 Spring and River Ecosystem ..... 74

5.5 Environmental Challenges to Aquatic Ecosystems ..... 75

5.6 Addressing Challenges to Aquatic Ecosystems ..... 76

Summary ..... 77

Self-Assessment Questions ..... 78

# INTRODUCTION

Aquatic ecosystems are the world's most significant ecosystems. Owing to their volume and strong role in the world's biodiversity and circulation. Preliminary knowledge of the aquatic ecosystems is mandatory for every individual. In this unit, we will delve into the vast variety of aquatic ecosystems stretching from the vast oceans to the tranquil freshwater resources. We will study different aquatic ecosystems and how the organisms have developed their specific characteristics depending upon their specific habitat. We will study the components of different water ecosystems and their physical and biological characteristics. Also, we will go through the overview of different trophic levels in the aquatic ecosystems. By understanding the intricate connections of the aquatic ecosystems, we will get a better appreciation of the delicate balance of different species and water bodies that are vital for sustaining our future generations.

# OBJECTIVES

The objectives of the unit are to:

1. Provide students with an overview of different types of aquatic ecosystems
2. Equip the students with knowledge of different water forms and their significance
3. Provide students with enough knowledge about the threats and challenges to different water bodies

## AQUEOUS ECOSYSTEMS

Water is the most beautiful and powerful natural object. This nature's marvel is the most obvious and prominent creation on this planet earth. It is noteworthy that water comprises more than 71% of the earth's surface. Water is present in different forms on the planet. Comprising of solids that are glaciers and icebergs, and liquid and gas as water vapours. Water in its every form is vital for living organisms and is significant for sustaining the earth's ecosystem.

## 5.1 INTRODUCTION TO AQUEOUS ECOSYSTEMS

Aqueous ecosystems are fascinating diverse environments that include both freshwater and salt water resources. These ecosystems are vital for the health of the planet as they support an array of living organisms and govern major environmental phenomena. Particularly the aqueous ecosystems are pivotal for maintaining the earth's climate through their water cycle. The natural water is continuously replenished by nature's water cycle (Figure 5.1). Nature sustains a delicate balance of atmospheric water and land water to govern major environmental changes in the world.

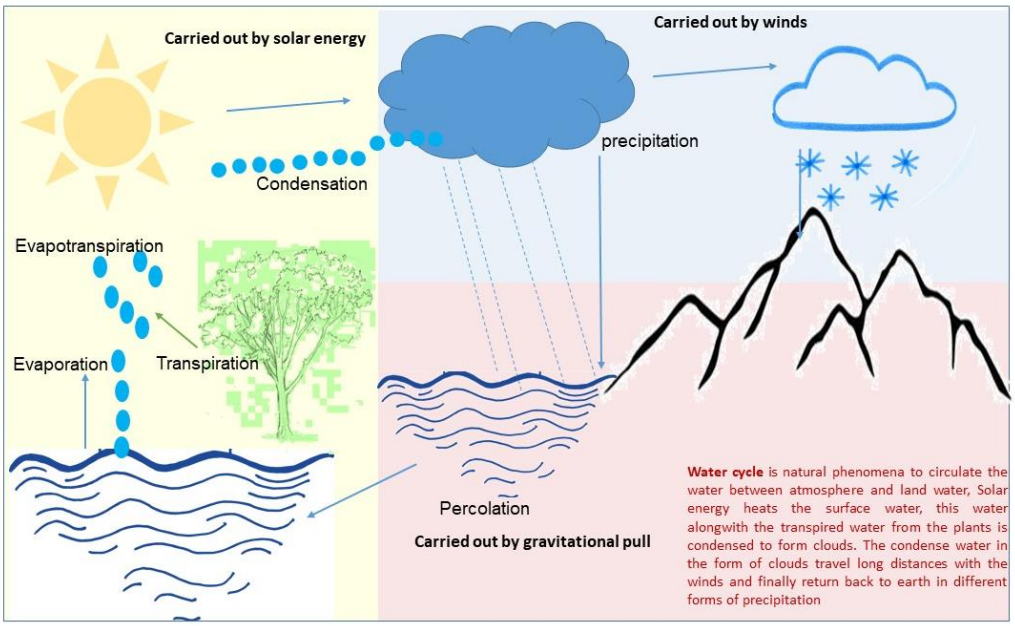


**Figure 5.1:** Water Cycle

Overall, the hydrosphere comprises 71% of the total earth's surface. Where this distribution is not even as 97% of the total water is present in the oceans. Among rest 3% of the water 69% is fixed in the form of glaciers and ice-caps and 30% is

underground water. Only 1% of the freshwater is present in rivers, lakes or ponds that can be used directly from the source. Therefore, the aqueous ecosystems are significant from their conservation perspective. Understanding that only 1% water is readily available for consumption by the users the significance of water conservation and management is further enhanced.

## 5.2 OCEAN ECOSYSTEMS

Knowing the fact that most of the earth's surface is covered by the oceans, oceanography* and marine biology have remained the most applied subjects of natural science. The ocean encompasses the most significant and complex ecosystems of the world. Around 97% of the water supply of the earth comes from the oceans. The marine ecosystems carry out the highest photosynthesis in the world. Therefore, produces the highest amount of oxygen that we breathe in. However, the distribution of the organisms in the ocean is greatly affected by light and their location.

**BOX 1:**

**Oceanography** *is the scientific study of the ocean and its processes. It encompasses various disciplines that investigate different aspects of the ocean, including its physical, chemical, biological, and geological properties.*

### 5.2.1 Distribution of Oceans in The World

The earth's oceans are divided into five major oceans present in different parts of the world. The largest ocean in the world is the Pacific Ocean which is estimated to cover 165.25 $km^2$of area. The Pacific Ocean extends between the Arctic Ocean in the north and the Antarctic in the south. Where it touches the major continents of the world including Asia and Australia in the west and America in the east. The second largest ocean is the Atlantic Ocean with 106.46 $km^2$ area. This ocean stretches from America in the west and Europe and America in the east. where the Arctic Ocean lies in the north and the Southern Ocean resides in the south of the Atlantic Ocean. The third largest ocean of the world is the Indian Ocean which roughly stretches over 70.56 $km^2$. The ocean is located between Africa, Africa, Asia and the Southern Ocean at west, east, north and south respectively. The Southern Ocean covers only a 20.33 $Km^2$ area. This ocean has unique characteristics in that it encircles Antarctica and has specific Antarctic Circumpolar currents. The smallest of all is the Arctic Ocean which covers only 13.98 $Km^2$, it is the smallest and also the shallowest ocean. It is located around the Arctic region bordered by Europe, Asia and North America.

### 5.2.2 Ocean as Habitat of Different Organisms

As the oceans of the world are different so is their biodiversity. This diversity is brought by the physical variations including light, temperature, depth and nutrient availability in different oceans. Also, within the large ocean bodies, the seas have different conditions that supports different life forms. So, each ocean and sea has a diverse group of distinctive organisms.

For instance, the Pacific Ocean being the largest and deepest of all provides vast stretches of variety of life. Most of the open ocean serves a variety of migratory species like whales, sharks and tuna. Where interestingly the tropical western Pacific includes the Great Barrier Reef of Australia and the Coral Triangle, which hosts a vast variety of species. Where the Mariana Trench is the deepest point in the world, therefore the organisms present here are very limited and adapted to extreme pressure and cold water. The Atlantic Ocean provides a habitat for a range of organisms. These habitats stretch from the temperate coastal water, tropical coral reefs and cold Arctic regions. The Atlantic corals are present in the southeastern Atlantic and Caribbean Sea. Where the North Atlantic has cold water, therefore, cold-adapted species like cods and krill are common in this area. Where Mid-Atlantic region has hydrothermal vents that host communities like worms and extremophiles. Where Indian Ocean also known as the warm water ocean is known for its year-round warm water and monsoon-driven climate. Therefore, it provides habitats in the form of coral reefs, temperate coasts and deep-sea habitats. Most of the corals are present in the Red Sea and islands of the Indian Ocean. Due to its - monsoon-driven climate, it provides a large amount of nutrients therefore it hosts a huge school of fish and marine mammals. On the other hand, the Southern Ocean has cold waters and seasonal ice covers, therefore it hosts a large amount of ice algae and krill. The ocean is beautifully adorned with some of the unique cold water species like penguins, seals and whales. The Arctic Ocean has characteristic cold water with ice for most of the time of the year, the ocean also has low light conditions. The organisms found in the Arctic Ocean are specifically adapted to this environment. This ocean hosts the world's most widely discussed animal and the largest bear in the world is a polar bear. The other species included are seals and cods. Thus, each ocean type is home to different types of organisms, that are well adapted to the specific characteristics of that particular ocean's physical conditions.

### 5.2.3 Light Penetration and Oceanic Zones

The life in the ocean is not only distributed by the type of ocean but also is greatly influenced by the light and pressure conditions. Depending upon the light penetration and the distance from the shore the ocean is divided into different zones. The ocean shore comprises the edge where the oceanic and continental plates meet, where all the open waters of the ocean are known as the pelagic realm. The Open

Ocean is further divided into 5 major zones depending upon the light penetration in the water.

**i. Epipelagic Zone (Euphotic Zone)**

This zone is enriched with solar radiation, from the surface this zone stretches up to 200 meters of vertical depth. The zone is well-lit with solar radiation and the light intensity is just perfect for photosynthesis. Since there is good light penetration therefore it is also warmer than the lower zones of the ocean. The primary producer in this zone is phytoplankton. This zone is enriched with zooplankton as well due to sufficient provision of food. The zone is food food-abundant zone and thus has a wide range of fishes ranging from anchovies to large sharks. This zone also hosts dolphins, whales and seals for feeding. In the shallow parts, corals are also present.

**ii. Mesopelagic Zone (Twilight Zone)**

Consider the evening time. This ocean zone has dim light penetration and thus is also known as the twilight zone of the ocean. This zone ranges between 200 meters to 1000 meters. The light is insufficient for photosynthesis. However, the zone is festooned with beautiful bio luminous organisms that can produce their light. The fishes like lantern fish and bristle mouth are well adapted to this region. Where marine mammals can come deep into this zone for hunting. This zone is specifically inhabited by cephalopods like octopuses and squids.

**iii. Bathypelagic Zone**

The zone lies between 1000 meters to 4000 meters. This zone is completely dark with no light penetration. The temperature of this zone is also low (usually ranges between 4°C to 10°C). This zone also has bioluminescent creatures. Where only specific types of deep-sea anglerfish and giant squids are present in this zone. However, invertebrates like deep-sea jellyfish and various crustaceans are present in this zone.

**iv. Abyssopelagic Zone (Abyssal zone)**

This zone lies from the depth of 4000 meters to 6000 meters. There is complete darkness and very low temperature in this zone. Since the conditions are very harsh, therefore, well adapted sparse biodiversity is present in this zone. Only abyssal Cusk eel and a few types of sea cucumbers are present here.

**v. Hadalpelagic Zone (Hadal zone)**

This zone lies below 6000 meters including the deep sea trenches and hidden zones. The conditions here are extremely harsh with no light and cold temperatures. Under such extreme conditions, only hadal snailfish and few invertebrates are found in this zone.

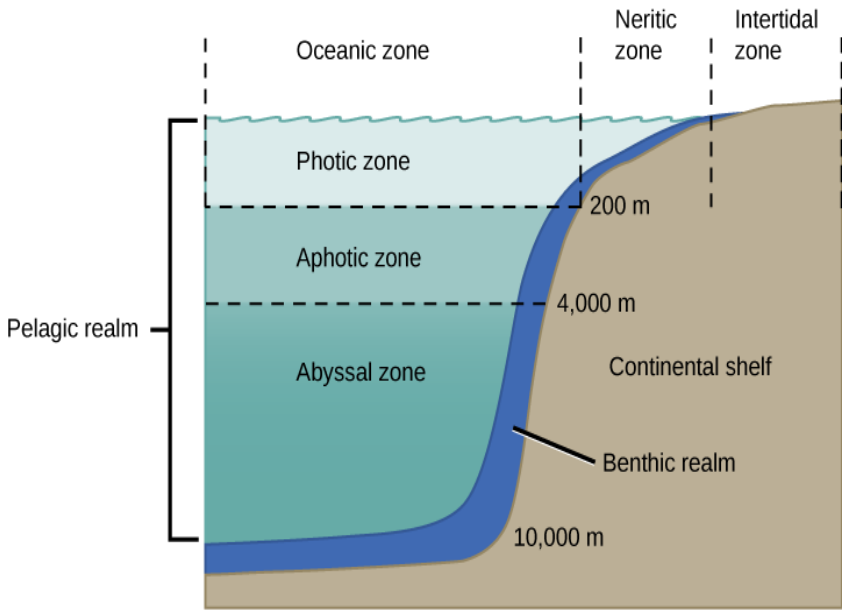


**Figure 5.2:** Different ocean zones along with ocean depths

**BOX 2:**

*Estuaries are vibrant and fecund coastal environments; these are formed where the freshwater from rivers and streams meets and mixes with saltwater from the ocean. These are the interim zones between land and ocean waters are characterized by a distinctive blend of salinity and nutrients, making them critical for both environmental and anthropogenic purpose.*

### 5.2.4 Different Ecosystems of the Ocean

Depending upon the vast biodiversity and physical characteristics of the ocean, the Ocean encompasses a few of the world's most unique ecosystems. These ecosystems are distinctive in their characteristics and host diverse forms of specific organisms. Each ecosystem is established by the intrinsic connections among organisms and their habitat. Few of the significant ocean ecosystems are:

**i. Coral Reefs**

Coral reefs are cliff-like ocean structures that are formed by the colonies of polyps. These structures are predominantly present in the tropical and sub-tropical regions of the world. Polyps are soft invertebrates that possess a calcareous *exoskeleton. These deposits of calcium carbonate form one of the ocean's most beautiful ecosystems, coral reefs. These coral reefs are mainly of two types hard corals and soft corals. The corals provide habitat for a wide range of marine life including

algae, fish invertebrates and marine mammals. The whole ecosystem consists of producers like algae to the top trophic-level predators including dolphins and sharks. Corals are the ocean's most significant contributors to sustaining marine biodiversity.

**BOX 3:**
*Interestingly the corals have also cleaning mechanism for the waste produced by the animals inhabiting them. Apart from the regular ocean currents Corals have cleaner fish like cleaner wrasses and gobies, similarly lysmata spp. a cleaner shrimp removes the parasites and dead tissues from the corals.*

### ii. Kelp Forest Ecosystems

Another interesting marine ecosystem is the Kelp Forest that are formed by the underwater brown algae. They can withstand cold temperatures and strong currents. The prime species of algae in the kelp forest is brown algae which are fast-growing algae where the marine animals usually present are rockfish, kelp bass, sea urchin abalone and sea otters. These kelp forests provide coastal protection and support to local fisheries.

### iii. Seagrass and Meadow Ecosystems

Seagrass and Meadows are underwater grasslands that are present in shallow warm waters, often they are present in lagoons. Eel grass and manatee grass are common grass species. Where the animals residing include seahorses, crabs, shrimps and dugongs. These seagrass forests play a significant role in carbon sequestration, coastal protection and nutrient filtration.

### iv. Mangrove Forest Ecosystems

These are coastal wetlands; this unique ecosystem consists of salt-tolerant trees and shrubs that can grow in the intertidal zones. They are found in mudflats, with brackish water in tropical and subtropical regions. The tree species are mainly red, white and black mangroves. These forests provide a habitat for the snapper and mangrove jackfishes, along with mangrove crabs, molluscs and kingfishers.

### v. Hydrothermal Vent Ecosystems

Hydrothermal vents are geographical formations that are found on the ocean floor where superheated, mineral-rich water is expelled with great force from the earth's crust. These are like underground hot geysers. Since these vents are present near the ocean beds they have specific biodiversity in the vent ecosystem. The primary producers here are chemotrophic bacteria where the animals are usually tube worms, giant clams and a few crustaceans.

### 5.2.5 Ocean and Environment

Oceans are interconnected with the natural ecosystems and thus provide specific environmental services. Some of the key environmental services that we get from the oceans are listed below:

i. Oceans absorb and store large amounts of solar heat, thus acting as a buffer to reduce the impacts of fluctuations
ii. The ocean regulates climate by absorbing heat and carbon dioxide
iii. Ocean influence global weather patterns by affecting rainfall, temperature and storm activities
iv. Ocean supports an array of biodiversity
v. Oceans are a source of food for billions of people around the globe
vi. Ocean ecosystems like corals and mangroves provide coastal protection against storms and high tides
vii. Ocean currents continuously circulate the nutrients
viii. The major source of evaporation from the earth is the ocean
ix. Oceanographic data are used for climate modelling

## 5.3 LAKE AND POND ECOSYSTEM

### 5.3.1 Lake Ecosystem

Similar to the marine ecosystems the lake ecosystem has also been defined into a different zone depending upon their specific biological and physical characteristics. The lake water is divided into the following zones:

**i. Littoral Zone**

This zone is a present nearshore area of a lake where solar light infiltrates the bottom of the water. Therefore, typically rooted aquatic plants are present in this region. This region holds plants like water lilies and reeds. Where the animals include bass and bluegills. This zone holds water beetles and frogs.

**ii. Limnetic Zone**

This is an open, well-lit area of the lake, that is far apart from the lake shore. Overall, all the open water of the lake where solar light can penetrate is included in this zone. This zone is the most biodiversity-rich zone of the lake and holds phytoplankton, zooplankton, and a variety of fish including trout perch and invertebrates.

**iii. Profundal Zone**

The zone is present below the limnetic zone and is characterized by the darker part of the lake. The temperature here drops, and oxygen levels are also low. However, this zone is rich in organic matter thus it has decomposers, some invertebrates and fish adapted to dim light conditions.

**iv. Benthic Zone**

This area comprises the lake bottom, spreading from the shoreline to the deepest parts of the lake. The organisms here are termed benthic organisms include snails, insects and detritivorous.

**v. Components of Lake Ecosystem**

The lake ecosystem consisted of the following components

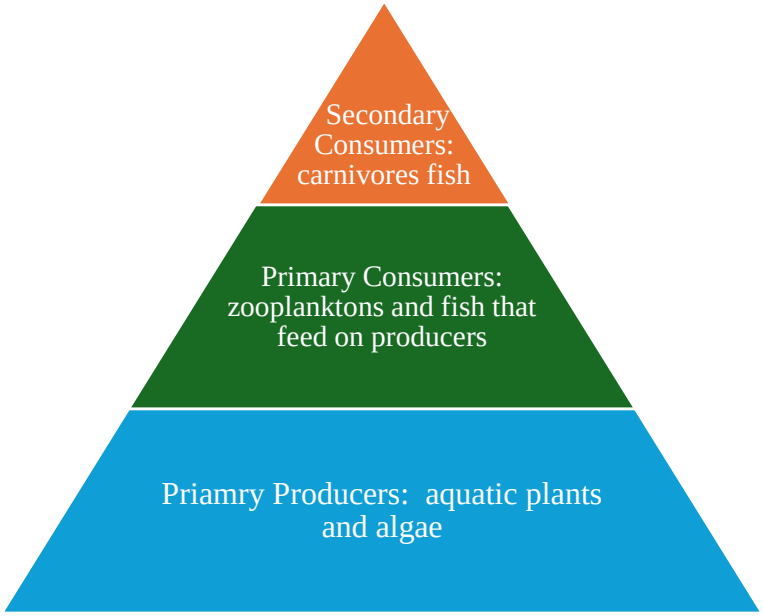


**Figure 5.3:** The distribution of organisms in the Lake ecosystem

### 5.3.2 Pond Ecosystem

Due to stagnant water, which is quite similar to the lake ecosystem the pond ecosystem shares many similar characteristics to the lake ecosystem. The unique characteristics of the pond ecosystem are presented below. The different zones of the pond ecosystem are presented in Figure 5.4.

**i.** **Shallow Depth**: Ponds are usually shallow enough for solar radiation to penetrate deep to the bottom of the pond. Thus, a pond ecosystem is ideal for the growth of aquatic plants throughout the pond.

**ii.** **Bounded Nature**: Unlike lakes, ponds are often quarantined, lacking a continuous flow of water in them. This can lead to varying levels of water quality and nutrient concentrations over the period.

**iii.** **Temperature Anomalies**: Ponds have more substantial temperature fluctuations compared to larger bodies of water due to their reduced size and shallowness.

iv. **Variable Oxygen Levels**: Oxygen levels in ponds can also fluctuate based on temperature, plant activity, and decomposition rates.

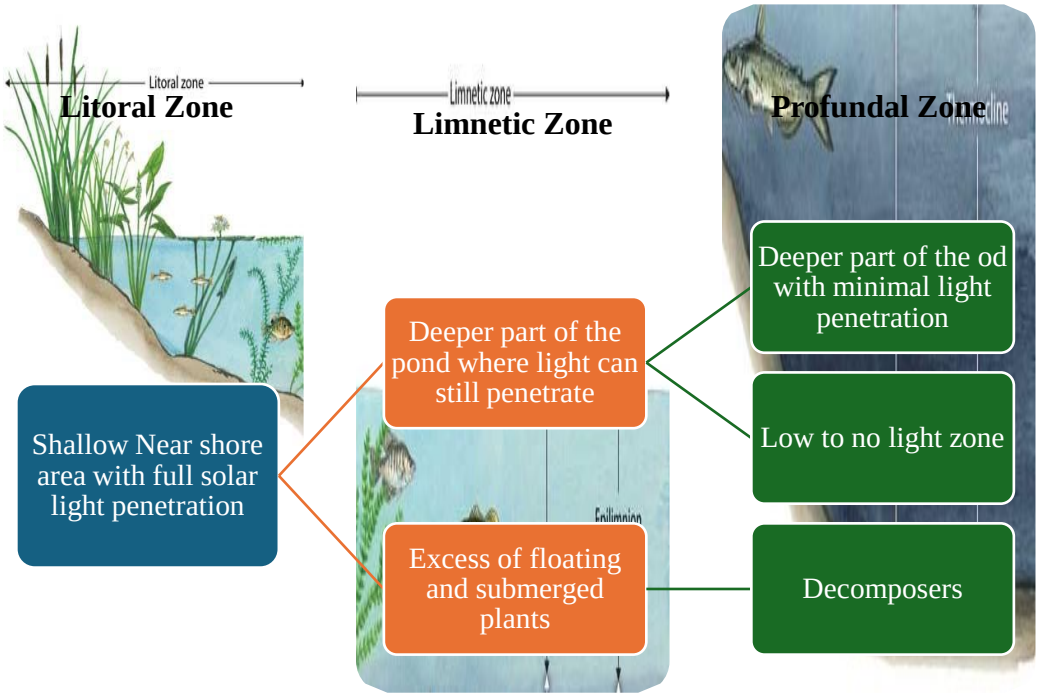


**Figure 5.4:** Zones of pond ecosystem

## 5.4 SPRING AND RIVER ECOSYSTEM

Springs and rivers are running fresh waters. Since these are in continuous movement therefore strong water currents shape their communities. The starting of the river and stream is usually termed as **headwaters**. These headwaters can be lakes, snow melts or even other springs. These water channels travel long distances from the headwaters to the mouth of the ocean. Since the water takes a long journey, therefore, the characteristics of rivers or streams change widely. This change in physical characteristics brought in the change in the biological diversity of the rivers and springs. For instance, at the headwaters the sea is usually cold, clear and oxygen-rich waters. But as they move downwards the temperature becomes moderate to normal supporting a large amount of biodiversity. To the mouth, the water tends to be more musky. Therefore, the organisms also change with the changing distance. In the start they are rich in phytoplankton and algae, freshwater fish such as trout can be found in these regions. Where the middle region is enriched with a variety of aquatic green plants and algae can be found. Where the mouth of

the river/stream usually carries organisms with low oxygen requirements that is catfish and crap. The organisms are modified to cope with the fast-moving water, for instance, they can have suckers or specified mouth parts to support them in strong currents, where they usually lay their larvae under the rocks. There are other freshwater ecosystems that are primarily wetlands, marshes and swamps. These are the transitional waters that allow the filtration of the excess nutrients before they enter the rivers or streams.

## 5.5 ENVIRONMENTAL CHALLENGES TO AQUATIC ECOSYSTEMS

Water is a non-renewable resource so are the water ecosystems. The world's freshwater resources only make up a minute part of the total available water. Therefore, there is continuous stress on the water resources. The water resources on the one hand are limited whereas on the other hand are under dire environmental stresses. Some of the key challenges that water ecosystems are facing today are given below:

**Climate Change:** Rising temperatures also raise the water temperature. The change in temperature and precipitation can change the physical characteristics of water. For instance, due to high temperature, there will be more ice melts resulting in more fresh water increasing the river flow and ultimately adding more to the marine water. Similarly, Ocean acidification due to high levels of carbon dioxide is also a serious threat to marine organisms, particularly coral reefs. Extreme temperatures can disrupt the breeding behaviour of many fishes.

**Pollution:** The present water resources are continuously being polluted by the bulk of nutrients from agricultural activities, heavy metals from industries, and persistent organic pollutants from pharmaceuticals, pesticides and other daily use items. One of the major challenges that currently lakes and ponds are facing is eutrophication. Another form of pollution that is paying havoc to aquatic ecosystems, particularly marine ecosystems is plastic pollution. It is estimated that tons of plastic are being added to oceans regularly, threatening the life inhabiting them.

**Overharvesting**: The ocean provides sufficient resources to be harvested but unsustainable practices may lead to depleted seafood. On the other hand, overconsumption limits their natural capacity to replenish and thus may endanger certain species. This can disturb the whole food web. On the other hand, apart from the biological resources humans are actively pumping freshwater resources from rivers, ponds and lakes, this practice limits water availability.

**Habitat Destruction:** Continues coastal development, dredging, construction of dams and other such activities can alter the habitat for various species the freshwater fishes may suffer from the low current that can alter their behaviour in the water reservoirs. Similarly, habitat destruction may destroy the conducive conditions for mangroves or corals.

**Invasive Species:** The foreign species introduced by human activities deliberately or accidentally can outcompete the natural species, or they can prey on the native species thus disrupting the whole ecosystem.

**Plastic Pollution:** One of the major threats to aqueous ecosystems is plastics. Plastic in practice has become an integral part of socie. This plastic particularly the single used plastics has become an environmental challenge worldwide. Most of the time the plastic ends up in water channels and ultimately ends up in the ocean. Where it can enter the bodies of aquatic animals. Many marine animals consider the plastic debris as food and thus eat them leading to death due to starvation of can get tangled to death.

> **Interesting fact:** *The ocean circulations known as north Pacific Gyre has circulating movement, this has resulted in the concentrated patch of plastic and its debris, known as great pacific garbage patch. This patch is so big that is stretches from the west coast of North America to the Japan.*

## 5.6 ADDRESSING CHALLENGES TO AQUATIC ECOSYSTEMS

To address the wide challenges posed to the aquatic ecosystem, well-coordinated efforts are required. The problem is global and therefore global efforts are required to resolve the issue. However, awareness regarding the extent and gravity of the problem can help in achieving the mitigation efforts. These efforts can be as follows but not necessarily limited to these:

**Pollution Control** ➔ Through law enforcement, technology innovation and awareness

**Climate Change Mitigation** ➔ Through Reduction in GHGs and ecosystem restoration activities

**Sustainable harvesting** ➔ Through well-monitored fishing, or can be diverted to aquaculture and fish farming

**Habitat Protection** ➔ Through marine protected areas

**Invasive Species Management** ➔ through early detection and monitoring

**Water Management** ➔ Through sustainable harvesting

**International Cooperation** ➔ Through global agreements on sustainable practices

## SUMMARY

Aqueous ecosystems (aquatic) ecosystems can broadly be classified into two main categories that are marine ecosystems and freshwater ecosystems. The marine ecosystem includes the oceans, seas and estuaries, where the freshwater ecosystems comprise vast extents of lakes, ponds, rivers and streams. Each waste body is unique in its characteristics and has diverse forms of life in it. Life in any ecosystem has been structured by its physical attributes and availability of nutrients. However, in ocean the light penetration and water pressure are the strong factors that affect the vertical distribution of the species. The waters of the world are under continuous pressure due to overconsumption, pollution and climate change. Coordinated efforts are required to tackle these challenges.

## SELF-ASSESSMENT QUESTIONS

1. What are aqueous ecosystems?
2. How is the ocean affected by the availability of light?
3. Discuss ocean types and biodiversity in them.
4. How freshwater ecosystems are different from the ocean ecosystems
5. What are the threats to aquatic ecosystems?
6. How can you play a role in maintaining aqueous ecosystems?
7. Draw and explain the water cycle.

**Activity**
Students can watch a video on the development of the great Pacific garbage patch.

**Unit–6**

# LAND ECOSYSTEMS

**Compiled by: Dr. Bibi Ilmas**

**Reviewed by: Dr. Muzammil Anjum**

# CONTENTS

***Page #***

Introduction ..... 81

Objectives ..... 81

6.1 Different Types of Land Ecosystems ..... 82

6.1.1 Tropical Rainforest Ecosystem ..... 83

6.1.2 Temperate Forest Ecosystem ..... 85

6.1.3 Shrubland Ecosystem ..... 85

6.1.4 Taiga or Boreal Ecosystem ..... 86

6.1.5 Tundra Ecosystem ..... 87

6.1.6 Desert Ecosystem ..... 88

6.1.7 Grassland Ecosystem ..... 89

6.2 Biogeographic Regions and Their Environmental Challenges ..... 90

Summary ..... 93

Self-Assessment Questions ..... 94

# INTRODUCTION

A community of organisms and their complex interaction with their physical environment is known as an 'Ecosystem. An ecosystem results from biotic (living organisms, including plants, animals, fungi, and microorganisms) and abiotic components (non-living elements such as soil, water, air, temperature, and sunlight). Both these components are inseparable but interrelated. These components are linked via nutrient cycles and energy flows.

Land ecosystems, or terrestrial ecosystems, are complex networks of organisms and their physical surroundings interacting on land surfaces. These ecosystems include diverse biomes, such as forests (including tropical, temperate, or boreal regions with dense tree cover and diverse species), grasslands (comprising savannas and prairies mainly covered by grasses with few trees), deserts (arid areas with scarce vegetation adapted to low water availability), and tundras (treeless, cold regions found in the Arctic and on high mountains). Land ecosystems support a wide range of species, contributing to global biodiversity. They also play a crucial role in climate regulation. For example, forests help mitigate climate change by sequestering carbon. Furthermore, these ecosystems provide essential resources such as food, medicine, and raw materials. Environmental problems vary across different biogeographic regions due to unique ecological, climatic, and human factors.

In this unit, you will learn about different land ecosystems and their environmental challenges.

# OBJECTIVES

Key objectives of the unit are:

1. To understand concepts of ecosystems, particularly land ecosystems
2. To understand the different forms/biomes of land ecosystems
3. To understand the Occurrence, species and environmental challenges of different biomes of the land ecosystems
4. To understand about different biogeographic regions of the world and their environmental challenges.

An ecosystem is a dynamic and intricate network comprising living organisms, including plants, animals, fungi, and microorganisms, that interact with non-living elements such as soil, water, air, temperature, and sunlight. This complex environment is interconnected through nutrient cycles and energy flows, as the biotic and abiotic components constantly influence and depend on each other for sustenance and survival.

Ecosystems vary in their size, some are small, while others are large enough to cover entire landscapes and regions. Terrestrial ecosystems, also known as land ecosystems, are intricate networks of organisms and their physical surroundings interacting on land surfaces.

## 6.1 DIFFERENT TYPES OF LAND ECOSYSTEMS

Different types of land ecosystems contain forests, deserts, grasslands, etc. Seven categories of land ecosystems are described below:

**Figure 6.1:** Types of Land Ecosystems

### 6.1.1 Tropical Rainforest Ecosystem

**(a) Occurrence**

Within the equatorial climate zone, the tropical rainforest biome is primarily located in a belt between 15° North and 15° South of the equator. The primary regions of the tropical rainforest ecosystem, which make up just 6% of the planet's surface, are found in the following parts of the world.

1. **Amazon**, the largest persisting rainforest, is mostly connected to Brazil but extends into portions of seven other countries.

2. **Central America,** comprises Belize, Costa Rica, Guatemala, Panama, Honduras, EI Salvador and Nicaragua

3. **Central Africa** is comprised of Congo, Central African Republic, Gabon, Cameroon, Guinea

4. **Indo-Malaysia** comprises Malaysia, Indonesia, and several other Southeast Asian nations.

**(b) Key Characteristics**

Latosols make up a large portion of the rainforest's soils. The soils are not very productive because of leaching and the quick uptake of nutrients by plants. The topsoil has the highest concentration of nutrients. The mean monthly temperature and year-round heat fall between 26 and 28°C. Because of the warm, damp climate, growth occurs all year round. Many different types of plants offer a variety of food sources and habitats. Rapid cycling of nutrients promotes plant development. The tropical rainforest's flora and fauna have evolved to thrive in the region's climate.

**(c) Structure of the Tropical Rainforest**

The tropical rainforest has five distinct layers.

1. Ground layer (0m)

2. Shrub layer (3-4m)

3. Under canopy (15m)

4. Canopy (30m)

5. Emergent (45-55m)

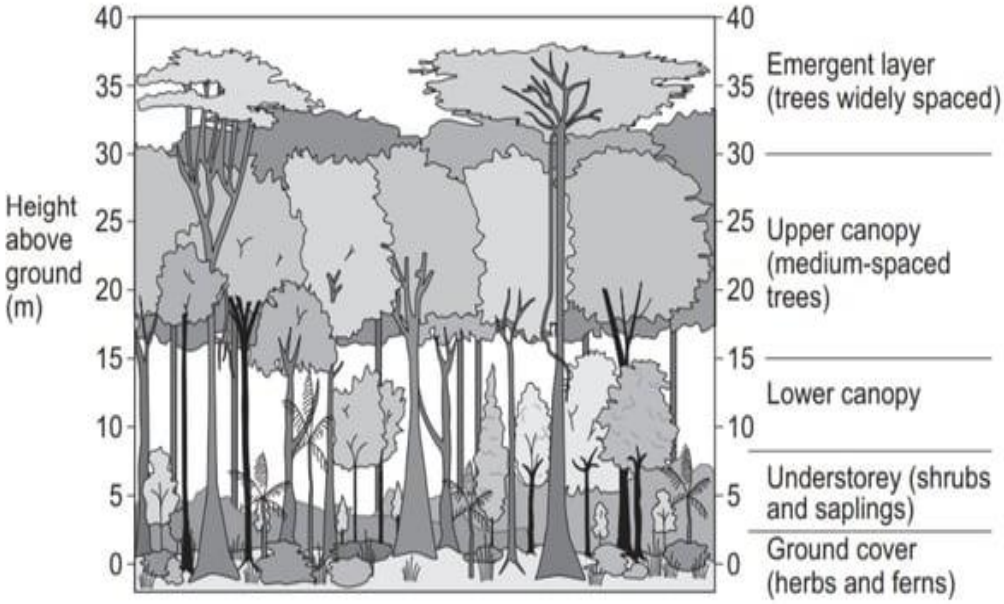


**Figure 6.2:** A typical rainforest ecosystem
(Source: https://www.savemyexams.com/gcse/geography)

**(d) Species**

On Earth, tropical rainforests are home to the highest diversity of flora and fauna, with an estimated 50% to 80% of all plant and animal species worldwide. Approximately 1,500 flowering plants, 750 tree species, 400 bird species, and 150 butterfly species are found there.

**(e) Challenges**

Human activity is the primary cause of the rising threats to the biodiversity of the rainforest, which include:

1. Farming, especially extensive slash and burn
2. Mining
3. Hydroelectric power
4. Logging
5. Development of roads and settlements
6. Wildfires (while they are naturally occurring, climate change is connected to their increased frequency and intensity.)

Because the components of the rainforest are interdependent, changes to one element also influence the others. A decrease in one species has the potential to cause a reduction in other species, particularly if it is a keystone species.

### 6.1.2 Temperate Forest Ecosystem

**(a) Occurrence**

Lush and vibrant, temperate forests are primarily located in the Northern Hemisphere and represent the second-largest terrestrial biome, encompassing a quarter of Earth's forested areas. These diverse ecosystems can be found in regions such as Eastern Asia, Central and Western Europe, and the Eastern United States.

**(b) Key Characteristics**

The characteristics of temperate woods include high amounts of precipitation, dampness, and a diversity of deciduous trees. In the fall, shorter days and dropping temperatures result in less photosynthesis from plants. The trees lose their leaves in autumn and grow new ones in spring as the weather warms and daylight hours increase. Typical summertime highs are around 86 degrees Fahrenheit, while wintertime lows can drop to -22 degrees Fahrenheit. Rainfall in temperate woods is usually between 20 to 60 inches per year and falls in the form of snow and rain.

**(c) Species**

Temperate woodlands have great biodiversity, including deciduous ones like oak, maple, beech, and hickory, and evergreen ones like pine, spruce, and hemlock. Fauna includes birds, reptiles, amphibians, mammals such as foxes, bears, deer, and raccoons, and a variety of reptiles like snakes and salamanders, temperate woodlands are also home to numerous birds including warblers, owls, and woodpeckers.

**(d) Structure**

The vegetation in temperate forests is divided into different layers.

1. Forest canopy, which is made up of birch, maple, and walnut trees.
2. Small tree tier: includes Shadbush, dogwoods, and redbuds species.
3. Shrub tier: includes huckleberries, azaleas, and mountain laurel species.
4. Herb tier: includes Indian cucumber, wild sarsaparilla, and blue bead lily species.
5. Ground level: Mosses and lichens.

**(e) Challenges**

Most of the world's population resides in the mid-latitudes, also home to temperate forests. These forests have been cleared for urban development i.e., New York, London, Paris, Tokyo, etc. as well as for agricultural purposes. Additionally, this biome has been harmed by pollution, hunting, logging, mining, deforestation, and habitat loss.

### 6.1.3 Shrubland Ecosystem

An ecosystem known as shrubland is distinguished by an abundance of shrubs and plants that resemble shrubs. Grass, bushes, and other herbaceous plants are among the other vegetation that can be found in shrubland environments.

**(a) Occurrence**

Shrublands are ecosystems found in coastal regions between 30° and 40° North and South latitudes such as Southern California, Chile, Mexico, the Mediterranean Sea region, and the southwestern regions of Africa and Australia. These areas are typically located adjacent to grasslands and deserts.

**(b) Key Characteristics**

The driest places are covered with shrubland; in even less rainy seasons, the individual bushes spread out until the region is better described as a desert. The shrubs belong to numerous species and families and range in height from 10 cm to 2 m.

**(c) Species**

Shrublands are found all across the world, but their biodiversity varies greatly from place to place. Shrublands of South Africa and Australia are among the world's most varied. Because the shrubland ecosystems in these countries have been there for a long time and have thus fully evolved their habitat, a wide range of species is present. 8,500 plant species have been found in South Africa's southwestern shrublands. Due to their adaptation and evolution to the limited nutrient availability in the local soil, many of these plants are indigenous to the area. Here, heathers, beans, daisies, orchids, and grasses are a few of the most prevalent plant species.

**(d) Challenges**

The main environmental risks that shrublands are currently facing include agriculture, cattle, other human activities, and biodiversity loss. Local human communities frequently use shrublands as pasture or grazing areas for their animals. The flora found here are destroyed by large livestock animals far more quickly than the environment can keep up. Furthermore, native species are frequently substituted with more lucrative crops like corn and wheat. The ecological damage brought about by these changes stops the shrubland from flourishing and expanding. In several instances, overhunting of native animal species in shrublands by humans has led to an imbalance in the natural food chain. These elements have caused irreversible harm to shrublands worldwide as well as a decline in biodiversity.

### 6.1.4 Taiga or Boreal Ecosystem

The word “taiga” is Russian for “land of little sticks”, and refers to the thick, icy forests that stretch over the high northern latitudes. The word "boreal" derives from the Greek word "Boreal," which means "north wind," reflecting the biome’s location in the Northern Hemisphere.

**(a) Occurrence**

The taiga biome encircles the Northern Hemisphere, covering these regions:

- North America: Alaska, Canada, and parts of the northern contiguous United States.
- Europe: Scandinavia (Norway, Sweden, Finland), and parts of Scotland.
- Asia: Russia (Siberia), northern Kazakhstan, Mongolia, and northern Japan.

**(b) Key Characteristics**

Winters(-50°C) in the taiga are harsh and long with 20 hours of darkness, whereas summers (10°C to 20°C) are brief and moderate with 20 hours of daylight. In taiga, the growing season lasts for 50 to 100 days. The life cycles of plants that inhabit the boreal forest are quickly completed. Because of the buildup of organic matter and the sluggish rates of decomposition, taiga soils are typically thin, nutrient-poor, and acidic. Usually, a layer of permafrost blankets the earth. About 30 to 85 centimetres (12 and 33 inches) of precipitation fall as snow in the taiga each year. Because evaporation is restricted by the frigid temperatures, the taiga is classified as a moist biome while having relatively little precipitation. Because of its high latitude, the daylight hours in the taiga vary greatly.

**(c) Species**

The taiga biome is characterized by an extremely cold environment and a limited variety of vegetation. The most common tree species in this biome are coniferous trees, which bear cones. The main types of conifers found in the taiga biome are spruce, evergreen, pine, and fir. Another common coniferous tree is the tamarack, which is a deciduous tree. Some broad-leafed trees, such as aspens and birches, have also adapted to survive in this harsh climate. The low temperatures restrict the number of animal species that can live year-round in the taiga biome. Fauna in this biome include bears, deer, and mice along with moles, squirrels, chipmunks, bobcats, and ermine.

**(d) Challenges**

More destruction has been done to this biome by modern people in the last 400 years than by hunters in the previous 1000 years. Certain creatures in the Taiga biome are in danger of going extinct due to widespread hunting and trapping.

### 6.1.5 Tundra Ecosystem

The word "tundra" is derived from the Finnish word "tunturi," which means "plain without trees."

**(a) Occurrence**

The tundra biome covers 10% of Earth's land, located between 60° and 70° North latitude in the Arctic, including Asia, Europe, and North America. Antarctic tundra can be found in the Antarctic Peninsula and a few sub-Antarctic islands. Alpine tundra exists above the tree line on high mountain tops worldwide, with the tree line in the tropics typically ranging from about 2,400 meters (8,000 feet) to 1,000 meters (3,300 feet) in higher latitudes.

**(b) Key Characteristics**

The absence of trees, a brief growth season, and bitterly low temperatures are the main features of the tundra. There are high winds, little precipitation, and low soil nutrients in this hostile environment. Despite these difficulties, a range of life forms that are specially adapted to exist in such conditions can be found on the tundra. Winters on the tundra are cold, but summers are short and warm enough for flowers and bugs. In

summers temperature ranges from -3°C and 12°C while in winters it ranges from -28°C and -34°C. In addition to being deficient in nutrients and frequently wet, tundra soils frequently have permafrost, which inhibits root development.

**(c) Species**

Mosses, lichens, grasses, sedges, and dwarf shrubs thrive in the tundra. Bearberry, Labrador tea, and Arctic moss are examples of common plants. There are many different kinds of birds, insects, and mammals among the tundra's fauna.

**(d) Challenges**

The tundra is seriously threatened by human activities including mining, infrastructural expansion, and the exploitation of oil and gas.

### 6.1.6 Desert Ecosystem

A community of living and non-living organisms in a severe, arid environment is called a desert ecosystem.

(a) **Occurrence**

Approximately one-fifth of the Earth's surface is covered by the desert biome. There are four different types of deserts: hot and dry, semi-arid, coastal, and cold. Deserts vary greatly in temperature and meteorological conditions. Every continent has desert in any form as detailed below.

| Sr. No. | Continent | Desert |
|---|---|---|
| 1. | North America | Mojave (Hot) desert |
| 2. | Asia | Gobi (Cold) desert |
| 3. | Africa | Namib (Coastal) desert |
| 4. | Australia | Gibson Desert (Semi-Arid) desert |
| 5. | Antarctica | Antarctica (Cold) desert |
| 6. | Europe | Tabernas (Semi-Arid) desert |
| 7. | South America | Atacama (Cold) desert |

**(b) Key Characteristics**

In different types of deserts, the soil layer can be stony, gravelly, or sandy. Deserts typically receive up to 50 centimetres (20 inches) of rainfall annually, and the plants and creatures that live there have adapted to this extremely dry environment by storing water.

**(c) Species**

Deserts are home to a wide variety of plant and animal species that have evolved to thrive in extreme temperatures and arid conditions. Within these harsh environments, xerophytes, such as squirrels, oryx, rats, and numerous other flora and fauna, have adapted to conserve water and endure the challenging climate.

**(d) Challenges**

The introduction of invasive species, over-cultivation, improperly drained irrigation systems, inadequate water management, and fossil fuel excavation are only a few of the environmental issues that humans have caused in desert biomes.

### 6.1.7 Grassland Ecosystem

The term "grassland ecosystem" refers to the terrestrial environment where grasses and herbaceous plants predominate.

**(a) Occurrence**

Grasslands are home to the largest biomes on Earth, making up almost 10% of the planet's surface. The climates of grasslands differ, but in general, all major areas of naturally occurring grasslands are hot and dry—though not as much as deserts. This is especially true in the summer. Savannas and temperate grasslands are two types of grasslands.

**(b) Key Characteristics**

The preservation of grasslands is maintained through animal grazing, wildland fires, and limited rainfall. The climate in grassland areas is conducive to the growth of grass, as the small amounts of precipitation are enough to support grasses but not forests of trees. Additionally, grassland ecosystems rely on regular fires, which grasses are well-suited to recovering from.

**(c) Species**

A grassland ecosystem consists of small herbs, weeds, grass, trefoil, shrubs, dicotyledonous plants, and other leguminous species. This habitat provides a rich variety of plant and animal species, serving as a vital source of sustenance for a diverse array of domesticated and wild herbivores such as mules, deer, goats, donkeys, sheep, zebras, and cattle. Economically, the grassland ecosystem is important as these creatures provide transportation, milk, wool, and food.

**(d) Challenges**

Due to changes in land usage, overgrazing, and hunting, humans have had a significant impact on the grassland biome. Elephants are among the many huge creatures that have disappeared from the African savannas as a result of illicit poaching. By trampling trees and shrubs, the elephants defend the savanna's grasslands. Savannas can easily transition into forests when there are no large animals around to trample down the trees and instead overpower the grasses. Less food would be available for grazing animals as a result of the grass dying off.

## 6.2 BIOGEOGRAPHIC REGIONS AND THEIR ENVIRONMENTAL CHALLENGES

The goal of bioregionalism is to raise awareness of natural resources and cascading effects among the people living in the bioregion. This will encourage them to take a more positive view of the environment and cherish their local natural capital. A biogeographic region, then, is a distribution area for plants and animals that has consistent or shared features across it. A broadly accepted sense of regional identity is recreated by the bioregional perspective, which is based on a revitalized critical understanding and respect for the integrity of our biological communities.

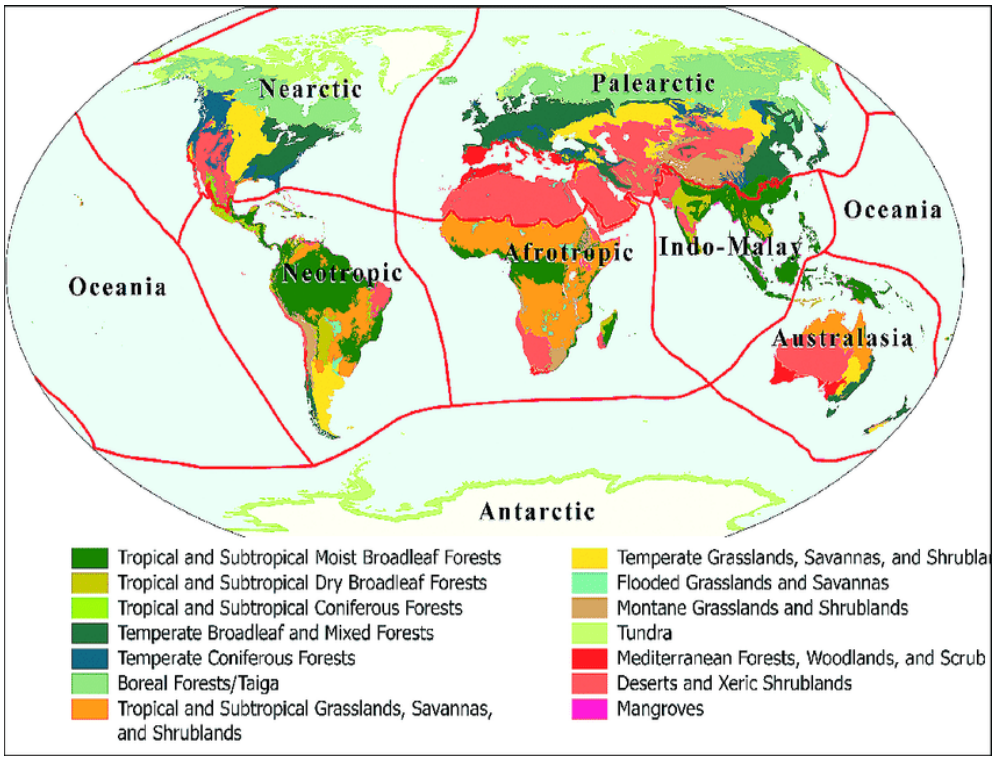


**Figure 6.3:** Biogeographic regions of the world
(Source: https://www.researchgate.net/profile/Eric-Wikramanayake/publication/216340317)

Due to distinct ecological, climatic, and human causes, environmental issues differ between biogeographic regions. Here are some of the most important problems in different biogeographic areas. Although the challenges faced by each biogeographic region are distinct, many interconnected issues need global cooperation to successfully address.

Eight (08) Biogeographic regions of the world along their environmental challenges are described below.

**Table 6.1: Biogeographic regions of the world and their Environmental Challenges**

| S.no | Biogeographic regions | Geographic presence | Environmental challenges of the region |
|---|---|---|---|
| **1.** | **Neoretic bioregion** | This region includes Greenland, the majority of North America, and the hilly parts of Mexico. | 1. Urbanization, agriculture, and deforestation result in habitat loss.<br>2. Industrial activities lead to air and water pollution. |
| **2.** | **Paleartic bioregion** | It is the largest bioregion of the earth that encompasses Asia, Europe, North Africa, the Northern Himalayas, and the Arabian Peninsula. | 1. Industrial pollutants cause adverse air quality.<br>2. Climate Change has altered the ecosystems and endangers species. |
| **3.** | **Afrotropical bioregion** | This region includes southern Africa, the Arabian Peninsula (southern regions), the Sahara (southern and eastern regions), Iran, southwest of Pakistan, and the Indian Ocean(southwest). | 1. Desertification is caused by Overgrazing and poor land management turning fertile land into desert.<br>2. Illegal hunting and timer logging have threatened the biodiversity |
| **4.** | **Antarctic bioregion** | Antarctica, on the other hand, is a mountainous continent near the South Pole, covered in a thin layer of ice, and has an average temperature of -57 degrees Celsius. Climate change is also mentioned as a potential implication for these regions. | Climate change has led to the melting of ice caps and glaciers, which poses a threat to polar species and global sea levels. Additionally, persistent organic pollutants and microplastics accumulate in polar environments, leading to pollution. |
| **5.** | **Australian Bioregion** | The Australian bioregion comprises Australia, the island of New Guinea, and the Indonesian Archipelago (eastern region). It includes the Maluku Islands, Timor, | 1. Non-native species or invasive species disrupt the local ecosystems.<br>2. Increased frequency and intensity of bushfires/forest fires |

| | | | |
|---|---|---|---|
| | | Sumbawa, Lombok, Sulawesi and Sumba. Additionally, several groups of Pacific islands are part of this bioregion. | due to climate change is the key challenge. |
| **6.** | **Bioregion Indomalaya** | It includes Southeast Asia, Afghanistan, Pakistan, the Indian subcontinent, southern China, Indonesia, Borneo, Java, Bali, the Philippines, Taiwan, and the Japanese Ryukyu Islands. The region was originally referred to as the eastern region and is divided from the Australian bioregion by Wallace's line. | Logging and agricultural expansion have led to severe deforestation, while industrial and agricultural runoff has caused water pollution in the region's rivers and lakes. |
| **7.** | **Oceanic bioregion** | It is the smallest and youngest ecosystem on Earth and does not include any continental land mass. This bioregion encompasses most of Polynesia (except New Zealand), the Pacific Ocean, the Micronesia Islands, and the Fiji Islands. | • Climate change leads to higher sea levels, which threaten coastal communities. Warmer ocean temperatures cause coral bleaching, leading to reef destruction. |
| **8.** | **Neotropical bioregion** | It is formed by southern America, Central America, Mexico, South Florida, and the Caribbean Islands | • Logging and agriculture, especially in the Amazon, result in significant habitat loss. Species go extinct due to habitat destruction and climate change. |

## SUMMARY

1. Land ecosystems, known as terrestrial ecosystems, are intricate systems of organisms and their physical surroundings on land. They include diverse biomes like forests, grasslands, deserts, and tundras.

2. These ecosystems support biodiversity and play a crucial role in climate regulation, while also providing essential resources.

3. Environmental challenges vary across regions due to unique ecological, climatic, and human factors.

## SELF-ASSESSMENT QUESTIONS

1. Discuss different types of land ecosystems based on their occurrence, key characteristics, and environmental challenges.

2. "Explore the impact of environmental conditions on species' adaptation for survival. Back your answer with three compelling examples."

3. Explain the different bioregions of the world.

4. What are the environmental challenges of the main bioregions of the world?

5. Give brief answers:
   i. Draw the canopy structure of the tropical rainforest.

   ii. Define ecosystems.

   iii. Deserts can be found in any form on every continent. Provide the names of the deserts located on each continent.

   iv. What is the biogeographic region where Pakistan is located?

   v. Name the mountainous biogeographic region near the south pole characterized by coverage of a thin ice layer.

**Unit–7**

# INTRODUCTION TO ENVIRONMENTAL CHEMISTRY

**Compiled by: Dr. Bibi Ilmas**

**Reviewed by: Dr. Azeem Khalid**

# CONTENTS

***Page #***

Introduction ..... 97

Objectives ..... 97

7.1 Chemicals in the Environment ..... 98

7.1.1 Pollutants/Chemicals in Air ..... 98

7.1.2 Contaminants/Chemicals in Water ..... 99

7.1.3 Contaminants/Chemicals in Soil ..... 100

7.2 Chemical-Associated Issues in the Environment ..... 101

7.2.1 Biological Toxicology of Pollutants or Chemicals ..... 101

7.2.2 Environmental and Health Impacts ..... 102

7.3 Green Chemicals/Chemistry ..... 103

7.3.1 Principles of Green Chemistry ..... 103

7.3.2 Green Chemicals ..... 104

Summary ..... 105

Self-Assessment Questions ..... 106

# INTRODUCTION

Environmental chemistry involves studying chemical processes occurring in the environment, which are influenced by both human activities and natural phenomena. It covers the chemical processes in air, water, soil, and living organisms, examining how substances move, react, and impact these environments. Key matters in environmental chemistry include atmospheric chemistry (the study of chemical reactions in the Earth's atmosphere, including pollution and climate change), aquatic chemistry (examination of chemical processes in water bodies such as rivers, lakes, and oceans), soil chemistry (analysis of chemical interactions in soils, including nutrient cycles and contamination), and pollutants and contaminants (understanding sources, reactions, and impacts of pollutants like heavy metals, pesticides, and industrial chemicals).

Environmental chemists use various analytical techniques to monitor and mitigate the effects of pollutants, aiming to protect and improve environmental health.

This unit will help you understand the basics of the environmental chemistry of air, water, and soil, and the chemicals present in them as pollutants.

# OBJECTIVES

The main objectives of the unit are:

1. To understand the concept of 'Environmental Chemistry'

2. To understand the chemicals, present in the environment (air, water, and soil)

3. To recognize the impacts of chemicals on human and environmental health

4. To comprehend green chemicals/ chemistry or the principles of green chemistry

## 7.1 CHEMICALS IN THE ENVIRONMENT

Air, water, and soil are fundamental components of our environment. They interact in various ways to support life on Earth. Any unwanted material that causes any sort of harmful impact on the natural environment and human health is a pollutant. Chemical substances/pollutants in the environment can be solids, liquids, dust, vapours, gases, mists, and fumes.

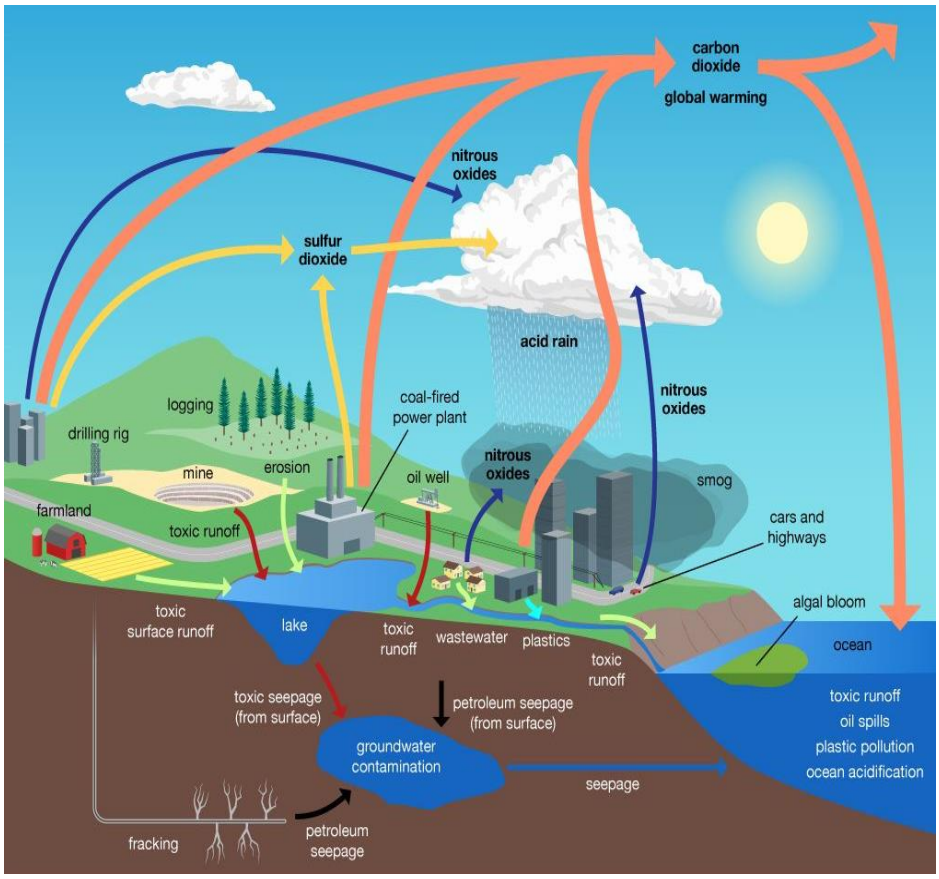


**Figure 7.1:** Pollution in the environment (Source: https://cdn.britannica.com)

### 7.1.1 Pollutants/Chemicals in Air

"Air pollutants that cause air pollution" are substances or airborne particles that have the potential to negatively impact the health of people, animals, and plants. Each year, approximately 7 million premature deaths are attributed to air pollution, making it the most significant environmental hazard to public health on a global scale.

These contaminants include;

**1. Nitrogen Oxides (NOx)**

NOx gases come from vehicle emissions, power plants, and industrial processes. $N_2O$ can lead to the formation of ground-level ozone and smog, which are responsible for causing respiratory problems such as asthma and other health issues.

**2. Sulfur Oxides (SOx)**

SOx gases are emitted from burning fossil fuels and industrial processes, leading to acid rain that can harm health, buildings, and ecosystems.

**3. Particulate Matter (PM)**

Particulate matter (PM) comes from sources like construction sites, unpaved roads, agricultural fields, industrial smokestacks, and fires. When inhaled, these particles can cause severe health problems, especially for the heart and lungs. $PM_{2.5}$, made up of particles smaller than 2.5 micrometres, is especially concerning due to its ability to deeply penetrate the lungs and cause significant health issues.

**4. Methane ($CH_4$)**

Methane, a significant greenhouse gas, is emitted from various sources including the waste sector, as well as during the production and transportation of coal, oil, and natural gas, from livestock and other agricultural practices. Methane is significantly contributing to global warming and climate change due to greater global warming potential.

**5. Water Vapors**

Water vapour is naturally present in the atmosphere as part of the water cycle. Human activities such as combustion and irrigation can lead to higher levels of water vapour. Water vapour is a significant greenhouse gas that amplifies the effects of other greenhouse gases, even though it is not hazardous in itself.

### 7.1.2 Contaminants/Chemicals in Water

Numerous chemicals have the potential to contaminate water, which can be extremely dangerous to human health. These are a few typical pollutants that are present in water.

**1. Biological pollution**

Harmful microorganisms like bacteria, viruses, and parasites that contaminate water are referred to as biological pollution. These can result in illnesses like giardiasis, dysentery, and cholera. Animal waste, agricultural runoff, and sewage discharge are typical sources.

**2. Organic chemicals**

Pesticides, herbicides, and industrial solvents are among the many organic chemicals found in water. Agricultural runoff, industrial discharges, and improper household chemical disposal can all lead to the introduction of these chemicals into water bodies. They can be toxic to both aquatic life and humans, potentially leading to cancer, reproductive issues, and other health problems.

**3. Inorganic chemicals**

Heavy metals (lead, mercury, and arsenic), nitrates, and phosphates are examples of inorganic chemical compounds. These can be caused by industrial processes, mining operations, or agricultural runoff. Inorganic chemicals can be highly toxic, leading to various health issues nephrological, neurological, and developmental problems in children.

### 7.1.3 Contaminants/Chemicals in Soil

Soil contamination/land pollution can be a serious issue, and it frequently involves both organic and inorganic chemicals.

Organic chemicals are carbon-based compounds that may include:

1. Pesticides are used to control pests in agriculture, but they can persist in the soil and harm non-target organisms.

2. Organic chemicals are carbon-based compounds that could include:

3. Petroleum hydrocarbons from oil spills or leaks can be toxic to plants and animals.

4. Polychlorinated biphenyls/PCBs are the industrial sourced persistent chemicals that are harmful to human health.

Inorganic chemicals: These commonly include:

1. Heavy metals such as Arsenic, Lead, Mercury, and Cadmium are detrimental to plants, animals, and humans.

2. Excessive fertilizer use can cause salt accumulation, reducing soil structure and fertility.

3. Nitrates and phosphates are commonly found in agricultural runoff and can cause nutrient imbalances and water pollution.

4. Acids from Industrial activities can degrade soil pH, affecting plant growth and soil microorganisms.

## 7.2 CHEMICAL-ASSOCIATED ISSUES IN THE ENVIRONMENT

Anthropogenic and naturally occurring chemicals have negative impacts on hurting living organisms and ecosystem processes.

**Table 7.1: Major environmental pollutants and impact on human health**

| Pollutants | Types | Impacts on human health |
|---|---|---|
| **Heavy metals** | Lead | • Kidney, brain, liver damage<br>• Neurobehavioral changes<br>• Abnormalities in fertility and pregnancy |
| | Mercury | • Hypertension<br>• Myocardial infarction<br>• Proteinuria<br>• Cardiovascular diseases |
| | Arsenic | • Respiratory Cancer<br>• Dermatomes<br>• Genetic toxicity |
| **Particulate matters** | $PM_{2.5}$, $PM_{10}$ | • Chronic Pulmonary disease<br>• Bronchitis<br>• Cardiovascular illness<br>• Stroke<br>• Change in blood pressure |
| **Pesticides** | Organochlorine compound (Dichloro-diphenyl-trichloroethane, DT, Dichlorodiphenyldichloroethane, Dicofol, Eldrin, Dieldrin) | • Damage human liver, kidney, neural and immune systems<br>• Birth defects<br>• Cancer<br>• Neurotoxicity<br>• Reproductive toxicity<br>Inflammation of the upper respiratory tract and bronchitis |
| **Plastics** | Polyvinyl chloride | Respiratory damage<br>immune system damage |
| **Plastic-Additives** | Polycyclic aromatic hydrocarbons (PAHs) | Developmental and reproductive toxicity |
| | Phthalates | Endocrine disruptor Interference with testosterone, sperm motility, testicular cancer |
| | Polychlorinated biphenyls (PCBs) | Thyroid hormone disruption |

### 7.2.1 Biological Toxicology of Pollutants or Chemicals

The most hazardous environmental properties of chemicals include persistence, bioaccumulation in organisms, human toxicity, and ecotoxicity to living organisms. Chemicals enter the body to cause health issues, and once inside, some chemicals can travel through the bloodstream to the target organs (i.e., kidneys, lungs, liver, nervous system, etc.)

There are three primary routes of exposure;

1. Inhalation
2. Ingestion
3. Direct contact (i.e., skin/eye)

After entering the body and target organ, factors such as potency, persistence, solubility, bioaccumulation, and bio-magnification determine the level of harm or toxicity being caused.

### 7.2.2 Environmental and Health Impacts

Exposure to pollutants poses a significant threat to human and environmental health. The environmental implications of chemicals and pollutants are complicated and encompass various concerns:

1. Surface and groundwater degradation of the quality and quantity of the water, deterioration of the aquatic biota, and enrichment of the organic nutrients; all these factors contribute to overall water degradation and pose great impacts on the overall environment.

2. Soil/land pollution may involve the destruction of soil structure, salinity, soil erosion, soil acidification, leaching of nutrients, deterioration of the soil biota, etc.

3. The presence and exposure of synthetic chemicals, heavy metals, herbicides, CFCs, and pesticides across any environmental compartments have the potential to cause ecosystem disturbances such as biodiversity loss, ozone depletion, bioaccumulation of hazardous materials, alterations of biogeochemical cycles and human health effects like various respiratory, reproductive, nervous, cardiac and immune system diseases.

4. Biogeochemical cycle changes can lead to further changes in the water cycle, acid rain and acid deposition, particulate pollution, and outbreaks of pests and pathogens

5. Climate change is also posing great impacts due to the rise in GHG emissions ($CO_2$, $CH_4$, $N_2O$, etc.) concentrations, alteration of airflow patterns, and precipitation, increased global temperature, compromised human health, and shifts within global ecosystems.

## 7.3 GREEN CHEMICALS/ CHEMISTRY

Green chemistry or sustainable chemistry aims to design chemical products or processes that can reduce/alternate the usage and production of hazardous chemicals or substances by deploying innovative techniques while lessening negative impacts on human health and the planet.

### 7.3.1 Principles of Green Chemistry

Twelve (12) principles of green chemistry are intended to make chemical processes more sustainable and environmentally friendly.

**Figure 7.2:** Principles of green chemistry

### 7.3.2 Green Chemicals

Green chemicals are chemicals that have a low environmental impact and are frequently derived from renewable resources. Here are a few examples.

1. Supercritical Carbon Dioxide ($scCO_2$) is often used as a green solvent in various industrial processes, such as in the decaffeination of coffee and dry cleaning.

2. Bio-based Polymers like polylactic acid (PLA) are obtained from renewable resources like maize starch and are biodegradable.

3. Ionic Liquids are salts in a liquid state that can be used as solvents and catalysts, providing a safer alternative to common organic solvents.

4. Hydrogen Peroxide is used for bleaching and purification, separating into water and oxygen while emitting only water and oxygen, making it safe for the environment.

5. Limonene is a solvent derived from citrus fruits, used in cleaning products and as a replacement for more hazardous solvents.

These chemicals exemplify green chemistry principles by reducing toxicity, increasing efficiency, and utilizing renewable resources.

## SUMMARY

1. Human survival relies on having clean air, water, and food, as well as ensuring the well-being of plants and animals.

2. However, both man-made and naturally occurring chemicals can harm living organisms and ecological processes.

3. Green chemicals are specifically created to have minimal impact on the environment. These chemicals follow the principles of green chemistry by reducing toxicity, enhancing efficiency, and utilizing renewable resources.

## SELF-ASSESSMENT QUESTIONS

1. What are the major pollutants of the air and their impacts on the environment and health?

2. What are the primary water pollutants and their impacts on the environment and health?

3. What are the main pollutants found in soil and how do they impact the environment and human health?

4. What is green chemistry? Discuss its key principles.

5. Give brief answers:
   i. Define environmental chemistry.
   ii. Define green chemistry.
   iii. Differentiate between pollutants and contaminants.
   iv. What is pollution?
   v. List 5 major air pollutants.

6. What is the importance of green chemistry?

**Unit–8**

# INTRODUCTION TO ENVIRONMENTAL PHYSICS

**Compiled by: Dr. Sapna**

**Reviewed by: Dr. Rohama Gill**

# CONTENTS


***Page #***

Introduction ..... 109

Objectives ..... 109

8.1 What is Environmental Physics? ..... 110

8.2 Physics in the Atmosphere ..... 110

8.2.1 The Atmosphere ..... 110

8.2.2 Sunlight and the Earth ..... 110

8.2.3 Atmospheric Processes ..... 111

8.3 Physics in the Hydrosphere ..... 111

8.3.1 Water in Motion ..... 111

8.3.2 Dynamics of Fluid ..... 111

8.3.3 Water Cycle ..... 112

8.3.4 Wave Motion ..... 113

8.3.5 Tides and Currents ..... 113

8.4 Physics in the Biosphere ..... 113

8.4.1 Flow of Energy in the Ecosystem ..... 113

8.4.2 Application of Thermodynamic Laws in Ecosystem ..... 113

8.4.3 Physical Adaptations: Thermoregulation ..... 114

8.4.4 Environmental Changes ..... 114

8.5 Physics in the Geosphere ..... 114

8.5.1 The Earth's Structure ..... 115

8.5.2 Energy Inside Earth ..... 115

8.5.3 Rocks and Minerals ..... 115

Summary ..... 116

Self-Assessment Questions ..... 116

## INTRODUCTION

Physics is everywhere and has its role in the environment as well. This unit is designed to highlight the various ways physics plays a role in environmental processes, both in our daily lives and in natural phenomena. Environmental Physics explains the relationship between land, atmosphere, oceans, soil, and vegetation. In this unit, Physics' role in the atmosphere, hydrosphere, biosphere, and geosphere will be discussed in detail.

## OBJECTIVES

After reading this unit, you will be able to:

1. Understand the basics of Environmental Physics
2. Know the important role of Physics in the atmosphere
3. Discover the influence of Physics in the hydrosphere
4. Discuss the Physics in the biosphere
5. Illuminate the geophysical forces shaping our world

## 8.1 WHAT IS ENVIRONMENTAL PHYSICS?

The environment can be defined as a medium in which any entity finds itself. For example, for clouds their environment will be their atmosphere, for the whales it will be the sea in which they live and for plants it will be a field in which they grow. Therefore, it is informative to discuss things in the context of the surroundings in which they live.

Environmental Physics is an interdisciplinary subject that applies the principles and concepts of Physics to understand and solve environmental problems. It combines the physical processes in the atmosphere, hydrosphere, biosphere, and geosphere. It explains solar radiation and climate, wind energy, water cycle and hydrology, ocean currents and thermal dynamics, atmospheric physics and weather forecasting, greenhouse gas emissions, noise pollution, soil physics, and remote sensing in environmental monitoring.

## 8.2 PHYSICS IN THE ATMOSPHERE

### 8.2.1 The Atmosphere

Earth's surface is surrounded by the atmosphere and its gravity holds it in its place. It helps to sustain life on Earth by providing important gases, protecting life from harmful solar radiation, trapping heat to warm the planet, and helping regulate the temperature. Layers of the atmosphere are five including troposphere, mesosphere, thermosphere, stratosphere, and exosphere. The temperature of each layer is different.

### 8.2.2 Sunlight and the Earth

Solar radiation is the main source of energy for most processes on Earth and it drives the energy budget of Earth. The solar radiation reaches the top of the Earth's atmosphere by travelling at the speed of light. These radiations are electromagnetic radiations which have visible light which we can see and invisible infrared light which we cannot see. It consists of a small portion of ultraviolet light as well which we cannot see. Some light reaches as solar wind which consists of gamma rays and x-rays which are also invisible.

Sun's energy is very important for life on Earth. It is responsible for natural processes such as water and air that create the weather, animals' vision, and photosynthesis in plants. Plants need sunlight to grow and if there was no sunlight, there would be no plants and therefore no oxygen for our breathing. Infrared radiation keeps the Earth warm and without infrared radiation, the Earth would freeze. There would be no air, ocean currents, or clouds to transport water. Solar radiation is used by humans for solar energy to heat their homes and enjoy electricity.

### 8.2.3 Atmospheric Processes

**Weather and Climate**

The weather is the current condition of the atmosphere starting from a few days to a few months. Elements of weather are wind, temperature, humidity, precipitation, atmospheric pressure, cloud type and cover, and sunshine. At least 30 years of average weather conditions define the climate of any region. Uneven heating of the Earth's surface causes differences in weather and climate of the regions. Different locations have different weather and climate conditions.

**Air in Motion: Wind**

The wind is air in motion. The movement of air molecules is from high-pressure locations to low-pressure locations. On a theoretical basis, cold dense air from polar areas should travel towards the equator but the spinning of Earth exerts force on these air currents, therefore deflecting them to the left side in the northern hemisphere and the north side in the southern hemisphere. This is known as the Coriolis effect. Wind speed is affected by some other factors as well, which are the unequal Earth's surface heating and the availability of mountains. An anemometer is utilized for the measurement of wind speed and a wind vane or wind sock for the wind direction.

**Ozone Production from Ultraviolet (UV)**

Ozone is produced naturally in our environment every day. When we hear this word, we imagine the ozone layer in the atmosphere. This ozone and the natural ozone (the one produced on the ground) are both produced from UV light. On the electromagnetic spectrum, this light has higher frequencies than visible light. Light ranging from 160nm to 140nm will play its role in the formation of ozone from oxygen. It is created by the process of photolysis of oxygen molecules. This disrupts the molecule and creates valent oxygen atoms that later attach with any individual oxygen molecule to form ozone.

## 8.3 PHYSICS IN THE HYDROSPHERE

### 8.3.1 Water in Motion

Water in motion is the study of how water moves through the hydrosphere including rivers, lakes, oceans, and underground aquifers. This movement of water is driven by several physical forces and processes which are very important for the understanding of environmental processes such as the water cycle, erosion, and weather patterns.

### 8.3.2 Dynamics of Fluid

The dynamics of fluid deals with how the fluids move. In the water context, it includes how water moves through rivers, streams, and oceans. The water flow is studied by using the continuity (the conservation of mass) equation, Bernoulli's principle, and the Navier-Stokes equations.

### 8.3.3 Water Cycle

The water cycle describes the movement of water. In this process, water changes only its state from one form to another, but the total number of particles remains the same. Firstly, the water evaporates from the Earth's surface, then the evaporated water rises into the atmosphere, and further, the rising water condenses or cools forming clouds leading to rain or snow.

Evaporation, transpiration and sublimation are the included processes. The Sun heats water at the surface of Earth. The **evaporated** water then turns from liquid to gas, and transforms into water vapor, rising in the atmosphere. Also, **transpiration** processes occur in which water vapours are released into the atmosphere by plants and trees. The state of water normally changes in a single stage to either solid, liquid, or gas but sometimes skipping a stage may happen and the water converts directly to water vapour by **sublimation**. All water vapours will then be forced to **condense** from a gas to liquid water droplets, often forming clouds. Within clouds, the liquid water droplets accumulate over time and make large water droplets which cannot stay because of heaviness and fall from the clouds. This leads to precipitation in the form of either snow or sleet, rain, and hail. When it reaches the Earth's surface, it often flows into rivers, streams, and further into the oceans. It may soak into the soil which will still move into the rivers but will take a long time to do so. Further, it may also soak into the plants' roots to help them grow. Figure 8.1 shows the water cycle.

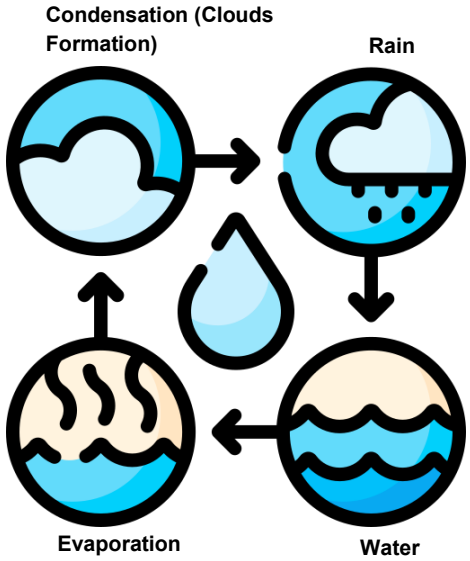


**Figure 8.1:** Water cycle showing the movement of water

### 8.3.4 Wave Motion

Wind blowing across the surface of water forms ocean waves. Their size and energy depend on the wind speed, the duration for which it is blowing, and the distance over which the wind blows. Wavelength, frequency, amplitude, and speed are important parameters for wave motion, and these are significant in the comprehension of coastal erosion, the effect of waves on marine structures, and navigation.

### 8.3.5 Tides and Currents

The gravitational forces exerted by the Sun and moon on Earth cause periodic rise and fall of sea levels which are known as tides. Tides have significant effects on the shipping and fishing activities of humans. Larger movements of water driven by the moon and the sun's gravitational pull, distinctions in water density due to salinity and temperature, the Coriolis effect, and wind are known as ocean currents. These currents play an important role in controlling the climate of Earth by redistributing heat from the equator to the polar regions and therefore affect weather conditions and marine life distribution.

## 8.4 PHYSICS IN THE BIOSPHERE

The biosphere is one of the five major environmental spheres. It is a global ecosystem that is comprised of living (biota) and non-living (abiotic) organisms which provide them with nutrients and energy. Physics has an important role in energy flow through ecosystems, organisms' adaptation to the environment, and changes in the environment's impact on living organisms.

### 8.4.1 Flow of Energy in the Ecosystem

The primary source of energy for the survival of such a large number of organisms is solar energy. Less than 50 percent of solar radiation has been received by our Earth which is an effective radiation, means that can be utilized by plants for photosynthesis.

Food chains and food webs are responsible for the flow of energy. During the flow of energy, the plants are producers which absorb energy from sunlight and some of this energy is transformed into chemical energy during photosynthesis which is further consumed by primary, secondary and tertiary consumers.

### 8.4.2 Application of Thermodynamic Laws in Ecosystem

The thermodynamics law in ecosystems describes the flow of energy at every trophic level. According to the first law, there is no creation or destruction of energy but only transformation happens from one form to another. In the case of energy flow in the ecosystem, this law holds. In the case of the second law, at each step of

the energy flow, the energy is lost. In ecology, this law holds as a decrease in energy at every trophic level has been observed.

### 8.4.3 Physical Adaptations: Thermoregulation

Various organisms regulate the temperature of their bodies and have developed physical adaptations. For example, warm-blooded animals maintain a continuous internal temperature through the production of metabolic heat, while cold-blooded animals depend on external heat sources. The physics of heat transfer such as convection, conduction, radiation, and evaporation help organisms in managing the heat gain and loss for survival in distinct environments.

### 8.4.4 Environmental Changes

**Climate Change Impacts**

Climate change is due to a rise in the concentration of greenhouse gas emissions which change the temperature and precipitation patterns, sea-level rise, and the persistent happening of extreme weather events. Physics helps in modelling these changes and predicting their effects on the distribution of species and ecosystems. Temperature variations can lead to changes in migration patterns, species survival, reproduction, metabolic rates, and shifts in the dynamics of the ecosystem.

**Habitat Destruction and Fragmentation**

Deforestation, agriculture, and urbanization are causing habitat destruction and fragmentation and in turn, reduce the space for species living and reproduction. Population dynamics physics can be utilized for the prediction of the effects of these alterations on species survival and biodiversity.

**Ocean Acidification**

Ocean acidification is due to the increase in the absorption of $CO_2$ by the oceans that are produced by increased human activities. The availability of calcium carbonate is affected by this change in pH which is very important for the marine organisms' skeletons and shell formation. Chemical physics explains these phenomena and further investigates how these alterations are affecting ecosystems.

## 8.5 PHYSICS IN THE GEOSPHERE

**The Geosphere**

Solid Earth is known as the geosphere which is the greatest natural capital source. It is a platform for food and is the plant, fossil fuels, and construction materials source of fertilizers. It includes the elements that form the core and crust of the Earth. Starts from the molten rock and heavy metals deep inside to the sand on beaches and mountain peaks.

### 8.5.1 The Earth's Structure

Earth's physical structure is important in studying the activities of volcanoes, plate tectonics, and earthquakes. Earth's layers are crust, core, and mantle.

**The crust**

The layer was made up of solid rock including basalt and granite. Two kinds of crusts are oceanic and continental. Oceanic is thin as well as dense and is composed of basalt while continental is less dense and thick which is composed of granite.

**The mantle**

Below the crust is a mantle which is composed of dense, hot, solid iron and magnesium-rich rock. The thickness of the mantle is up to 2900 km. The mantle's upper part and the crust both form the lithosphere, that is broken into large and small plates.

**The core**

Earth's center is the core which is formed by two parts that is outer core (liquid) and the inner core (solid). Temperature can reach up to 50,000 $^{C}$ in the outer core, which is composed of iron, Nickel, and molten rock.

Knowing about the formation and mechanics of these layers is important to analyze the energy storage and then release as volcanic eruptions and earthquakes which have great impacts on the environment, for example, tsunamis, changing landscapes, and affecting the conditions of the atmosphere. In addition to this, the Earth's physical structure has an impact on the natural resources' distribution such as fossil fuels and minerals which when extracted can have direct effects on environmental sustainability.

### 8.5.2 Energy Inside Earth

The energy of the Earth is represented in volcanic eruptions, movements of the tectonic plates, and geothermal activities. Environmental physics studies these energy flows to know the effect of Earth's structure and its atmosphere including their roles in climate change patterns, contribution to natural hazards, and shaping the landscape.

### 8.5.3 Rocks and Minerals

The study of rocks and minerals tells us about the history of the Earth, how and when it was formed, and which processes have shaped its surface this way. Physical conditions such as temperature, presence of fluids, and pressure have impacted their breakdown, composition as well as transformation. Understanding these conditions is important for the formation of soil, sediment movement, and erosion which are significant for effectively managing natural resources and reducing the environmental effects.

## SUMMARY

This unit introduced the field of Environmental Physics. The first section describes the basics of Environmental Physics. The second section explores the physics involved in the atmosphere and the processes in the atmosphere. The third section focuses on the physics role in the hydrosphere. The fourth section explains the physics role in the biosphere and the final section discusses the physics in the geosphere.

## SELF-ASSESSMENT QUESTIONS

1. What is the difference between weather and climate?

2. What do you know about the water cycle?

3. How do the thermodynamic laws play their role in the ecosystem?

4. What is thermoregulation?

5. Explain the three layers of Earth's surface.

**Unit–9**

# ENVIRONMENTAL CONSERVATION

**Compiled by: Dr. Samia Qadeer**

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# CONTENTS


***Page #***

Introduction ..... 120

Objectives ..... 120

9.1 What is Environmental Conservation? ..... 121

9.1.1 Significance of Environmental Conservation ..... 121

9.1.2 Historical Outlook ..... 121

9.2 Fundamental Environmental Problems ..... 122

9.2.1 Habitat Destruction ..... 122

9.2.3 Environmental Pollution ..... 123

9.2.3 Climate Change ..... 123

9.2.4 Resource Overexploitation ..... 123

9.2.5 Ecological Invaders ..... 124

9.3 Natural Resource Management ..... 124

9.3.1 Principles of Sustainable Resource Management ..... 124

9.3.2 Specific Resource Management ..... 125

9.4 Natural Resource Management ..... 126

9.4.1 Principles of Sustainable Resource Management ..... 126

9.4.2 Targeted Resource Management ..... 127

9.5 Conservation Strategies for Environmental Degradation ..... 127

9.5.1 In-Situ Conservation Practices ..... 128

9.5.2 Ex-Situ Conservation Practices ..... 129

9.6 Biological Conservation Strategies ..... 131

9.6.1 Species Conservation ..... 131

9.6.2 Genetic Conservation ..... 132

9.7 Sustainable Practices ..... 132
9.7.1 Sustainable and Smart Agriculture ..... 133
9.7.2 Sustainable Forest Management ..... 133
9.7.3 Sustainable Fisheries ..... 133
9.7.4 Alternate and Renewable Energy Sources ..... 134
9.7.5 Waste Management and Recycling ..... 134
9.8 Environmental Policies and Legislation ..... 134
Summary ..... 136
Self-Assessment Questions ..... 136
Practical ..... 138
References ..... 139

# INTRODUCTION

This chapter "Environmental Conservation" will provide undergraduate students with a comprehensive understanding of the principles and practices involved in preserving the natural environment. Environmental conservation is essential for maintaining the health and stability of ecosystems, which in turn supports human well-being and economic growth. Through this chapter, students will explore the historical context of conservation efforts, from early resource management practices to the modern environmental movement. They will learn about the significance of conserving biodiversity, protecting natural habitats, and promoting sustainable resource use. By understanding the major threats to the environment, such as habitat destruction, pollution, and climate change, students will be better equipped to address these challenges in their future careers.

The objective of this chapter is to equip students with the knowledge and tools necessary to contribute to sustainable environmental practices and inspire them to become active participants in global conservation efforts. By the end of this chapter, students will have a deeper appreciation for the intricate balance of ecosystems and the urgent need to protect them. They will also gain practical insights into various conservation strategies, sustainable practices, and the role of policy and legislation in environmental protection. This foundational knowledge will be crucial for their academic and professional development in the field of environmental sciences.

# OBJECTIVES

- Understanding the basic concepts and importance of environmental conservation, including its historical development and current relevance
- Recognize the primary threats to the environment, such as habitat destruction, pollution, and climate change
- To Learn about different conservation strategies, both in-situ and ex-situ, and their applications in real-world scenarios
- Understand sustainable practices in agriculture, forestry, fisheries, and energy, and how they contribute to conservation efforts
- Gain insights into international and national policies, laws, and the role of various organizations in environmental conservation

## 9.1 WHAT IS ENVIRONMENTAL CONSERVATION?

With the growing environmental concerns, environmental conservation has gained momentous significance in the past few years. *Environmental conservation is a specific term used for the activities and techniques particularly designated for the sustained environment, its resources and ecological balance*. The scope of environmental conservation ranges from preserving natural habitats, ecosystems, and ecological services to the restoration of degraded environments. Environmental conservation revolves around the following key areas.

i. **P**reservation
ii. **P**rotection
iii. **C**onservation
iv. **R**ehabilitation
v. **R**estoration

### 9.1.1 Significance of Environmental Conservation

The environment though appears as a static entity, but it is a widely misunderstood concept. The environment and the ecosystems are continuously changing entities, these changes are nature's drivers for the provision of a vast range of services, including clean water, clean air fertile land, weather patterns and lots of other resources. Environment and particularly ecosystems provide a wide range of services to mankind. Conservation practices are the efforts that we put forward for the protection of environment against the global challenges like climate change, pollution load and unsustainable practices. The conservation can shortly be identified as the practices that promote sustainable resource consumption and avoid environmental pollution whole.

### 9.1.2 Historical Outlook

The history of environmental conservation has continuously evolved over the period. The pre-historic environmental conservation was driven by human needs; therefore, the initial conservation efforts were to preserve the forest for wood or to protect the wildlife from hunting. The era of modern conservation started in the 19$^{th}$ century, some noteworthy names of that time were John Muir and Gifford Pinchot who widely advocated the protection of the landscapes and drove the cause of sustainable resource consumption. The most significant milestones achieved in that period were the establishment of national parks and wildlife reserves. The era of 1960 to 1970 was identified as the era when most of the environmental challenges were identified, challenges like environmental pollution and biodiversity loss were greatly recognized and with due efforts international agreements focusing on conservation and sustainability were formulated. In recent years the advancements are being conducted with more advanced and integrated approaches. At present the

conservation activities are more focused on species protection, habitat conservation and ecosystem restoration. Advanced tools like environmental DNA and satellite imagery with community inclusion have further helped in achieving the conservation objectives. Current environmental conservation strategies follow the interconnected and holistic approach presented in Figure 1.

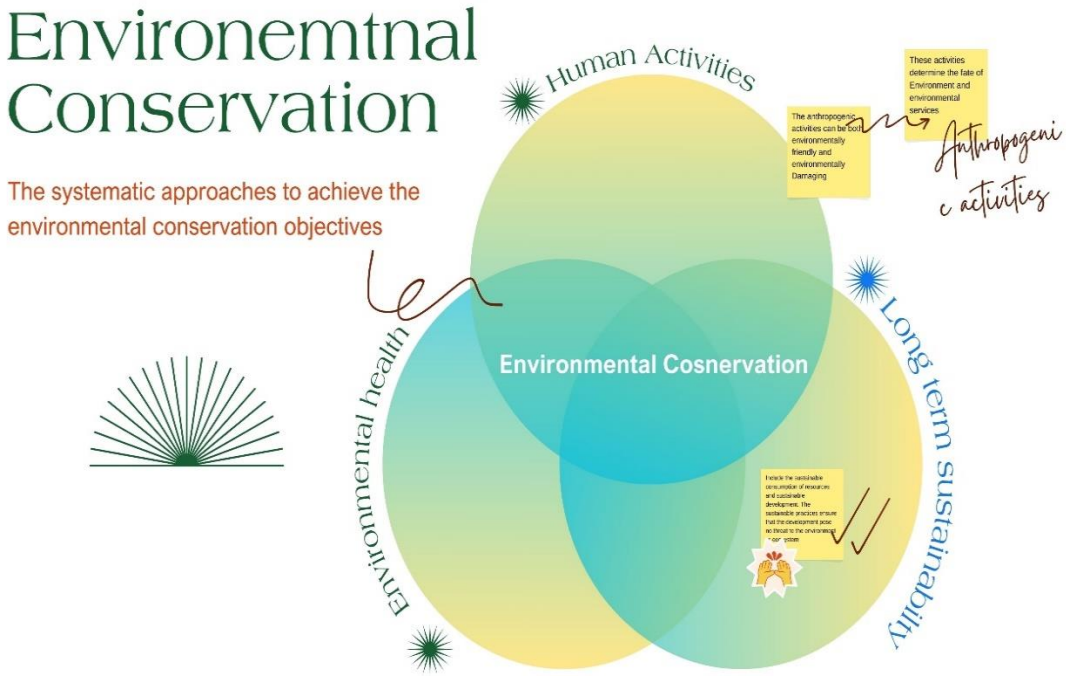


**Figure 9.5:** Interconnected and holistic concept for environmental conservation

## 9.2 FUNDAMENTAL ENVIRONMENTAL PROBLEMS

There is a continuous threat to the environment. Environment and environmental systems are under continuous pressure therefore the stability of the environment is at risk. The environment faces dire challenges posed by humans, and to mitigate these problems knowledge of environmental problems is mandatory.

### 9.2.1 Habitat Destruction

Every living being requires a proper habitat for its developmental needs. So one of the most challenging issues is habitat destruction. Human activities like deforestation, urbanization and continuous land acquisition for agricultural and construction activities continuously decrease the habitat of wild animals. When the animals lose their space they not only lose living space but also their feeding and breeding space such species are pushed towards the verge of extinction. Along with

habitat destruction, habitat fragmentation is also a great threat, that produces the isolation of the population and reduces the genetic diversity.

### 9.2.3 Environmental Pollution

Environmental pollution is broadly classified as air, water and soil pollution. These sources of pollution are serious threats to the environment and environmental systems. The sources of this pollution include different point and non-point sources. For instance, air pollution is caused by vehicular exhaust, agricultural practices, biomass burning and climate change. Similarly, water pollution is caused by untreated sewage, industrial effluents, sedimentation, and agricultural runoff. Where land pollution is caused by improper waste disposal, chemical contamination and degraded lands.

### 9.2.3 Climate Change

Over the last periods, humans have exploited long-term environmental resources. Specifically, fossil fuel burning is linked with long-term changes in the climate. The carbon dioxide from fossil fuels is a well-known culprit that absorbs huge amounts of heat thus the atmospheric temperature rises resulting in global warming. Global warming alters weather patterns and global precipitations. This not only results in extreme weather events but also disrupts the ecosystem. Climate change affects the environment in two ways one by the altered weather patterns and second by the effects of carbon dioxide on various species and environmental balance. Climate change has the potential to affect the environment directly and indirectly. Climate change is a feedback mechanism, where the root of the problem is supplemented by the effects and thus creates a continuous loop that produces more causes of climate change. With the advent of the 21$^{st}$ century climate change has been identified as a major challenge that drives species loss due to human activities. This put 25% of the world's species on the verge of extinction.

**BOX 1**: *The species extinction rate at current era is 10,000 times faster than previous decades, this has placed more than 25% of the global specie at the risk of early extinction compared to their natural rate (Wang et al, 2024).*

### 9.2.4 Resource Overexploitation

Natural resources are developed over a long period, some of these resources are developed over thousands of years. Trees, fossil fuels and other biological resources can take centuries-long duration. When a resource is consumed at a pace faster than it is replenished, the resource is termed a non-renewable resource. Unsustainable resource extraction also termed as resource exploitation like overfishing, deforestation, over-harvesting extraction of minerals, and extraction of fossil fuels

continuously being used much faster rate than their background production rate. This has put continuous pressure not only on the resources but also on the biodiversity and ecological balances. Such resources if not managed will be lost forever and our future generations will have to live without these resources.

### 9.2.5 Ecological Invaders

Some species of plants or animals when deliberately or accidentally introduced into any ecosystem, can outcompete the natural flora and fauna of the system. This introduction can be on purpose or sometimes can be accidental. By altering the habitat, the species can find a more conducive environment, since their natural competitors are missing in that habitat they are in a better position to fight with the naturally present species. Such favours can support them and soon a species can appear as an ecological invader, decreasing the natural organisms. Sometimes these species can carry potential diseases for humans or other inhabitants and thus can be devastating to the ecosystem health.

Since the problem has high gravity, therefore, collective efforts are required for sustainability promotion and to protect the planet's natural resources. A holistic approach with the integration of local perspectives on global issues can result in better environmental management.

## 9.3 NATURAL RESOURCE MANAGEMENT

Natural resource management is defined as the sustainable use and conservation of natural resources like water, minerals, forests, and marine habitats. It seeks to create a balance between human requirements and ecosystem health in order to preserve the long-term availability and quality of resources. Natural resource management is essential because it helps preserve biodiversity, provides ecosystem services, and ensures the long-term development of communities that rely on these resources.

### 9.3.1 Principles of Sustainable Resource Management

Sustainable management of natural resources is associated with several key principles:

1. **Ecological Balance**: Maintaining ecological balance by limiting the consumption of resources to the natural renewal rate.

2. **Economic Viability**: Economic viability refers to managing resources in a way that promotes economic growth while preserving future resources.

3. **Social Equity**: Providing equitable access to resources for all segments of society to meet the requirements of current and future generations.

4. **Adaptive Management**: Employing dynamic strategies to adapt the changing scenarios and emerging information.

5. **Integrated management**: It involves the management of different inter-connected resources and ecosystems holistically.

### 9.3.2 Specific Resource Management

Some natural resources are more important than others because of their impact on the environment and climate. These include water resources, minerals, forests, and marine and coastal environments.

**Water Resources**

Effective water resource management entails the sustainable distribution and use of freshwater for household, industrial, and agricultural applications. Modern approaches such as rainfall collecting, watershed management and the installation of efficient irrigation systems can contribute to improving water availability. Moreover, the protection of water quality by using pollution control technologies and sustainable sanitation systems is also very essential.

**Mineral Resources**

In the case of mineral resources, sustainable management involves controlling environmental impacts while acquiring minerals from nature for economic development and requirements of any country. This includes the use of rehabilitating mine sites, cleaner mining technology and implementation of legislation to prevent illegal mining. Furthermore, it is also important to consider that mining operations not only support the country's requirements but also significantly support the development of local communities.

**Forest Resources**

To achieve a balance between the conservation of forest ecosystems and the need for forest products, sustainable forest management approaches play a crucial role. Modern approaches such as replanting, selective logging and community-based forest management can help to preserve biodiversity and forest health. The forests are operated as substantial carbon sinks therefore, forest protection is also essential to mitigate climate change impacts. In Pakistan, the total tree cover is about 5% of the total land area, with diverse ecosystems ranging from coniferous forests in the northern regions to mangrove forests along the coastal areas of Balochistan and Sindh.

In the current scenario, the country faces several challenges such as encroachment and illegal logging which threaten its forest cover. Conservation efforts are ongoing, including the Billion Tree Tsunami Project, which aims to plant and protect trees across the country to combat deforestation and promote sustainable forest management.

**Marine and Coastal Resources**

The objective of sustainable marine resource management is to maintain marine biodiversity and support the livelihoods of coastal communities. This involves the establishment of marine protected zones, reducing pollution from land-based sources and regulating fishing techniques to minimize the impacts of overfishing. The activities such as fishing, tourism and industrial development can be carried out sustainably through integrated coastal zone management. By using this concept, natural resource management can ensure that the environment remains healthy and productive for future generations while serving the current requirements of society. In the case of Pakistan, the protection of marine biodiversity is crucial for the sustainable management of marine and coastal resources. For instance: (1) establishing marine protected areas like the Astola Island Marine Protected Area, (2) regulating fishing techniques to prevent overfishing, and (3) reducing land-based pollutants like industrial runoff and plastic garbage. Furthermore, the Integrated Coastal Zone Management (ICZM) framework of Pakistan has been developed to ensure the sustainability of activities such as tourism, fishing, and industrial growth.

## 9.4 NATURAL RESOURCE MANAGEMENT

Natural Resource Management refers to the sustainable utilization and conservation of natural resources such as water, forests, minerals, and marine ecosystems. It aims to balance human needs with the health of ecosystems to ensure the long-term availability and quality of these resources. NRM is critical because it helps maintain biodiversity, supports ecosystem services, and ensures the sustainable development of communities dependent on these resources.

### 9.4.1 Principles of Sustainable Resource Management

Sustainable resource management is grounded in several key principles:

1. **Ecological Balance:** Ensuring that resource use does not exceed the natural replenishment rate to maintain ecological integrity.
2. **Social Equity:** Ensuring fair access to resources for all segments of society, addressing the needs of current and future generations.
3. **Economic Viability:** Managing resources in a way that supports economic growth and development without depleting future resources.
4. **Integrated Management:** Considering the interconnectedness of different resources and ecosystems and managing them holistically.
5. **Adaptive Management:** Implementing flexible strategies that can adjust to changing conditions and new information.

### 9.4.2 Targeted Resource Management

**Water Resources**

Effective water resource management involves the sustainable distribution and use of freshwater for agricultural, industrial, and domestic purposes. Techniques such as watershed management, rainwater harvesting, and the implementation of efficient irrigation systems help maintain water availability. Protecting water quality through pollution control measures and sustainable sanitation practices is also essential.

**Mineral Resources**

The sustainable management of mineral resources involves minimizing environmental impacts while extracting minerals necessary for economic development. This includes adopting cleaner mining technologies, rehabilitating mining sites, and implementing regulations to prevent illegal mining activities. Ensuring that mining operations contribute to local communities' development is also important.

**Forest Resources**

Sustainable forest management aims to balance the need for forest products with the preservation of forest ecosystems. Practices such as selective logging, reforestation, and community-based forest management contribute to maintaining forest health and biodiversity. Forest conservation is also crucial for mitigating climate change, as forests act as significant carbon sinks.

**Marine and Coastal Resources**

Sustainable management of marine and coastal resources focuses on protecting marine biodiversity and supporting the livelihoods of coastal communities. This includes establishing marine protected areas, regulating fishing practices to prevent overfishing, and controlling pollution from land-based sources. Integrated coastal zone management ensures that activities such as tourism, fisheries, and industrial development are conducted sustainably.

By adopting these principles and practices, natural resource management can help ensure that the environment remains healthy and productive for future generations while meeting the current needs of society.

## 9.5 CONSERVATION STRATEGIES FOR ENVIRONMENTAL DEGRADATION

The loss and damage that we have put forth to the environment require rehabilitation and mitigation strategies. The conservation strategies are the protection strategies that can help in the mitigation of environmental degradation

and help preserve the biological diversity of the area. These strategies are divided into two types of approaches.

i. In-situ Conservation
ii. Ex-situ Conservation

Each of these approaches is specifically articulated depending upon the specific protection needs.

### 9.5.1 In-Situ Conservation Practices

The conservation efforts that are put forth at the specific location without much disturbance to the species or their habitats are termed in-situ conservation places. The **in-situ** means "at the original place", so the approach follows the practices and strategies that can be used in place to protect a natural resource. A few examples of in-situ conservation are provided below:

**Protected Areas (National Parks, Wildlife Reserves)**

Protected areas consist of national parks and wildlife reserves. These areas are significant for the protection of natural habitats for the wild flora and fauna. There are no construction or technical testing is allowed in national parks and wildlife reserves. These areas are maintained at their natural threshold, or close to the natural conditions. These protected areas are like refugees for the species under pressure where they can thrive better as these areas are protected by human encroachment in any form. Protection of these areas not only helps species to grow better but also helps to maintain ecological services like water purification, carbon sequestration, noise reduction, soil fertility and air purification. A well-protected area will involve enforced regulations, ecological monitoring and community conservation efforts. Pakistan has also different types of protected areas presented in Table 01 below (data extracted from the data based on the July 2018 version of the World Database on Protected Areas (WDPA).

**Table 1:** United Nations list of protected areas of Pakistan

| S. No. | Type of Protected Area | Number |
|---|---|---|
| 1 | Ramsar Site, Wetland of International Importance | 19 |
| 2 | UNESCO-MAB Biosphere Reserve | 01 |
| 3 | Game Reserve | 66 |
| 4 | National Park | 14 |
| 5 | Private Reserve | 01 |
| 6 | Other Area | 07 |
| 7 | Wildlife Sanctuary | 70 |

**Community-based Conservation**

Another type of in-situ conservation is community-based conservation. This method involves incorporating the local communities to achieve conservation goals like sustainable resource management and biodiversity conservation. Along with the awareness strategies about the significance of the protection these efforts empower the local communities considering them integral stakeholders who are involved in the decision-making process for the conservation efforts. This two-way benefit provides the conservation and reserve for the local community along with the promotion and protection of local knowledge. Thus, the strategy not only offers incentives for the locals but also helps the science of conservation through significant addition of local knowledge**.** One of the most successful examples of community-based conservation efforts is the protection of the national animal of Pakistan, the Markhor, a species once declared endangered but now under a low-threat list of conservation efforts.

### 9.5.2 Ex-Situ Conservation Practices

Sometimes onsite conservation efforts are hard to conduct, in that situation the species or resource has to be extracted from their natural habitat and protected outside. This type of protection effort is termed an ex-situ conservation practice. The term **ex-situ** means "outside" thus the protection strategies or practices conducted outside the natural habitat. A few examples of ex-situ conservations are given below:

**Zoos and Botanical Gardens**

The most well-known examples of ex-situ conservation are botanical gardens and zoos. This conservation effort is providing ideal conditions outside and thus propagating the organisms under controlled conditions. This type of conservation effort is suggested for the species when they are critically endangered or their natural habitat is destructed beyond the rehabilitation (for instance oil spills, chemical spills, or radioactive waste). Such institutes play a vital role in providing educational and research platforms as well. The animals are maintained in captivity and after a specific period, they are re-introduced into the natural habitat.

**Seed Banks and Gene Banks**

Another advanced form of conservation strategy is the establishment of seed and gene banks. Such reserves help in restoring the species that are on the verge of biological extinction. These banks are designated to store genetic material from plants, animals and microorganisms thus they can be used for conservation practices through research and development. The seed banks are very important in developing stress-resistant and disease-resistant crop varieties. Genetic repositories serve as storage centres for natural or wild varieties and thus can be used in times of need to develop resistant varieties. Other benefits that can be achieved from the

seed banks include restoring the degraded varieties, developing new breeds and propagation of significant genes. Thus these seed and gene banks are significant contributors in ensuring food security worldwide.

| **Difference between in-situ and ex-situ conservation practices** | |
|---|---|
| **In-situ Conservation** | **Ex-situ Conservation** |
| In the home approach, conserve the species in their natural habitat. | Species are extracted from their natural habitat and are protected in artificial environments. |
| No dislocation required | Require dislocation |
| Requires efforts to protect the natural habitat | No requirement for the protection of natural habitats on the spot, however, conservation is required before releasing the species into wild |
| Offers additional benefit of ecosystem protection and thus associated ecological services | Offers the benefit of protecting endangered species with a greater success rate |
| Wild protection doesn't offer genetic or germplasm storage | Offers genetic reserve that can help in addressing future challenges as well |
| No need to create any habitat or control any physical or biological practice | Requires strictly controlled habitat provision with all the services |
| More cost-effective | Comparatively costly |

For successful conservation efforts integration of both conservation practices along with stakeholder involvement is necessary. There is a strong link between the conservation and protection of the environment the sustainable development healthily and productively.

**BOX 2:** *The primary seed and gene banks in Pakistan that are essential to the preservation of plant genetic resources are:*

- *National Agricultural Research Centre (NARC) Seed and Gene Bank*
- *Pakistan Agricultural Research Council (PARC) Gene Bank*
- *Punjab Seed Corporation (PSC) Seed Bank*
- *Sindh Seed Corporation (SSC) Seed Bank*

## 9.6 BIOLOGICAL CONSERVATION STRATEGIES

The prime purpose of this protection strategy is to protect biological diversity. The conservation efforts mainly target the conservation species or species diversity.

### 9.6.1 Species Conservation

**Endangered and Threatened and High-risk Species**

The species whose population is declining at an alarming pace and can soon become endangered are termed threatened species. Where the species with a low population and is soon to be extinct is termed an endangered species. Some species have only a limited number of individuals alive such species are under dire pressure of extinction and require emergency steps for protection, such species are called high-risk species. Conservation efforts are targeted to identify and protect precarious habitats with strict monitoring and rehabilitation strategies. The International Union for Conservation of Nature provides a list of species based on their current situation and categorizes them accordingly.

**Captive Breeding Programs**

Captive breeding programs are ex-situ conservation strategies that include zoos, aquariums, and specialized breeding centres. These are used for endangered and high-risk species. The species under the threat of imminent extinction is protected and provided with ideal conditions to grow, develop and breed. Thus, when a considerable number is achieved the species can be released into the wild.

**Habitat Conservation**

The habitat conservation strategy is key to the rehabilitation of any speciesit encompasses the restoration of ecology and wildlife corridors.

**Restoration Ecology:** involves repairing the natural ecosystem and thus bringing it back to its natural state. The most widely applied techniques in restoration ecology are reforestation, restoration of wetlands and protection its services. Projects like the restoration of the Aral Sea wetlands and Yellow Stone National Park exhibit the potential of restoration ecology in reversing environmental damage and supporting local communities.

**Wildlife Corridors**: are the pathways that connect the fragmented habitat, these habitat connections help the wildlife to migrate, and disperse across the corridors. Such strategies help the species to adapt to the changing climate. These corridors are structurally engineered designs that allow wildlife to move freely without any disturbance across the two fragmented habitats. The threats and challenges faced due to isolated populations are greatly reduced by the provision of wildlife

corridors. One of the successful examples is the Yellowstone to Yukon Conservation Initiative, which helps in providing the habitat connection for species like grizzly bears and wolves.

### 9.6.2 Genetic Conservation

The conservation efforts are incomplete without securing the future of any population or species, therefore genetic conservation is an integral part of any protection effort. Conserving genetic diversity inside species is critical for the lasting endurance and adaptation to environmental challenges of the species. These efforts involve conserving gene pools using seed banks, gene banks, and breeding programs. Genetic diversity is significant in increasing resilience against diseases and other environmental stresses that maintain diverse genetic traits. This diversity ensures resilience against diseases, climate fluctuations, and other stressors.

**Genetic Engineering and Biotechnology**

The recent developments with the advent of molecular and genetic science have provided new grounds for conservation strategies. For instance, genetic engineering and biotechnology can offer innovative solutions to conservation strategies. These new advancements offer disease resistance in endangered species similarly genetically modified organisms (GMOs) can provide a solution as genetic refugees for rescue operations.

Overall by adopting the key biodiversity conservation practice we can safeguard our environment, restore ecosystems and conserve the species providing a substantial contribution in paving wat to inherit natural heritage to our future generations.

## 9.7 SUSTAINABLE PRACTICES

The conservation and preservation practices alone don’t serve the purpose of long-term environmental stability goals. Therefore, to ensure the long-term goals sustainable practices are pivotal for balancing human development and ecosystems. These practices involve a broad scope ranging from sustainable practices in food production (i:e agriculture and aquaculture), to sustainable harvesting of resources from forests and other natural resources like water, soil and grasslands. Sustainable practices ensure the judicial use of resources without compromising future needs and developments. Sustainable practices though are required in every field of life here we will discuss a few of them.

### 9.7.1 Sustainable and Smart Agriculture

This practice involves the standard land harvesting protocols with the viable use of chemicals. The practice emphasized crop rotation, organic farming, and integrated pest control. Thus, it promotes avoiding the use of pesticides and chemical fertilizers. The realization of the damaging consequences of the use of chemicals in agriculture has been recognized by the global efforts to shift towards organic agriculture. It has been mentioned in the Global Organic Agriculture Performance Report that organic agriculture in 2021 surged up to 75.8 million hectares of agricultural land. This also includes techniques like, precision agriculture, precision farming and remote sensing data for optimal use of resources. The use of AI for precision monitoring of soil health and water requirements has broken new ground in achieving smart agriculture objectives with greater accuracy and resource optimization. Similarly, integrated smart agriculture practices, including automated irrigation systems and data-driven pest management, can provide more accurate information that can be used to increase agricultural productivity and ecosystem balance. These tools allow agriculturalists to articulate conversant choices that decrease the dependence on chemical fertilizers, thus, decreasing greenhouse gas production, and preserving biodiversity by abating the ecological footprint of agriculture.

### 9.7.2 Sustainable Forest Management

Forests are home to lots of resources and services to society. Therefore, the forests of the world are under continuous threat of loss. These forests are home to a wide variety of living organisms providing significant services to the ecosystem and mankind. Therefore, sustainable forest management also promotes the judicial use of forests and their resources. Such a practice is mandatory if we want to have a continuous supply of resources from the forest ecosystem. Benefits. Forest Stewardship Council (FSC) certification confirms that forest goods arise from sensible forest management. As of 2021, FSC-certified woods cover about 210 million hectares globally. Reforestation goals, such as the Trillion Trees Initiative, intend to reinstate 1 trillion trees by 2030 to fight deforestation, climate change and global warming.

### 9.7.3 Sustainable Fisheries

The fishery is one of the most primitive businesses of mankind. Humans have used the ocean products and fishes since ancient times however the overharvesting of the fish from the ocean or river not only put the hunted species at risk of extinction but also the other interrelated species become threatened. The Marine Stewardship Council (MSC) certification endorses viable casting techniques, to add to the success of MSC over 20.4% of marine fishery programs were MSC certified till 2021, and 20.4% of global marine fisheries were MSC certified. Moreover,

aquaculture has grown significantly, providing a sustainable alternative to wild capture. The Food and Agriculture Organization (FAO) indicates that aquaculture created 82.1 million tons of fish in 2020.

### 9.7.4 Alternate and Renewable Energy Sources

The prime sustainable practice required at the current time is to focus on more renewable energy resources. Since fossil fuel consumption has been known for its prime role in the global climate crisis, finding other energy resources can help conserve resources indirectly by minimizing calamities. A wide range of renewable resources are available naturally including wind, solar, hydro and tidal power. According to the International Energy Agency (IEA), the use of renewable energy has grown considerably for instance in 2021 28% of the global electricity was generated by renewable resources, and each sector like solar grew by 21% and wind by 11%. However, in developing countries, the transition is relatively slow.

### 9.7.5 Waste Management and Recycling

Efficient waste management and recycling activities are important for lessening environmental effects and preserving natural resources. According to the World Bank in 2020 only 17% of the waste produced worldwide was recycled, where the rest ended up in landfills or open dumps, Heaps of solid waste are dumped in water bodies and water channels. Approaches like reducing waste generation, improving recycling structure, and stimulating circular economy models are required for sustainable waste management. One of the successful examples of waste recycling circular economy is the European Union's Circular Economy Action Plan which aims to grow recycling rates and thus reduce waste to the other channels, aiming at 65% recycling for municipal waste by 2035.

## 9.8 ENVIRONMENTAL POLICIES AND LEGISLATION

The conservation efforts require legal backup to support in different fields. Some of the noteworthy efforts are detailed below:

**National Policies and Laws:** National policies and laws are structured frameworks to address the particular environmental needs of any nation. In Pakistan, the key policies related to the environment include the Pakistan Environmental Protection Act (1997), which offers a base for environmental regulation and the constitution of the Pakistan Environmental Protection Agency, the Second significant one is the National Climate Change Policy (2012) that outlines the approaches for moderating

and adapting to climate change impacts. The Forest Policy (2015) of Pakistan is also an addition to secure sustainable forest management and land use plans. The policy also promotes afforestation efforts to fight deforestation and habitat loss.

**BOX 3:** *List of Existing Environmental Policies of Pakistan*

- *Pakistan Environmental Protection Act (1997)*
- *National Environmental Policy (2005)*
- *Pakistan Climate Change Act (2017)*
- *National Climate Change Policy (2012)*
- *National Forest Policy (2015)*
- *Pakistan Clean Green Initiative (2018)*
- *The Punjab Water Act (2019)*
- *Pakistan's Alternative and Renewable Energy Policy (2019)*
- *National Energy Efficiency and Conservation Policy (2020)*
- *National Adaptation Plan (NAP) (2021)*
- *National Clean Air Policy (2023)*

**International Agreements:** Various international agreements are also in line to strengthen conservation efforts. International agreements like the Convention on Biological Diversity (CBD) established in 1992 played a pivotal role in the promotion of sustainable development through ecological and biological conservation. The CBD also provides guidelines to structure the national policy guidelines.

**Role of Government and Non-Governmental Organizations (NGO):** The synergy of the government and NGOs plays a significant role in achieving the objective of sustainable resource consumption. In Pakistan, the government implements policies and regulations, where NGOs, like the Worldwide Fund for Nature (WWF) Pakistan and the Indus Earth Trust, are involved in common maintenance projects, educate the masses, and promote policy expansions.

## SUMMARY

This chapter covers the fundamental environmental challenges that are being governed by human activities. The chapter also focuses on the threats challenges, strategies and policies to address these challenges. Natural resource management highlights sustainable practices to preserve significant resources. These practices encompass sustainable agriculture, precision agriculture, organic farming, sustainable harvest of resources, reforestation, forest management, forest and marine certification and shift to renewable energy resources to minimize the environmental impacts.

The conservation efforts need to integrate the in-situ and ex-situ management practices that can ensure long-term environmental conservation goals. These goals can only be achieved through some legal support. Thus, national and international agreements can help to cope with the environmental challenges and thus can pave the way to better environmental health.

## SELF-ASSESSMENT QUESTIONS

1. Fill in the blanks
   a. The is the specific term used for the activities and techniques particularly designated for the sustained environment its resources and ecological balance.

   b. The species extinction rate at present is _____times faster than in previous decades

   c. The conservation efforts that are put forth at the specific location without much disturbance to the specie or their habitats, are termed as _ _____conservation.

   d. The areas consist of national parks and wildlife reserves.

   e. The species whose population is declining at an alarming pace and can soon become endangered are termed _____species.

2. What are the fundamental environmental challenges to the environment?
3. Explain different types of conservation practices with example
4. How NGOs and government can play a role in environmental conservation

5. What are sustainable conservation practices?
6. What are different types of conservation practices?
7. How smart agriculture can play a role in sustainability?

**ACTIVITY**

Keeping in mind any particular animal or bird species of your area prepare a conservation plan mentioning the conservation-specific strategies.

## PRACTICAL

1. Enlist the major environmental challenges of your area.
2. How clear is the water of your area, compare it with the turbidity chart provided at the end?
3. Is there winter smog in your area? What is the color of the smog? (hint: it can be grey haze or brown haze)
4. Prepare a table mentioning the temperature of your specific area for the month of June and January (hint: you can use weather app of your cell phone).

Reference question 2:

| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

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